

ISO15765 Protocol

This document is written to satisfy the general user who has an understanding of the ISO15765 protocol, a common standard for petrol cars and small cars, and provides a manual capture of ISO15765 protocol data and manual parsing of a method. This is a way to get started and learn about automotive CAN communication.

The OBDII interface of petrol cars and small cars generally has ISO15765 protocol, and the acquisition of ISO15765 data is to acquire data by sending corresponding request commands, and the car will not respond to the corresponding data without sending request commands. (This is very obvious for Volkswagen cars. Between the OBD interface of the Volkswagen and the CAN network of the car, there is an OBD gateway which blocks and filters the data of the CAN network of the car, there will be no signals, waveforms or data in the OBD port. (At this point, if you want to get the data, you must request the data by sending a specific request command through the standard specified in the ISO15765 protocol). For other models, the OBD generally has a certain periodicity of data that is actively sent out.

ISO15765 has two forms of CAN data, standard CAN and extended CAN, regardless of which form of CAN data your vehicle is currently using. Whichever form of CAN data your vehicle is currently using, apart from the different CAN identifiers (i.e. the standard CAN command identifier is 0x07DF and the extended CAN command identifier is 0x18DB33F1), the other formats and algorithms are all identical.

Whether the vehicle is standard or extended, it is necessary to send a standard frame request command and an extended frame command to see if the device has a response.

OBD equipment on the market, OBD box, Volkswagen 5053 line, automotive diagnostic equipment, etc. are automatically completed by the software to obtain and parse the data. The principle is the same as that described in this article, but it is done automatically by the software. If you need to analyse the data automatically, you can buy our USBCAN-OBD

model!



In addition to the ISO15765 protocol, the car also uses the manufacturer's customised protocol, which is not public. ISO15765 can therefore only access a small part of the car's data, and some of the data in ISO15765 is not accessible, depending on the model.

The following are examples of two CAN data acquisition commands

Standard CAN (More Common)

Send Request Command:

ID=0x000007DF Standard Frame, Data Frame, Data Bits: 02 01 0C 00 00 00 00 00

Receive Response Data:

ID=0x000007E8 Standard Frame, Data Frame, Data Bits: 04 41 0C 0B 18 AA AA

Extended CAN (less common)

Send request command:

ID=0x18DB33F1 Extended Frame, Data Frame, Data Bits: 02 01 0C 00 00 00 00 00

Receive Response Data:

ID=0x18DAF1XX Extended Frame, Data Frame, Data Bits: 04 41 0C 0B 18 AA AAAA

1. Request Data Flow

Data stream request command format (in standard frames, for example)

Send request command:

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                                BYTE  0  1  2  3  4  5  6  7
ID=0x000007DF      Standard Frame, Data Frame, Data Bit: 02 01 XX 00 00 00 00 00

```

In the above instruction, the third byte BYTE2 is XX, which is a variable called PID, which can be interpreted as an index number of a physical parameter. If you need any parameter, you can check the table and fill in the corresponding value (hexadecimal). Other values are fixed.

Response data is received:

```

                                BYTE  0  1  2  3  4  5  6  7
ID=0x000007E8      Standard Frame, Data Frame, Data Bit: 06 41 XX FF FF FF FF 00

```

Above this frame instruction, the third byte BYTE2 for XX, for the variable, called PID, can be understood as a physical parameter of an index number, the request instruction request what PID, this place will return the corresponding PID, the next few bytes is the corresponding physical parameter lookup table, manual parsing.

Usage

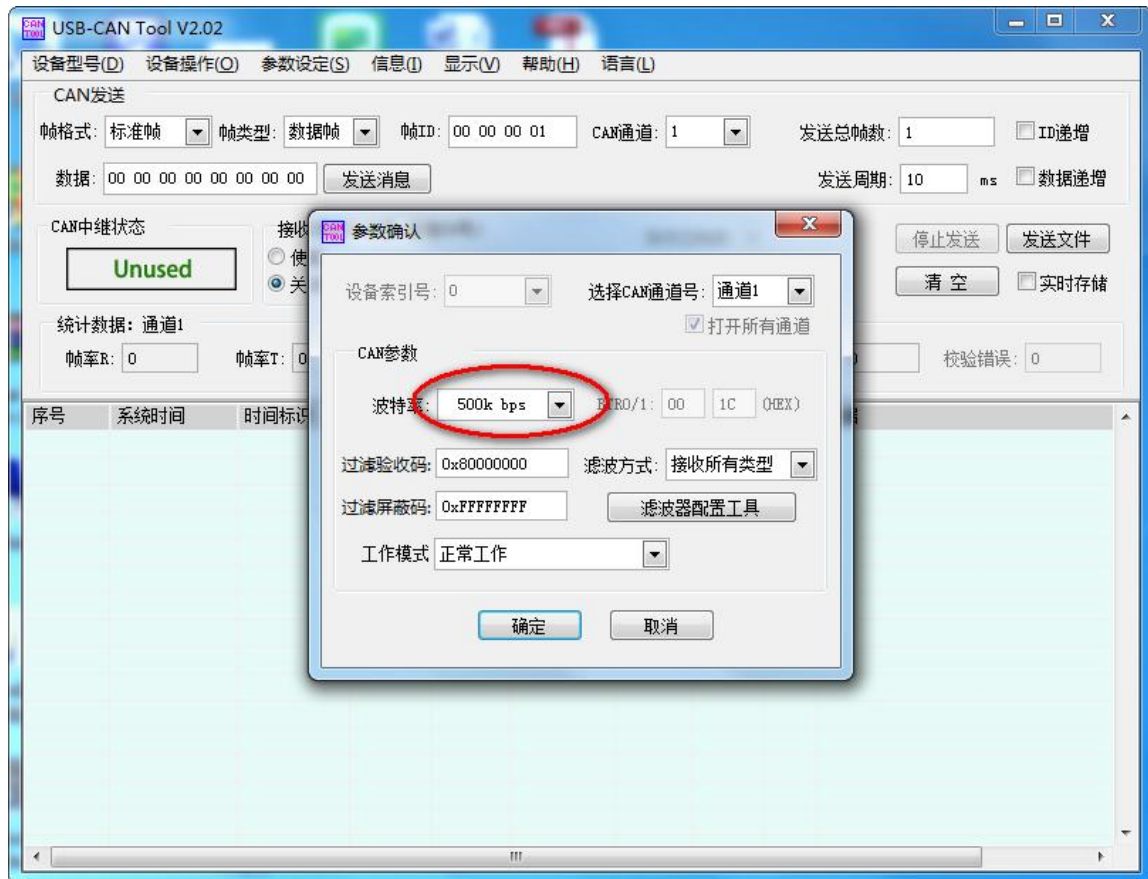
If you have purchased the version with OBD adapter, you can use the matching OBD adapter directly, the OBD male connector is directly plugged into the female connector of the car OBD, the OBD adapter lead red (internally connected to the OBD pin 6) connects to the CANH of the USBCAN device; the OBD adapter lead blue (internally connected to the OBD pin 14) connects to the CANL of the USBCAN device; the other pins are not connected. The built-in resistor of the USBCAN device is not connected to the bus (USBCAN models R+ and R- are left blank, and the dip switches of the CANalyst-II analyzer are all set to the upper OFF position).

If you purchase a version without an OBD adapter, you can lead the CAN cable yourself. Pin 6 of the female automotive OBD header connects to CANH of the

USBCAN device; pin 14 of the female automotive OBD header connects to CANL of the USBCAN device.



Start the device, select 500K for the baud rate and leave all other parameters unchanged.

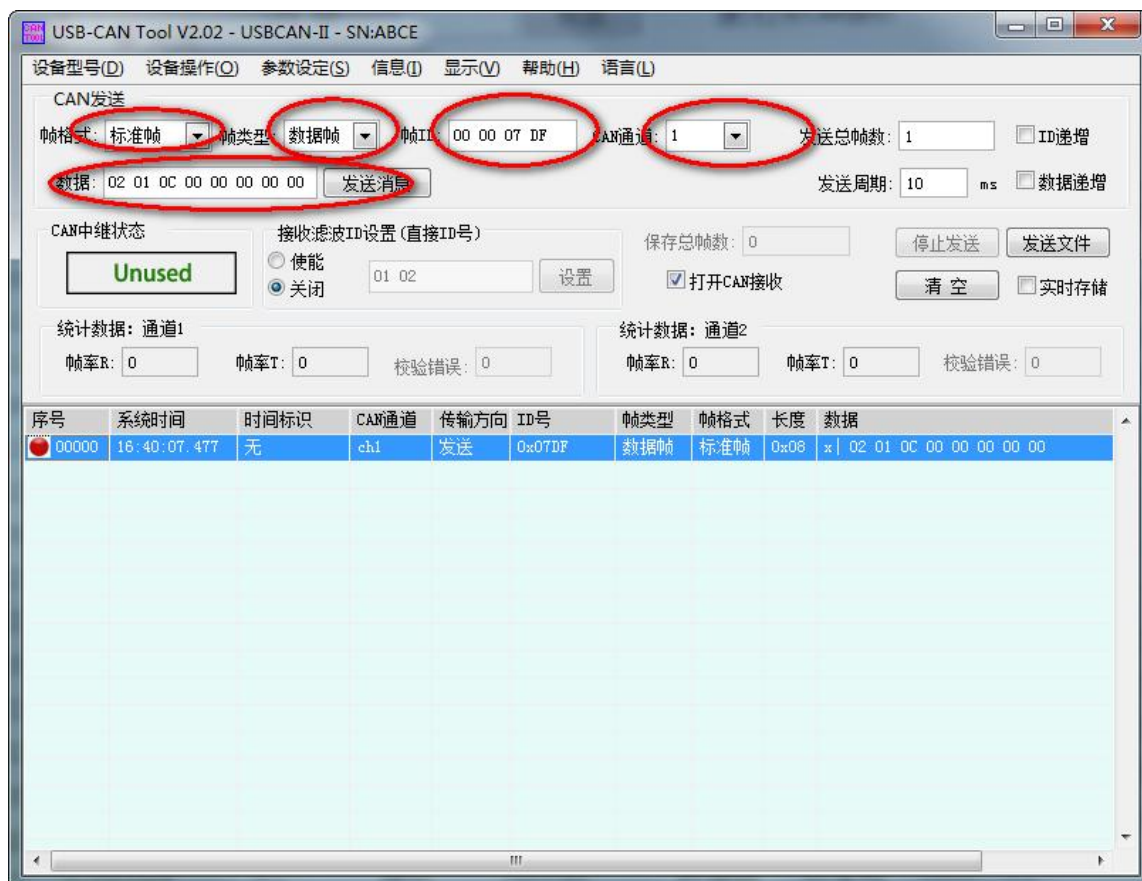


Here, a request for engine speed is made and ignition is required.

By checking the PID correspondence table below, the engine speed corresponds to a PID value of 0x0C, so that the

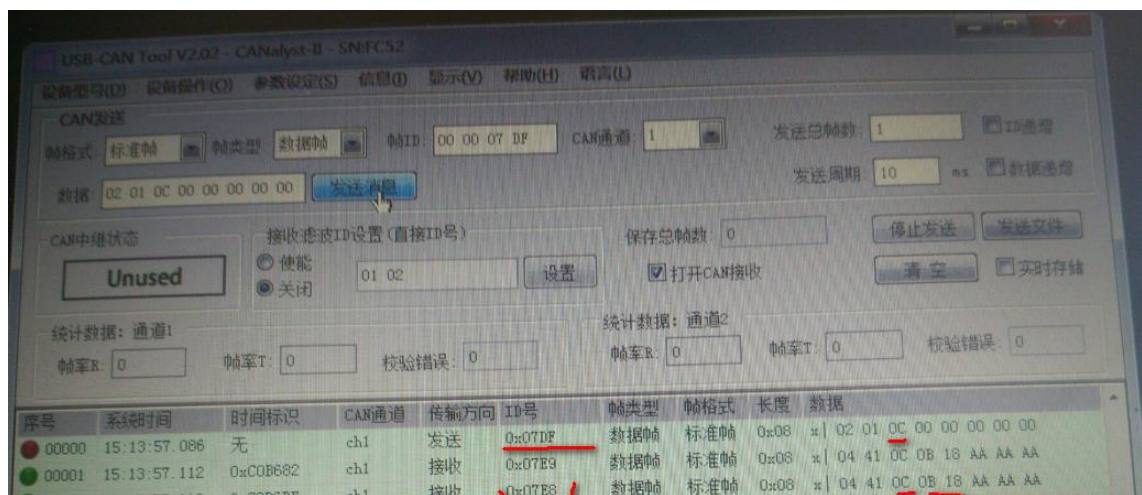
Send the request command:

ID=0x000007DF Standard frame, data frame, data bit: 02 01 0C 00 00 00 00 00 00



If the connection is normal, you can get the response data.

Response data is received:



BYTE 0 1 2 3 4 5 6 7
ID=0x000007E8 Standard Frame, Data Frame, Data Bits: 04 41 **0C** **0B** 18 AA AA AA

Receive data with ID other than ID=0x000007E8 and ignore it. Focus only on data frame with ID=0x000007E8, BYTE2=0C

Check the PID table

Engine Speed	0C	BYTE3,4/4	RPM
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BYTE3,4/4 means that BYTE3 and BYTE4 form a 16-bit hexadecimal number. Then divide by 4 to get the speed RPM as shown above, BYTE3=0x0B, BYTE4=0x18, RPM=0x0B18/4=2840/4=710 rpm.

2、Read the Fault Code

Fault code request command format (for example standard frames)

Send request command:

BYTE 0 1 2 3 4 5 6 7

ID=0x000007DF Standard Frame, Data Frame, Data Bits: 01 03 00 00 00 00 00 00

All values are fixed.

Response data is received:

BYTE 0 1 2 3 4 5 6 7

ID=0x000007E8 Standard Frame, Data Frame, Data Bits: 06 43 02 03 04 05 06 07

Above this frame of instructions, the third byte BYTE2 is the number of fault codes, and every two bytes after that represent the code number. The above data code number is P0304, P0506, which can be interpreted as an index number of the fault code, look up the table, manual parsing.

3、Clear the Fault Code

Fault code request command format (for example standard frames)

Send request command:

BYTE 0 1 2 3 4 5 6 7

ID=0x000007DF Standard Frame, Data Frame, Data Bits: 01 04 00 00 00 00 00 00

All values are fixed.

Response data is received:

BYTE 0 1 2 3 4 5 6 7

ID=0x000007E8 Standard Frame, Data Frame, Data Bits: 01 04 00 00 00 00 00 00

4、Parameter PID Control Table

Entry	Nibblesxx (PID)	Regulation	Unit
Number of Fault Code Storage	01	BYTE3&0x7FF	
Supports Fire Detection	01	BYTE4.bit0 0-- (Not Supported) 1--(Supported)	
Supports Integrated Component Monitoring	01	BYTE4.bit2 0-- (Not Supported) 1--(Supported)	
Fuel System Monitoring Ready	01	BYTE4.bit5 0--(Complete or N/A) 1--Incomplete(Supported)	
Support:Catalytic Converter	01	BYTE5.BIT0 0--(Not Supported) 1--(Supported)	
Support:Fuel Evaporative Emission System	01	BYTE5.BIT2 0--(Not Supported) 1--(Supported)	
Support:Air Conditioning Refrigerant	01	BYTE5.BIT4 0--(Not Supported) 1--(Supported)	
Support:Oxygen Sensor Heater	01	BYTE5.BIT6 0--(Not Supported) 1--(Supported)	
Catalytic Converter Status	01	BYTE6.BIT0 0-- (Complete or N/A) 1--Incomplete(Supported)	
Fuel Evaporative Emissions System Status	01	BYTE6.BIT2 0-- (Complete or N/A) 1--Incomplete(Supported)	
Air Conditioning Refrigerant Status	01	BYTE6.BIT4 0-- (Complete or N/A) 1--Incomplete(Supported)	
Oxygen Sensor Heater Status	01	BYTE6.BIT6 0-- (Complete or N/A) 1--Incomplete(Supported)	
Fault Indicator Status	01	BYTE3.bit7 0--OFF 1--ON	

Supports Fuel System Monitoring	01	BYTE4.bit1 0-- (Not Supported) 1--(Supported)	
Fire Monitoring Ready	01	BYTE4.bit4 0-- (Complete or N/A) 1--Incomplete(Supported)	
Comprehensive Component Monitoring Ready	01	BYTE4.bit6 0-- (Complete or N/A) 1--Incomplete(Supported)	
Support:Heater Catalytic Converter	01	BYTE5.BIT1 0--(Not Supported) 1--(Supported)	
Support:Secondary Air Injection System	01	BYTE5.BIT3 0--(Not Supported) 1--(Supported)	
Support: Oxygen Sensor	01	BYTE5.BIT5 0--(Not Supported) 1--(Supported)	
Support: Exhaust Gas Recirculation System	01	BYTE5.BIT7 0--(Not Supported) 1--(Supported)	
Heated Catalytic Converter Status	01	BYTE6.BIT1 0-- (Complete or N/A) 1--Incomplete(Supported)	
Secondary Air Injection System Status	01	BYTE6.BIT3 0-- (Complete or N/A) 1--Incomplete(Supported)	
Oxygen Sensor Status	01	BYTE6.BIT5 0-- (Complete or N/A) 1--Incomplete(Supported)	
Exhaust Gas Recirculation System Status	01	BYTE6.BIT7 0--(Complete or N/A) 1--Incomplete(Supported)	
Error Codes That Generate Freeze	02	byte3,4 Hexadecimal	
Fuel System Status B1	03	BYTE3 01:OBD II(California ARB) 02:OBD(Federal EPA) 03: OBD and OBD II 04:OBD I 05:Not intended to meet	
Fuel System Status B2	03	BYTE4 B1:S---- B2:S---- Each of the 4 '-'s after "B1:S" represents one bit (lower 4 bits), when the corresponding bit is 1, the corresponding number is displayed, e.g., "1234", the higher 4 bits are the same.	
Calculated Load Value	04	BYTE3*100/255	%
Engine Coolant Temperature	05	BYTE3-40	°C
Short-Term Fuel Correction (Cylinder Group 1)	06	BYTE3*100/128-100	%

Long-Term Fuel Correction (Block 1)	07	BYTE3*100/128-100	%
Short-Term Fuel Correction (Cylinder Bank 2)	08	BYTE3*100/128-100	%
Long-Term Fuel Correction (Block 2)	09	BYTE3*100/128-100	%
Fuel Pressure	0A	BYTE3*3	kPaG
Absolute Intake Manifold Pressure	0B	BYTE3	kPaA
Engine RPM	0C	BYTE3,4/4	RPM
Speed	0D	BYTE3	km/h
Ignition Timing	0E	BYTE3/2-64	°
Inlet Temperature	0F	BYTE3-40	°C
Air Flow	10	BYTE3,4*0.01	g/s
Throttle Positions	11	BYTE3*100/255	%
Secondary Air Requests	12	BYTE3 0:0 bit0=1:UpStream 1 CAT.C bit1=1:DownStream 1 CAT.C bit2=1:Atmosphere/off Other: 0 Display low when two or more are present at the same time	
Oxygen Sensor Location	13	BYTE3 B1:S---- B2:S----Each of the 4 '-'s after "B1:S" represents one bit (lower 4 bits). When the corresponding bit is 1, the corresponding number is displayed, e.g. "1234". Higher 4 bits are the same as "1234".	
Oxygen Sensor Output Voltage (Cylinder Group 1, Sensor 1) Sensor 1)	14	BYTE3*0.005	V
Short Term Fuel Correction (Cylinder Group 1, Sensor 1) Sensor 1)	14	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 1, Sensor 2) Sensor 2)	15	BYTE3*0.005	V
Short-Term Fuel Correction (Cylinder Group 1, Sensor 2) Sensor 2)	15	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 1, Sensor 3) Sensor 3)	16	BYTE3*0.005	V

Short Term Fuel Correction (Cylinder Group 1, Sensor 3) Sensor 3)	16	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 1, Sensor 4) Sensor 4)	17	BYTE3*0.005	V
Short Term Fuel Correction (Cylinder Group 1, Sensor 4) Sensor 4)	17	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 2, Sensor 1)	18	BYTE3*0.005	V
Short Term Fuel Correction (Cylinder Block 2, Sensor 1) Sensor 1)	18	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 2, Sensor 2)	19	BYTE3*0.005	V
Short Term Fuel Correction (Cylinder Group 2, Sensor 2) Sensor 2)	19	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 2, Sensor 3)	1A	BYTE3*0.005	V
Short-Term Fuel Correction (Cylinder Block 2, Sensor 3) Sensor 3)	1A	BYTE4*100/128-100	%
Oxygen Sensor Output Voltage (Cylinder Group 2, Sensor 4)	1B	BYTE3*0.005	V
Short Term Fuel Correction (Cylinder Block 2, Sensor 4) Sensor 4)	1B	BYTE4*100/128-100	%
Vehicle or Engine Certified OBD Requirements	1C	BYTE3 01:OBD II(California ARB) 02:OBD(Federal EPA) 03: OBD and OBD II 04:OBD I 05:Not intended to meet	
Oxygen Sensor Location	1D	BYTE3 B1:S---- B2:S----Each of the 4 '-'s after "B1:S" represents one bit (lower 4 bits), when the corresponding bit is 1, the corresponding number is displayed, e.g., "1234", the higher 4 bits are the same. "B1:S"	

PTO (Power Take-Off) Status	1E	BYTE3.bit0 0:PTO not active 1:PTO active	
Time After Engine Start	1f	BYTE3,4 Two-byte decimal Display	sec
Driving Distance After the Malfunction Indicator Light Is on Distance	21	BYTE3,4 Two-byte decimal Display	Km
Relative Fuel Pressure	22	BYTE3,4*5177.27/65535	kPa
Fuel Rail Pressure	23	BYTE3,4*10	kPa
Equivalent ratio (λ) (cylinder group 1, sensor 1)	24	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 1, Sensor 1) Sensor 1)	24	BYTE5,6*7999/65535	mV
Equivalent ratio (λ) (cylinder group 1, sensor 2)	25	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 1, Sensor 2) Sensor 2)	25	BYTE5,6*7999/65535	mV
Equivalent ratio (λ) (cylinder group 1, sensor 3)	26	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 1, Sensor 3) Sensor 3)	26	BYTE5,6*7999/65535	mV
Equivalent ratio (λ) (block 1, sensor 4)	27	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 1, Sensor 4) Sensor 4)	27	BYTE5,6*7999/65535	mV
Equivalent ratio (λ) (block 2, sensor 1)	28	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 2, Sensor 1) Sensor 1)	28	BYTE5,6*7999/65535	mV
Equivalent ratio (λ) (cylinder group 2, sensor 2)	29	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 2, Sensor 2) Sensor 2)	29	BYTE5,6*7999/65535	mV
Equivalent ratio (λ) (block 2, sensor 3)	2A	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 2, Sensor 3) Sensor 3)	2A	BYTE5,6*7999/65535	mV

Equivalent ratio (λ) (block 2, sensor 4)	2B	BYTE3,4*1.999/65535	
Oxygen Sensor Voltage (Cylinder Group 2, Sensor 4) Sensor 4)	2B	BYTE5,6*7999/65535	mV
Exhaust Gas Recirculation Directive	2C	BYTE3*100/255	%
Exhaust Gas Recirculation Error	2D	BYTE3*100/128-100	%
Fuel Vapor Purification	2E	BYTE3*100/255	%
Fuel Level Input	2F	BYTE3*100/255	%
Number of Warm-Ups After Clearing Fault Codes	30	BYTE3 Decimal System	
Distance Traveled After Clearing Fault Codes	31	BYTE3,4(DEC)	
Fuel vapor system pressure	32	BYTE3,4*1/4	Pa
Atmospheric Pressure	33	BYTE3 Decimal System	kPa
Equivalent Ratio (λ) (Cylinder Group 1, Sensor 1)	34	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 1, Sensor 1) Sensor 1)	34	BYTE5,6*256/65535-128	mA
Equivalent Ratio (λ) (Cylinder Group 1, Sensor 2)	35	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 1, Sensor 2) Sensor 2)	35	BYTE5,6*256/65535-128	mA
Equivalent ratio (λ) (cylinder group 1, sensor 3)	36	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 1, Sensor 3) Sensor 3)	36	BYTE5,6*256/65535-128	mA
Equivalent ratio (λ) (block 1, sensor 4)	37	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 1, Sensor 4) Sensor 4)	37	BYTE5,6*256/65535-128	mA
Equivalent ratio (λ) (block 2, sensor 1)	38	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 2, Sensor 1) Sensor 1)	38	BYTE5,6*256/65535-128	mA
Equivalent ratio (λ) (cylinder group 2, sensor 2)	39	BYTE3,4*1.999/65535	

Oxygen Sensor Current (Cylinder Group 2, Sensor 2) Sensor 2)	39	BYTE5,6*256/65535-128	mA
Equivalent ratio (λ) (block 2, sensor 3)	3A	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 2, Sensor 3) Sensor 3)	3A	BYTE5,6*256/65535-128	mA
Equivalent ratio (λ) (block 2, sensor 4)	3B	BYTE3,4*1.999/65535	
Oxygen Sensor Current (Cylinder Group 2, Sensor 4) Sensor 4)	3B	BYTE5,6*256/65535-128	mA
Catalyst Temperature (Cylinder Group 1, Sensor) 1)	3C	BYTE3,4/10-40	°C
Catalyst Temperature (Cylinder Group 2, Sensor) 1)	3D	BYTE3,4/10-40	°C
Catalyst Temperature (Cylinder 1, Sensor 2) 2)	3E	BYTE3,4/10-40	°C
Catalyst Temperature (Cylinder Group 2, Sensor) 2)	3F	BYTE3,4/10-40	°C
Fire Monitoring	41	byte4.bit0 0-- (Not Supported) 1--(Supported)	
Monitoring of the Fuel System	41	byte4.bit1 0-- (Not Supported) 1--(Supported)	
Integrated Component Monitoring	41	byte4.bit2 0-- (Not Supported) 1--(Supported)	
Fire Monitoring Completed	41	byte4.bit4 0--(Complete or N/A) 1--(Supported)	
Fuel System Monitoring Completed	41	byte4.bit5 0--(Complete or N/A) 1--(Supported)	
Component Completion of Integrated Monitoring	41	byte4.bit6 0--(Complete or N/A) 1--(Supported)	
Catalyst Monitoring	41	byte5.bit0 0--NO 1--YES	
Heating Catalyst Monitoring	41	byte5.bit1 0--NO 1--YES	
Monitoring of Evaporation Systems	41	byte5.bit2 0--NO 1--YES	
Monitoring of Secondary Air Systems	41	byte5.bit3 0--NO 1--YES	

A/C System Refrigerant Monitoring	41	byte5.bit4	0--NO 1--YES	
Oxygen Sensor Monitoring	41	byte5.bit5	0--NO 1--YES	
Oxygen Sensor Heater Monitoring	41	byte5.bit6	0--NO 1--YES	
Monitoring of the EGR System	41	byte5.bit7	0--NO 1--YES	
Catalyst Monitoring Completed	41	byte6.bit0	0--(Complete or N/A) 1--Incomplete(Supported)	
Heating Catalyst Monitoring Completed	41	byte6.bit1	0--(Complete or N/A) 1--Incomplete(Supported)	
Completion of Monitoring of the Evaporation System	41	byte6.bit2	0--(Complete or N/A) 1--Incomplete(Supported)	
Monitoring of secondary air systems completed	41	byte6.bit3	0--(Complete or N/A) 1--Incomplete(Supported)	
A/C System Refrigerant Monitoring Completed	41	byte6.bit4	0--(Complete or N/A) 1--Incomplete(Supported)	
Oxygen Sensor Monitoring Complete	41	byte6.bit5	0--(Complete or N/A) 1--Incomplete(Supported)	
Heater Monitoring Oxygen Sensor Complete	41	byte6.bit6	0--(Complete or N/A) 1--Incomplete(Supported)	
Completion of Monitoring of the EGR System	41	byte6.bit7	0--(Complete or N/A) 1--Incomplete(Supported)	
Control Module Voltage	42	byte3,4/1000		
Absolute Load Value	43	byte3,4*100/255		
Equivalent Ratio of Fuel/Air Instructions	44	byte3,4*2/65535		
Throttle Relative Position	45	byte3*100/255		%
Ambient Air Temperature	46	byte3-40		°C
Throttle Absolute Position A	47	byte3*100/255		%
Throttle Absolute Position B	48	byte3*100/255		%
Throttle Absolute Position C	49	byte3*100/255		%
Throttle Absolute Position D	4A	byte3*100/255		%
Commanded Throttle Actuator Control	4C	byte3*100/255		%
Engine Running Time When the Malfunction Indicator Light Is On	4d	byte3,4		min
Engine Running After Clearing Fault Codes	4e	byte3,4		min
Maximum Value of the Equivalence Ratio	4f	byte3		
Maximum Voltage Value of the Oxygen Sensor	4F	byte4		V

Maximum Current Value of the Oxygen Sensor	4F	byte5	mA
Maximum Absolute Intake Manifold Pressure	4f	Byte4*10	kPa
Air in Air Mass Flow Sensor Flow Rate Maximum	50	byte3*10	g/s

5、 Fault Code Library

0x0001:P0001:Fuel Level Regulator Control Circuit/Open Circuit
0x0002:P0002:Fuel Level Regulator Control Circuit Range/Performance
0x0003:P0003:Fuel Level Regulator Control Circuit Low Voltage
0x0004:P0004:Fuel level Regulator Control Circuit Voltage High
0x0005:P0005:Fuel Shutoff Valve A Control Circuit/Open Circuit
0x0006:P0006:Fuel Shutoff Valve A Control Circuit Voltage Low
0x0007:P0007:Fuel Shutoff Valve Control Circuit Voltage High
0x0008:P0008:Engine Position System Performance (BANK 1)
0x0009:P0009:Engine Position System Performance (BANK 2)
0x000A:P000A:A Slow Camshaft Position Response (BANK 1)
0x000B:P000B:B Slow Camshaft Position Response (BANK 1)
0x000C:P000C:A Slow Camshaft Position Response (BANK 2)
0x000D:P000D:B Slow Camshaft Position Response (BANK 2)
0x0010:P0010:A Camshaft Position Actuating Circuit/Open Circuit (BANK 1)
0x0011:P0011:A Camshaft Position-Timing Too High or System Performance (BANK 1)
0x0012:P0012:A Camshaft Position-Timing Response Too Slow (BANK 1)
0x0013:P0013:B Camshaft Position Actuation Circuit/Open Circuit (BANK 1)
0x0014:P0014:B Camshaft Position-Timing Too High or System Performance (BANK 1)
0x0015:P0015:B Camshaft Position-Timing Response Too Slow (BANK 1)
0x0016:P0016:Crankshaft Position-Camshaft Position Correlation (BANK 1 SENSOR A)
0x0017:P0017:Crankshaft Position - Camshaft Position Correlation (BANK 1 SENSOR B)
0x0018:P0018:Crankshaft position - Camshaft position correlation (BANK 2 SENSOR A)
0x0019:P0019:Crankshaft Position - Camshaft Position Correlation (BANK 2

SENSOR B)

0x0020:P0020:Camshaft A position Actuator Circuit/Open Circuit (BANK 2)

0x0021:P0021:Camshaft A Position - Timing Too High or System Performance
(BANK 2)

0x0022:P0022:Camshaft A Position-Timing Response Too Slow (BANK 2)

0x0023:P0023:Camshaft B Position-Actuator Circuit/Open Circuit (BANK 2)

0x0024:P0024:Camshaft B Position-High Timing or System Performance (BANK 2)

0x0025:P0025:Camshaft B Position-Timing Response Too Slow (BANK 2)

0x0026:P0026:Intake Solenoid Control Circuit Range/Performance (BANK 1)

0x0027:P0027:Exhaust Solenoid Control Circuit Range/Performance (BANK 1)

0x0028:P0028:Intake Solenoid Control Circuit Range/Performance (BANK 2)

0x0029:P0029:Exhaust Solenoid Control Circuit Range/Performance (BANK 2)

0x0030:P0030:Oxygen Sensor Heater Control Circuit (BANK 1 SENSOR 1)

0x0031:P0031:Oxygen Sensor Heater Control Circuit Voltage Low (BANK 1
SENSOR 1)

0x0032:P0032:Oxygen Sensor Heater Control Circuit Voltage High (BANK 1
SENSOR 1)

0x0033:P0033:Turbocharger Bypass Valve Control Circuit

0x0034:P0034:Turbocharger Bypass Valve Control Circuit Voltage Low

0x0035:P0035:Turbocharger Bypass Valve Control Circuit Voltage Is High.

0x0036:P0036:Oxygen Sensor heater Control Circuit (BANK 1 SENSOR 2)

0x0037:P0037:Oxygen Sensor Control Circuit Voltage Low (BANK 1 SENSOR 2)

0x0038:P0038:Oxygen Sensor Control Circuit Voltage High (BANK 1 SENSOR 2)

0x0039:P0039:Turbo Bypass Valve Control Circuit Range/Performance

0x0040:P0040:Oxygen Sensor Signal Exchange BANK 1 SENSOR 1/BANK 2
SENSOR 1

0x0041:P0041:Oxygen Sensor Signal Exchange Group 1 Sensor 2/Group 2 Sensor 2

0x0042:P0042:Oxygen Sensor Heater Control Circuit (BANK 1 SENSOR 3)

0x0043:P0043:Oxygen Sensor Heater Control Circuit Low Voltage (BANK 1

SENSOR 3)

0x0044:P0044:Oxygen Sensor heater Control Circuit Voltage High (BANK 1

SENSOR 3)

0x0045:P0045:Turbo Boost Regulator Solenoid A Circuit/Open Circuit

0x0046:P0046:Turbo Boost Regulation Solenoid A Circuit Range/Performance

0x0047:P0047:Turbo Boost Regulator Solenoid A Circuit Low Voltage

0x0048:P0048:Turbo Boost Regulation Solenoid A Circuit Voltage High

0x0049:P0049:Turbocharger Turbine Overspeed

0x004A:P004A:Turbo Boost Control Solenoid B Circuit/Open Circuit

0x004B:P004B:Turbo Boost Control Solenoid B Circuit Range/Performance

0x004C:P004C:Turbo Boost Control Solenoid B Circuit Low Voltage

0x004D:P004D:Turbo Boost Control Solenoid B Circuit Voltage High

0x004E:P004E:Turbo Boost Control Solenoid A Circuit Intermittent/Unstable

0x004F:P004F:Turbo Boost Control Solenoid B Circuit Intermittent/Unstable

0x0050:P0050:Oxygen Sensor Heater Control Circuit (BANK 2 SENSOR 1)

0x0051:P0051:Oxygen Sensor Heater Control Circuit Low Voltage (BANK 2

SENSOR 1)

0x0052:P0052:Oxygen Sensor Heater Control Circuit Voltage High (BANK 2

SENSOR 1)

0x0053:P0053:Oxygen Sensor Heater Resistance (BANK 1 SENSOR 1)

0x0054:P0054:Oxygen Sensor Heater Resistance (BANK 1 SENSOR 2)

0x0055:P0055:Oxygen Sensor Heater Resistor (BANK 1 SENSOR 3)

0x0056:P0056:Oxygen Sensor Heater Control Circuit (BANK 2 SENSOR 2)

0x0057:P0057:Oxygen Sensor Heater Control Circuit Low Voltage (BANK 2

SENSOR 2)

0x0058:P0058:Oxygen Sensor Heater Control Circuit Voltage High (BANK 2

SENSOR 2)

0x0059:P0059:Oxygen Sensor Heater Resistance (BANK 2 SENSOR 1)

0x0060:P0060:Oxygen Sensor Heater Resistance (BANK 2 SENSOR 2)

0x0061:P0061:Oxygen Sensor Heater Resistor (BANK 2 SENSOR 3)

0x0062:P0062:Oxygen Sensor Heater Control Circuit (BANK 2 SENSOR 3)
0x0063:P0063:Oxygen Sensor Heater Control Circuit Low Voltage (BANK 2
SENSOR 3)
0x0064:P0064:Oxygen Sensor Heater Control Circuit Voltage High (BANK 2
SENSOR 3)
0x0065:P0065:Air Assist Injector Control Range/performance
0x0066:P0066:Air Assist Injector Control Circuit or Circuit Voltage Low
0x0067:P0067:Air Assist Injector Control Circuit Voltage High
0x0068:P0068:Intake Manifold Absolute Pressure/Air Flow Meter-Throttle Position
Correlation
0x0069:P0069:Intake Manifold Absolute Pressure - Atmospheric Pressure Correlation
0x006A:P006A:Intake Manifold Absolute Pressure-Air Flow Correlation
0x0070:P0070:Ambient Air Temperature Sensor Circuitry
0x0071:P0071:Ambient air Temperature Sensor Range/Performance
0x0072:P0072:Ambient Air Temperature Sensor Circuit Low Voltage
0x0073:P0073:Ambient Air Temperature Sensor Circuit Voltage High
0x0074:P0074:Ambient Air Temperature Sensor Circuit Intermittent

0x0075:P0075:Intake solenoid control circuit (BANK 1)
0x0076:P0076:Intake Solenoid Control Circuit Low Voltage (BANK 1)
0x0077:P0077:Intake Solenoid Control Circuit Voltage High (BANK 1)
0x0078:P0078:Exhaust Solenoid Control Circuit (BANK 1)
0x0079:P0079:Exhaust Solenoid Control Circuit Voltage Low (BANK 1)
0x0080:P0080:Exhaust Solenoid Control Circuit High Voltage (BANK 1)
0x0081:P0081:Intake Solenoid Control Circuit (BANK 2)
0x0082:P0082:Intake Solenoid Control Circuit Voltage Low (BANK 2)
0x0083:P0083:Intake Solenoid Control Circuit Voltage High (BANK 2)
0x0084:P0084:Exhaust Solenoid Control Circuit (BANK 2)
0x0085:P0085:Exhaust Solenoid Control Circuit Voltage Low (BANK 2)
0x0086:P0086:Exhaust Solenoid Control Circuit Voltage High (BANK 2)

0x0087:P0087:Fuel Rail/System Pressure - Too Low
0x0088:P0088:Oil Rail/System Pressure - Too High
0x0089:P0089:Fuel Pressure Regulator 1 Performance
0x0090:P0090:Fuel Pressure Regulator 1 Control Circuit
0x0091:P0091:Fuel Pressure Regulator 1 Control Circuit Low Voltage
0x0092:P0092:Fuel Pressure Regulator 1 Control Circuit High Voltage
0x0093:P0093:Fuel System Leak Detected - Long Term Leak
0x0094:P0094:Fuel System Leak Detected - Short Term Leak
0x0095:P0095:Intake Air Temperature Sensor 2 Circuit
0x0096:P0096:Intake Air Temperature Sensor 2 Circuit Range/Performance
0x0097:P0097:Intake Air Temperature Sensor 2 Circuit Low Voltage
0x0098:P0098:Intake Air Temperature Sensor 2 Circuit High Voltage
0x0099:P0099:Intake Air Temperature Sensor 2 Circuit Intermittent/Unstable
0x0100:P0100:Mass or Volume Air Flow A Circuit
0x0101:P0101:Mass or Volume Air Flow A Circuit Range/Performance
0x0102:P0102:Mass or Volume Air Flow A Circuit Low Input
0x0103:P0103:Mass or Volume Air Flow A Circuit High Inputs
0x0104:P0104:Mass or Volume Air Flow A Circuit Intermittent Interruption
0x0105:P0105:Intake Manifold Absolute/Atmospheric Pressure Circuit
0x0106:P0106:Intake Manifold Absolute/Atmospheric Pressure Circuit
Range/Performance
0x0107:P0107:Manifold Absolute/Atmospheric Pressure Circuit Low Input
0x0108:P0108:Manifold Absolute Pressure/Atmospheric Pressure Circuit High Input
0x0109:P0109:Manifold Absolute/Atmospheric Pressure Circuit Intermittent
Interrupts
0x010A:P010A: Mass/Flow B Circuit
0x010B:P010B:Mass/Flow B Circuit Range/Performance
0x010C:P010C:Mass/Flow B Circuit Low Input
0x010D:P010D:Mass/Flow B Circuit High Input
0x010E:P010E:Mass/Flow B Circuit Intermittent Interrupt/Unstable

0x010F:P010F:Mass/Flow Sensor A/B Correlation
0x0110:P0110:Temperature Sensor 1 Circuit
0x0111:P0111:Temperature Sensor 1 Circuit Range/Performance
0x0112:P0112:Temperature Sensor 1 Circuit Low Voltage
0x0113:P0113:Temperature Sensor 1 Circuit High Voltage
0x0114:P0114:Temperature Sensor 1 Circuit Intermittent Interruption
0x0115:P0115:Engine Coolant Temperature Sensor 1 Circuit
0x0116:P0116:Engine Coolant Temperature Sensor 1 Circuit Range/Performance
0x0117:P0117:Engine Coolant Temperature Sensor 1 Circuit Low Voltage
0x0118:P0118:Engine Coolant Temperature Sensor 1 Circuit High Voltage
0x0119:P0119:Engine Coolant Temperature Sensor 1 Circuit Intermittent Interruption
0x011A:P011A:Engine coolant temperature sensor 1/2 correlation
0x0120:P0120:Throttle/Pedal Position Sensor/Switch A Circuit
0x0121:P0121:Throttle/Pedal Position Sensor/Switch A Circuit Range/Performance
0x0122:P0122:Throttle/Pedal Position Sensor/Switch A Circuit Low Voltage
0x0123:P0123:Throttle/Pedal Position Sensor/Switch A Circuit Voltage High
0x0124:P0124:Throttle/Pedal Position Sensor/Switch A Circuit Intermittent Interruption
0x0125:P0125:Insufficient Coolant Temperature for Fuel Closed Loop Control
0x0126:P0126:Insufficient Coolant Temperature for Stabilized Operation
0x0127:P0127:Intake Temperature Too High
0x0128:P0128:Coolant Thermostat (Coolant Temperature Below Thermostat Regulation Temperature)
0x0129:P0129:Atmospheric Pressure Too Low
0x0130:P0130:Oxygen Sensor Circuit (BANK 1 SENSOR 1)
0x0131:P0131:Oxygen Sensor Circuit Low Voltage (BANK 1 SENSOR 1)
0x0132:P0132:Oxygen Sensor Circuit High Voltage (BANK 1 SENSOR 1)
0x0133:P0133:Oxygen Sensor Circuit Response Delayed (BANK 1 SENSOR 1)
0x0134:P0134:Oxygen Sensor Circuit Not Detecting Activity (BANK 1 SENSOR 1)
0x0135:P0135:Oxygen Sensor Heater Circuit (BANK 1 SENSOR 1)

0x0136:P0136:Oxygen Sensor Circuit (BANK 1 SENSOR 2)
0x0137:P0137:Oxygen Sensor Circuit Low Voltage (BANK 1 SENSOR 2)
0x0138:P0138:Oxygen Sensor Circuit High Voltage (BANK 1 SENSOR 2)
0x0139:P0139:Oxygen Sensor Circuit response Delayed (BANK 1 SENSOR 2)
0x0140:P0140:Oxygen Sensor Circuit Not Detecting Activity (BANK 1 SENSOR 2)
0x0141:P0141:Oxygen Sensor Heater Circuit (BANK 1 SENSOR 2)
0x0142:P0142:Oxygen Sensor Circuit (BANK 1 SENSOR 3)
0x0143:P0143:Oxygen Sensor Circuit Low Voltage (BANK 1 SENSOR 3)
0x0144:P0144:Oxygen Sensor Circuit high Voltage (BANK 1 SENSOR 3)
0x0145:P0145:Oxygen Sensor Circuit Response Delayed (BANK 1 SENSOR 3)
0x0146:P0146:Oxygen Sensor Circuit Not Detecting Activity (BANK 1 SENSOR 3)
0x0147:P0147:Oxygen Sensor Heater Circuit (BANK 1 SENSOR 3)
0x0148:P0148:Fuel Delivery Error
0x0149:P0149:Fuel Timing Error
0x0150:P0150:Oxygen Sensor Circuit (BANK 2 SENSOR 1)
0x0151:P0151:Oxygen Sensor Circuit Low Voltage (BANK 2 SENSOR 1)
0x0152:P0152:Oxygen Sensor Circuit high Voltage (BANK 2 SENSOR 1)
0x0153:P0153:Oxygen Sensor Circuit Response Delayed (BANK 2 SENSOR 1)
0x0154:P0154:Oxygen Sensor Circuit Not detecting Activity (BANK 2 SENSOR 1)
0x0155:P0155:Oxygen Sensor Heater Circuit (BANK 2 SENSOR 1)
0x0156:P0156:Oxygen Sensor Circuit (BANK 2 SENSOR 2)
0x0157:P0157:Oxygen Sensor Circuit Low Voltage (BANK 2 SENSOR 2)
0x0158:P0158:Oxygen Sensor Circuit High Voltage (BANK 2 SENSOR 2)
0x0159:P0159:Oxygen Sensor Circuit Response Delayed (BANK 2 SENSOR 2)
0x0160:P0160:Oxygen Sensor Circuit Not Detecting Activity (BANK 2 SENSOR 2)
0x0161:P0161:Oxygen Sensor Heater Circuit (BANK 2 SENSOR 2)
0x0162:P0162:Oxygen Sensor Circuit (BANK 2 SENSOR 3)
0x0163:P0163:Oxygen Sensor Circuit Low Voltage (BANK 2 SENSOR 3)
0x0164:P0164:Oxygen sensor circuit high voltage (BANK 2 SENSOR 3)
0x0165:P0165:Oxygen Sensor Circuit Response Delayed (BANK 2 SENSOR 3)

0x0166:P0166:Oxygen Sensor Circuit Not Detecting Activity (BANK 2 SENSOR 3)

0x0167:P0167:Oxygen Sensor Heater Circuit (BANK 2 SENSOR 3)

0x0168:P0168:Fuel Temperature Too High

0x0169:P0169:Fuel Composition Error

0x0170:P0170:Fuel Correction (BANK 1)

0x0171:P0171:System Too Thin (BANK 1)

0x0172:P0172:System Over density (BANK 1)

0x0173:P0173:Fuel Correction (BANK 2)

0x0174:P0174:System Over thinning (BANK 2)

0x0175:P0175:System Over density (BANK 2)

0x0176:P0176:Fuel Composition Sensor circuit

0x0177:P0177:Fuel Group Positive Sensor Circuit Range/Performance

0x0178:P0178:Fuel Group Sensor Circuit Low Voltage

0x0179:P0179:Fuel Group Sensor Circuit Voltage High

0x0180:P0180:Fuel Group Sensor A Circuit

0x0181:P0181:Fuel Composition Sensor A Circuit Range/Performance

0x0182:P0182:Fuel Temperature Sensor A Circuit Low Voltage

0x0183:P0183:Fuel Temperature Sensor A Circuit High Voltage

0x0184:P0184:Fuel Temperature Sensor A Circuit Intermittent Interruption

0x0185:P0185:Fuel Temperature Sensor B Circuit

0x0186:P0186:Fuel Temperature Sensor B Circuit Range/Performance

0x0187:P0187:Fuel Temperature Sensor B Circuit Low Voltage

0x0188:P0188:Fuel Temperature Sensor B Circuit High Voltage

0x0189:P0189:Fuel Temperature Sensor B Circuit Intermittent Interruption

0x018A:P018A:Fuel Pressure Sensor B Circuit

0x018B:P018B:Fuel Pressure Sensor B Circuit Range/Performance

0x018C:P018C:Fuel Pressure Sensor B Circuit Low Voltage

0x018D:P018D:Fuel Pressure Sensor B Circuit High Voltage

0x018E:P018E:Fuel Pressure Sensor B Circuit Intermittent/Unstable

0x018F:P018F:ISO/ASA Reserved

0x0190:P0190:Fuel Rail Pressure Sensor A Circuit
0x0191:P0191:Oil Rail Pressure Sensor A Circuit Range/Performance
0x0192:P0192:Oil Rail Pressure Sensor A Circuit Low Voltage
0x0193:P0193:Oil Rail Pressure Sensor A Circuit High Voltage
0x0194:P0194:Oil Rail Pressure Sensor A Circuit Intermittently Interrupted/Unstable
0x0195:P0195:Engine Oil Temperature Sensor
0x0196:P0196:Engine Oil Temperature Sensor Range/Performance
0x0197:P0197:Engine Oil Temperature Sensor Low Voltage
0x0198:P0198:Engine Oil Temperature Sensor Voltage High
0x0199:P0199:Engine Oil Temperature Sensor Intermittent Interruption
0x0200:P0200:Injector Circuit/Open Circuit
0x0201:P0201:Injector Circuit/Open Circuit-Cylinder 1
0x0202:P0202:Injector Circuit/Open Circuit-Cylinder 2
0x0203:P0203:Injector Circuit/Open Circuit-Cylinder 3
0x0204:P0204:Injector Circuit/Open Circuit-Cylinder 4
0x0205:P0205:Injector Circuit/Open Circuit-Cylinder 5
0x0206:P0206:Injector Circuit/Open Circuit-Cylinder 6
0x0207:P0207:Injector Circuit/Open Circuit-Cylinder 7
0x0208:P0208:Injector Circuit/Open Circuit-Cylinder 8
0x0209:P0209:Injector Circuit/Open Circuit-Cylinder 9
0x020A:P020A:Cylinder 1 Injection Timing
0x020B:P020B:Cylinder 2 Injection Timing
0x020C:P020C:Cylinder 3 Injection Timing
0x020D:P020D:Cylinder 4 Injection Timing
0x020E:P020E:Cylinder 5 Injection Timing
0x020F:P020F:Cylinder 6 Fuel Injection Timing
0x0210:P0210:Injector Circuit/Open Circuit-Cylinder 10
0x0211:P0211:Injector Circuit/Open Circuit-Cylinder 11
0x0212:P0212:Injector Circuit/Open Circuit-Cylinder 12
0x0213:P0213:Cold Start Injector 1

0x0214:P0214:Cold Start Injector 2

0x0215:P0215:Engine Shutdown Solenoid Valve

0x0216:P0216:Injector/Injection Timing Control Circuit

0x0217:P0217:Engine Coolant > Temperature Condition

0x0218:P0218:Transmission Fluid Greater Than Temperature Condition

0x0219:P0219:Engine Over speed Condition

0x021A:P021A:Cylinder 7 Injection Timing

0x021B:P021B:Cylinder 8 Injection Timing

0x021C:P021C:Cylinder 9 Injection Timing

0x021D:P021D:Cylinder 10 Jet Timing

0x021E:P021E:Cylinder 11 Jet Timing

0x021F:P021F:Cylinder 12 Jet Timing

0x0220:P0220:Throttle/Pedal Position Sensor/Switch B Circuit

0x0221:P0221:Throttle/Pedal Position Sensor/Switch B Circuit Range/Performance

0x0222:P0222:Throttle/Pedal Position Sensor/Switch B Circuit Low Voltage

0x0223:P0223:Throttle/Pedal Position Sensor/Switch B Circuit Voltage High

0x0224:P0224:Throttle/Pedal Position Sensor/Switch B Circuit Intermittent Interruption

0x0225:P0225:Throttle/Pedal Position Sensor/Switch C Circuit

0x0226:P0226:Throttle/Pedal Position Sensor/Switch C Circuit Range/Performance

0x0227:P0227:Throttle/Pedal Position Sensor/Switch C Circuit Low Voltage

0x0228:P0228:Throttle/Pedal Position Sensor/Switch C Circuit Voltage High

0x0229:P0229:Throttle/Pedal Position Sensor/Switch C Circuit Intermittent Interruption

0x022A:P022A:Intake Air Cooler Bypass Control A Circuit/Open Circuit

0x022B:P022B:Intake Air Cooler Bypass Control A Circuit Low Voltage

0x022C:P022C:Intake Air Cooler Bypass Control A Circuit High Voltage

0x022D:P022D:Intake Air Cooler Bypass Control B Circuit/Open Circuit

0x022E:P022E:Intake Air Cooler Bypass Control B Circuit Voltage Low

0x022F:P022F:Intake Air Cooler Bypass Control B Circuit High Voltage

0x0230:P0230:Fuel Pump Primary Circuit

0x0231:P0231:Fuel Pump Secondary Circuit Low Voltage

0x0232:P0232:Fuel Pump Secondary Circuit Voltage High

0x0233:P0233:Fuel Pump Secondary Circuit Intermittent Interruption

0x0234:P0234:Turbo Over boost Condition

0x0235:P0235:Turbo/Mechanical Supercharger Boost Sensor A Circuit

0x0236:P0236:Turbo/Mechanical Booster Boost Sensor A Circuit Range/Performance

0x0237:P0237:Turbo/Mechanical Booster Sensor A Circuit Low Voltage

0x0238:P0238:Turbo/Mechanical Booster Sensor A Circuit High Voltage

0x0239:P0239:Turbo/Mechanical Booster Booster Sensor B Circuit

0x023A:P023A:Intake Cooler Cooling Pump Control Circuit / Disconnect

0x023B:P023B:Intake Cooler Cooling Pump Control Circuit Low Voltage

0x023C:P023C:Intake Air Cooler Cooling Pump Control Circuit Voltage High

0x023D:P023D:Intake Manifold Absolute Pressure-Turbo/Mechanical Supercharger
Sensor A Correlation

0x023E:P023E:Intake Manifold Absolute Pressure-Turbo/Mechanical Supercharger
Sensor B Correlation

0x0240:P0240:Turbo/Mechanical Supercharger Sensor B Circuit Range/Performance

0x0241:P0241:Turbo/Mechanical Supercharger Sensor B Circuit Low Voltage

0x0242:P0242:Turbo/Mechanical Booster Sensor B Circuit High Voltage

0x0243:P0243:Turbo/Mechanical Supercharger Wastegate Solenoid A

0x0244:P0244:Turbo/Mechanical Supercharger Wastegate Solenoid A
Range/Performance

0x0245:P0245:Turbo/Mechanical Supercharger Wastegate Solenoid A Low Voltage

0x0246:P0246:Turbo/Mechanical Supercharger Wastegate Solenoid A High Voltage

0x0247:P0247:Turbo/Mechanical Supercharger Wastegate Solenoid B

0x0248:P0248:Turbo/Mechanical Supercharger Wastegate Solenoid B
Range/Performance

0x0249:P0249:Turbo/Mechanical Supercharger Wastegate Solenoid B Low Voltage

0x024A:P024A:Intake Cooler Bypass Control A Range/Performance

0x024B:P024B: Intake Cooler Bypass Control A Blocking

0x024C:P024C: Intake Cooler Bypass Position Sensor A Circuit

0x024D:P024D: Intake Cooler Bypass Position Sensor A Circuit Range/Performance

0x024E:P024E: Intake Cooler Bypass Position Sensor A Circuit Low Voltage

0x024F:P024F: Intake Cooler Bypass Position Sensor A Circuit High Voltage

0x0250:P0250: Turbo/Mechanical Supercharger Wastegate Solenoid B High Voltage

0x0251:P0251: Injection Pump Fuel Metering Control A (Camshaft/Rotor/Injector)

0x0252:P0252: Injection Pump Fuel Metering Control A Range/Performance
(Camshaft/Rotor/Injector)

0x0253:P0253: Injector Pump Fuel Metering Control A Low Voltage
(Camshaft/Rotor/Injector)

0x0254:P0254: Injector Pump Fuel Metering Control A High Voltage
(Camshaft/Rotor/Injector)

0x0255:P0255: Injector pump fuel metering Control A Intermittent
(Camshaft/Rotor/Injector)

0x0256:P0256: Injector Pump Fuel Metering Control B (Camshaft/Rotor/Injector)

0x0257:P0257: Injection Pump Fuel Metering Control B Range/Performance
(Camshaft/Rotor/Injector)

0x0258:P0258: Injector Pump Fuel Metering Control B Low Voltage
(Camshaft/Rotor/Injector)

0x0259:P0259: Injector Pump Fuel Metering Control B High
Voltage(Camshaft/Rotor/Injector)

0x0260:P0260: Injection Pump Fuel Metering Control B Intermittent
(Camshaft/Rotor/Injector)

0x0261:P0261: 1 Cylinder Injector Circuit Low Voltage

0x0262:P0262: 1 Cylinder Injector Circuit High Voltage

0x0263:P0263: 1 Cylinder Action/Balance

0x0264:P0264: 2 Cylinder Injector Circuit Voltage Low

0x0265:P0265: 2 Cylinder Injector Circuit Voltage High

0x0266:P0266: 2 Cylinder Action/Balance

0x0267:P0267:3 Cylinder Injector Circuit Voltage Low
0x0268:P0268:3 Cylinder Injector Circuit Voltage High
0x0269:P0269:3 Cylinder Action/Balance
0x0270:P0270:4 Cylinder Injector Circuit Voltage Low
0x0271:P0271:4 Cylinder Injector Circuit Voltage High
0x0272:P0272:4 Cylinder Action/Balance
0x0273:P0273:5 Cylinder Injector Circuit Voltage Low
0x0274:P0274:5 Cylinder Injector Circuit Voltage High
0x0275:P0275:5 Cylinder Action/Balance
0x0276:P0276:6 Cylinder Injector Circuit Voltage Low
0x0277:P0277:6 Cylinder Injector Circuit Voltage High
0x0278:P0278:6 Cylinder Action/Balance
0x0279:P0279:7 Cylinder Injector Circuit Voltage Low
0x0280:P0280:7 Cylinder Injector Circuit Voltage High
0x0281:P0281:7 Cylinder Action/Balance
0x0282:P0282:8 Cylinder Injector Circuit Voltage Low
0x0283:P0283:8 Cylinder Injector Circuit Voltage High
0x0284:P0284:8 Cylinder Action/Balance
0x0285:P0285:9 Cylinder Injector Circuit Voltage Low
0x0286:P0286:9 Cylinder Injector Circuit Voltage High
0x0287:P0287:9 Cylinder Action/Balance
0x0288:P0288:10 Cylinder Injector Circuit Voltage Low
0x0289:P0289:10 Cylinder Injector Circuit Voltage High
0x0290:P0290:10 Cylinder Action/Balance
0x0291:P0291:11 Cylinder Injector Circuit Voltage Low
0x0292:P0292:11 Cylinder Injector Circuit Voltage High
0x0293:P0293:11 Cylinder Action/Balance
0x0294:P0294:12 Cylinder Injector Circuit Voltage Low
0x0295:P0295:12 Cylinder Injector Circuit Voltage High
0x0296:P0296:12 Cylinder Action/Balance

0x0297:P0297:Vehicle Overspeed Condition
0x0298:P0298:Engine Fuel Over Temperature
0x0299:P0299:Turbo/Mechanical Supercharger Under Boost Condition
0x0300:P0300:Random/Detected Multi-Cylinder Misfires
0x0301:P0301:1 Cylinder Misfire Detected
0x0302:P0302:2 Cylinder Misfire Detected.
0x0303:P0303:3 Cylinder Misfire Detected.
0x0304:P0304:4 Cylinder Misfire Detected.
0x0305:P0305:5 Cylinder Misfire Detected.
0x0306:P0306:6 Cylinder Misfire Detected.
0x0307:P0307:7 Cylinder Misfire Detected.
0x0308:P0308:8 Cylinder Misfire Detected.
0x0309:P0309:9 Cylinder Misfire Detected.
0x0310:P0310:10 Cylinder Misfire Detected.
0x0311:P0311:11 Cylinder Misfire Detected.
0x0312:P0312:12 Cylinder Misfire Detected.
0x0313:P0313:Low Fuel Misfire Detected
0x0314:P0314:Single Cylinder Misfire (Cylinder Non-Specific)
0x0315:P0315:Crankshaft Position System Parameter Adjustment Not Learned
0x0316:P0316:Engine Misfire Detected at Start (Start Speed 1000)
0x0317:P0317:Uneven Road Hardware Not Present
0x0318:P0318:Uneven Road Sensor A Signal Circuit
0x0319:P0319:Uneven Road Sensor B
0x0320:P0320:Ignition/Distributor Engine Speed Input Circuit
0x0321:P0321:Ignition/Distributor Engine Speed Input Circuit Range/Performance
0x0322:P0322:Ignition/Distributor Engine Speed Input Circuit No Signal
0x0323:P0323:Ignition/Distributor Engine Speed Input Circuit Intermittent
Interruption
0x0324:P0324:Burst Control System Error
0x0325:P0325:Burst Sensor 1 Circuit (BANK 1 OR SINGLE SENSOR)

0x0326:P0326:Burst Sensor 1 Circuit Range/Performance (BANK 1 OR SINGLE SENSOR)

0x0327:P0327:Burst Sensor 1 Circuit Low Voltage (BANK 1 OR SINGLE SENSOR)

0x0328:P0328:Burst Sensor 1 Circuit Voltage High (BANK 1 OR SINGLE SENSOR)

0x0329:P0329:Burst Sensor 1 Circuit Input Intermittently Interrupted (BANK 1 OR SINGLE SENSOR).

0x0330:P0330:Burst Sensor 2 Circuit (BANK 2)

0x0331:P0331:Burst Sensor 2 Circuit Range/Performance (BANK 2)

0x0332:P0332:Burst Sensor 2 Circuit Low Voltage (BANK 2)

0x0333:P0333:Burst Sensor 2 Circuit Voltage High (BANK 2)

0x0334:P0334:Burst Sensor 2 Circuit Input Intermittent Interrupt (BANK 2)

0x0335:P0335:Crankshaft Position Sensor A Circuit

0x0336:P0336:Crankshaft Position Sensor A Circuit Range/Performance

0x0337:P0337:Crankshaft Position Sensor A Circuit Low Voltage

0x0338:P0338:Crankshaft Position Sensor A Circuit High Voltage

0x0339:P0339:Crankshaft Position Sensor A Circuit Intermittent Interruption

0x0340:P0340:Camshaft Position Sensor A Circuit (BANK 1 OR SINGLE SENSOR)

0x0341:P0341:Camshaft Position Sensor A Circuit Range/Performance (BANK 1 OR SINGLE SENSOR)

0x0342:P0342:Camshaft Position Sensor A Circuit Low Voltage (BANK 1 OR SINGLE SENSOR)

0x0343:P0343:Camshaft Position Sensor A Circuit Voltage High (BANK 1 OR SINGLE SENSOR)

0x0344:P0344:Camshaft Position Sensor A Circuit Intermittent Interrupt (BANK 1 OR SINGLE SENSOR)

0x0345:P0345:Camshaft Position Sensor A Circuit (BANK 2)

0x0346:P0346:Camshaft Position Sensor A Circuit Range/Performance (BANK 2)

0x0347:P0347:Camshaft Position Sensor A Circuit Low Voltage (BANK 2)

0x0348:P0348:Camshaft Position Sensor A Circuit High Voltage (BANK 2)

0x0349:P0349:Camshaft Position Sensor A Circuit Intermittent Interrupt (BANK 2)

0x0350:P0350:Ignition Coil Primary/Secondary Circuit
0x0351:P0351:Ignition Coil A Primary/Secondary Circuits
0x0352:P0352: Ignition Coil B Primary/Secondary Circuits
0x0353:P0353:Ignition Coil C Primary/Secondary Circuits
0x0354:P0354:Ignition Coil D Primary/Secondary Circuits
0x0355:P0355:Ignition Coil E Primary/Secondary Circuits
0x0356:P0356:Ignition Coil F Primary/Secondary Circuits
0x0357:P0357:Ignition Coil G Primary/Secondary Circuits
0x0358:P0358:Ignition Coil H Primary/Secondary Circuits
0x0359:P0359:Ignition Coil I Primary/Secondary Circuits
0x0360:P0360:Ignition Coil J Primary/Secondary Circuits
0x0361:P0361:Ignition Coil K Primary/Secondary Circuits
0x0362:P0362:Ignition Coil L Primary/Secondary Circuits
0x0363:P0363:Loss of Ignition Detected - Fuel Shutdown
0x0365:P0365:Camshaft Position Sensor B Circuit (BANK 1)
0x0366:P0366:Camshaft Position Sensor B Circuit Range/Performance (BANK 1)
0x0367:P0367:Camshaft Position Sensor B Circuit Low Voltage (BANK 1)
0x0368:P0368:Camshaft Position Sensor B Circuit High Voltage (BANK 1)
0x0369:P0369:Camshaft Position Sensor B Circuit Intermittent Interrupt (BANK 1)
0x0370:P0370:Timing Reference High Resolution Signal A
0x0371:P0371:Timing Reference High Resolution Signal A Excessive Pulses
0x0372:P0372:Timing Reference High-Resolution Signal A Too Few Pulses
0x0373:P0373:Timing Reference Hi-Res Signal A Intermittent Interrupt/Unstable Pulse
0x0374:P0374:Timing Reference High Resolution Signal A No Pulse
0x0375:P0375:Timing Reference High Resolution Signal B
0x0376:P0376:Timing Reference High-Resolution Signal B Excessive Pulses
0x0377:P0377:Timing Reference High Resolution Signal B Too Few Pulses
0x0378:P0378:Timing Reference High Resolution Signal B Intermittent Interrupt/Unstable Pulse

0x0379:P0379:Timing Reference High Resolution Signal B No Pulse

0x0380:P0380:Preheat Plug/Heat Circuit A

0x0381:P0381:Preheat Plug/Heat Indicator Circuit

0x0382:P0382:Preheat Plug/Heat Circuit B

0x0383:P0383:Preheat Plug Control Module Control Circuit Low Voltage

0x0384:P0384:Preheat Plug Control Module Control Circuit High Voltage

0x0385:P0385:Crankshaft Position Sensor Circuit B

0x0386:P0386:Crankshaft Position Sensor B Circuit Range/Performance

0x0387:P0387:Crankshaft Position Sensor B Circuit Low Voltage

0x0388:P0388:Crankshaft Position Sensor B Circuit High Voltage

0x0389:P0389:Crankshaft Position Sensor B Circuit Intermittent Interruption

0x0390:P0390:Crankshaft Position Sensor B Circuit (BANK 2)

0x0391:P0391:Crankshaft Position Sensor B Circuit Range/Performance (BANK 2)

0x0392:P0392:Crankshaft Position Sensor B Circuit Low Voltage (BANK 2)

0x0393:P0393:Crankshaft Position Sensor B Circuit High Voltage (BANK 2)

0x0394:P0394:Crankshaft Position Sensor B Circuit Intermittent Interrupt (BANK 2)

0x0400:P0400:Exhaust Recirculation Flow Rate

0x0401:P0401:Insufficient Exhaust Gas Recirculation Flow Detected

0x0402:P0402:Excessive Exhaust Gas Recirculation Flow Rate Detected

0x0403:P0403:Exhaust Gas Recirculation Control Circuit

0x0404:P0404:Exhaust Gas Recirculation Control Circuit Range/Performance

0x0405:P0405:Exhaust Gas Recirculation Sensor A Circuit Voltage Low

0x0406:P0406:Exhaust Gas Recirculation Sensor A Circuit Voltage High

0x0407:P0407: Circuit Voltage of Exhaust Gas Recirculation Sensor B Is Low.

0x0408:P0408:High Circuit Voltage for Exhaust Gas Recirculation Sensor B

0x0409:P0409: Circuit of Exhaust Gas Recirculation Sensor A

0x040A:P040A:Exhaust Gas Recirculation Temperature Sensor A Circuit

0x040B:P040B:Exhaust Recirculation Temperature Sensor A Circuit

Range/Performance

0x040C:P040C:Exhaust Gas Recirculation Temperature Sensor A Circuit Low Voltage

0x040D:P040D:Exhaust Gas Recirculation Temperature Sensor A Circuit High Voltage

0x040E:P040E:Exhaust Gas Recirculation Temperature Sensor A Circuit Intermittent/Unstable

0x040F:P040F:Exhaust Gas Recirculation temperature sensor A/B correlation

0x0410:P0410:Secondary Air Injection System

0x0411:P0411:Incorrect Secondary Air Injection System Flow Detected

0x0412:P0412:Secondary Air Injection System Switching Valve A Circuit

0x0413:P0413:Secondary Air Injection System Switching Valve A Circuit Disconnected

0x0414:P0414:Secondary Air Injection System On/Off Valve A Circuit Shorted

0x0415:P0415:Secondary Air Injection System On/Off Valve B Circuit

0x0416:P0416:Secondary Air Injection System On/Off Valve B Circuit Disconnected

0x0417:P0417:Secondary Air Injection System On/Off Valve B Circuit Shorted

0x0418:P0418:Secondary Air Injection Control A Circuit

0x0419:P0419:Secondary Air Injection System Control B Circuit

0x041A:P041A:Exhaust Recirculation Temperature Sensor B Circuit

0x041B:P041B:Exhaust Gas Recirculation Temperature Sensor B Circuit

Range/Performance

0x041C:P041C:Exhaust Gas Recirculation Temperature Sensor B Circuit Low Voltage

0x041D:P041D:Exhaust Gas Recirculation Temperature Sensor B Circuit High Voltage

0x041E:P041E:Exhaust Gas Recirculation Temperature Sensor B Circuit Intermittent Interrupt/Unstable

0x0420:P0420:Catalytic Unit Performance Below Limits (BANK 1)

0x0421:P0421:Warm-Up Catalyst Performance Below Limit (BANK 1)

0x0422:P0422:Main Catalyst Performance Below Limit (BANK 1)

0x0423:P0423:Warming Catalyst Performance Below Limit (BANK 1)

0x0424:P0424:Heated Catalyst Temperature Below Limit (BANK 1)

0x0425:P0425:Catalytic Converter Temperature Sensor Circuit (BANK 1 SENSOR1)

0x0426:P0426:Catalytic Converter Temperature Sensor Circuit Range/Performance
(BANK 1 SENSOR1)

0x0427:P0427:Catalytic Converter Temperature Sensor Circuit Low Voltage (BANK
1 SENSOR1)

0x0428:P0428:Catalytic Converter Temperature Sensor Circuit Voltage High (BANK
1 SENSOR1)

0x0429:P0429:Catalytic Converter Heating Control Circuit (BANK 1)

0x042A:P042A:Catalytic Converter Temperature Sensor Circuit (BANK 1
SENSOR2)

0x042B:P042B:Catalytic Converter Temperature Sensor Circuit Range/Performance
(BANK 1 SENSOR2)

0x042C:P042C:Catalytic Converter Temperature Sensor Circuit Low Voltage (BANK
1 SENSOR2)

0x042D:P042D:Catalytic Converter Temperature Sensor Circuit Voltage High
(BANK 1 SENSOR2)

0x0430:P0430:Catalytic Unit Performance Below Limit (BANK 2)

0x0431:P0431:Warm-up Catalyst Performance Below Limit (BANK 2)

0x0432:P0432:Residential Catalyst Performance Below Limit (BANK 2)

0x0433:P0433: Heating Catalyst Performance Below Limit (BANK 2)

0x0434:P0434: Heated Catalyst Temperature Below Limit (BANK 2)

0x0435:P0435:Catalyst Temperature Sensor Circuit (BANK 2 SENSOR1)

0x0436:P0436:Catalyst Temperature Sensor Circuit Range/Performance (BANK 2
SENSOR1)

0x0437:P0437:Catalyst Temperature Sensor Circuit Low Voltage (BANK 2
SENSOR1)

0x0438:P0438:Catalyst Temperature Sensor Circuit Voltage High (BANK 2
SENSOR1)

0x0439:P0439:Catalyst Heating Control Circuit (BANK 2)

0x043A:P043A:Catalyst Temperature Sensor Circuit (BANK 2 SENSOR2)

0x043B:P043B:Catalyst Temperature Sensor Circuit Range/Performance (BANK 2
SENSOR1)

0x043C:P043C:Catalyst Temperature Sensor Circuit Low Voltage (BANK 2
SENSOR1)

0x043D:P043D:Catalyst Temperature Sensor Circuit Voltage High (BANK 2
SENSOR1)

0x043E:P043E:Evaporative Discharge System Leak Detection Reference Orifice Low
Flow Rate

0x043F:P043F:Evaporative Emission System Leak Detection Reference Hole High
Flow Rate

0x0440:P0440:Evaporative Drain System

0x0441:P0441:Evaporative Emissions System Charcoal Canister Flow Error

0x0442:P0442:Evaporative Discharge System Leak Detection (Short Term Leak)

0x0443:P0443:Evaporative Emissions System Charcoal Canister Control Valve
Circuit

0x0444:P0444:Evaporative Emissions System Charcoal Canister Control Valve
Circuit Disconnected

0x0445:P0445:Evaporative Emissions System Charcoal Canister Control valve
Circuit Shorted

0x0446:P0446:Evaporative Emissions System Emissions Control Circuit

0x0447:P0447:Evaporative Emissions System Emissions Control Circuit Broken
Circuit

0x0448:P0448:Evaporative Emission System Emission Control Circuit Short Circuit

0x0449:P0449:Evaporative Emissions System Discharge Valve/Solenoid Circuit

0x0450:P0450:Evaporative Emissions System Pressure Sensor/Switch

0x0451:P0451:Evaporative Emissions System Pressure Sensor/Switch
Range/Performance

0x0452:P0452:Evaporative Emissions System Pressure Sensor/Switch Low Voltage

0x0453:P0453:Evaporative Emission System Pressure Sensor/Switch Voltage High

0x0454:P0454:Evaporative Emission System Pressure Sensor/Switch Intermittent

Interruption

0x0455:P0455:Evaporative Emission System Leak Detection (Long Term Leaks)

0x0456:P0456:Evaporative Emission System Leak Test (Small Short-Term Leak)

0x0457:P0457:Evaporative Emission System Leak Test (Loose/Disconnected Fuel Cap)

0x0458:P0458:Evaporative Emissions System Charcoal Canister Control Valve Circuit Low Voltage

0x0459:P0459:Evaporative Emissions System Charcoal Canister Control Valve Circuit High Voltage

0x0460:P0460:Fuel Level Sensor A Circuit

0x0461:P0461:Fuel Level Sensor A Circuit Range/Performance

0x0462:P0462:Fuel Level Sensor A Circuit Low Voltage

0x0463:P0463:Fuel Level Sensor A Circuit High Voltage

0x0464:P0464:Fuel Level Sensor A Circuit Intermittent Interruption

0x0465:P0465:Fuel Evaporative Emission Control Dirt Flow Sensor Circuit

0x0466:P0466:Fuel Evaporative Emission Control Dirt Flow Sensor Circuit Range/Performance

0x0467:P0467:Fuel Evaporative Emission Control Dirt Flow Sensor Circuit Low Voltage

0x0468:P0468:Fuel Evaporative Emission Control Dirt Flow Sensor Circuit Voltage High

0x0469:P0469:Fuel Evaporative Emission Control Dirt Flow Sensor Circuit Intermittent Interruption

0x0470:P0470:Exhaust Pressure Sensor A Circuit

0x0471:P0471:Exhaust Pressure Sensor A Circuit Range/Performance

0x0472:P0472:Exhaust Pressure Sensor A Circuit Low Voltage

0x0473:P0473:Exhaust Pressure Sensor A Circuit High Voltage

0x0474:P0474:Exhaust Pressure Sensor A Circuit Intermittent/Unstable

0x0475:P0475:Exhaust Pressure Control Valve

0x0476:P0476:Exhaust Pressure Control Valve Range/Performance

0x0477:P0477:Exhaust Pressure Control Valve Low Voltage
0x0478:P0478:Exhaust Pressure Control Valve High Voltage
0x0479:P0479:Exhaust pressure Control Valve Intermittent Interruption
0x047A:P047A:Exhaust Pressure Sensor B Circuit
0x047B:P047B:Exhaust Pressure Sensor B Circuit Range/Performance
0x047C:P047C:Exhaust Pressure Sensor B Circuit Low Voltage
0x047D:P047D:Exhaust Pressure Sensor B Circuit High Voltage
0x047E:P047E:Exhaust Pressure Sensor B Circuit Intermittent/Unstable
0x0480:P0480:Cooling Fan 1 Control Circuit
0x0481:P0481:Cooling Fan 2 Control Circuit
0x0482:P0482:Cooling Fan 3 Control Circuit
0x0483:P0483:Cooling Fan Reasonability Check
0x0484:P0484:Cooling Fan Circuit High Current
0x0485:P0485:Cooling Fan Power Mode/Ground Circuit
0x0486:P0486:Exhaust Gas Recirculation Sensor B circuit
0x0487:P0487:Exhaust Recirculation Throttle Position Control Circuit
0x0488:P0488:Exhaust Recirculation Throttle Position Control Range/Performance
0x0489:P0489:Exhaust Gas Recirculation Control Circuit Low Voltage
0x0490:P0490:Exhaust Gas Recirculation Control Circuit Voltage High
0x0491:P0491:Insufficient Flow to Secondary Air Injection Unit
0x0492:P0492:Insufficient Flow to Secondary Air Injection Unit
0x0493:P0493:Cooling Fan Overspeed
0x0494:P0494:Cooling Fan Speed Low
0x0495:P0495:Cooling Fan Speed High
0x0496:P0496:Evaporative Discharge System Purge Airflow High
0x0497:P0497:Evaporative Discharge System Purge Airflow Low
0x0498:P0498:Evaporative Emission System Vent Valve Control Circuit Voltage Low
0x0499:P0499:Evaporative Emission System Vent Valve Control Circuit Voltage High
0x0500:P0500:Vehicle Speed Sensor A
0x0501:P0501:Vehicle Speed Sensor A Range/Performance

0x0502:P0502:Vehicle Speed Sensor A Circuit Input Voltage Low
0x0503:P0503:Vehicle Speed Sensor A Intermittent Interrupt/Unstable/High Voltage
0x0504:P0504:Brake Switch A/B Correlation
0x0505:P0505:Idle Air Control System
0x0506:P0506:Idle Air Control System RPM Lower Than Expected
0x0507:P0507:Idle Air Control System Speed Higher Than Expected
0x0508:P0508:Idle Air Control System Circuit Voltage Low
0x0509:P0509:Idle Air Control System Circuit Voltage High
0x050A:P050A:Cold Start Idle Control System Performance
0x050B:P050B:Cold Start Ignition Timing Performance
0x050C:P050C:Cold Start Engine Coolant Temperature Performance
0x050D:P050D:Cold Start Rough Idling
0x0510:P0510:Throttle Position Switch
0x0511:P0511:Idle Air Control Circuit
0x0512:P0512:Starter Request Circuit
0x0513:P0513:Incorrect Stop Unit Key
0x0514:P0514:Battery Temperature Sensor Circuit Range/Performance
0x0515:P0515:Battery Temperature Sensor Circuit
0x0516:P0516:Battery Temperature Sensor Circuit Low Voltage
0x0517:P0517:Battery Temperature Sensor Circuit Voltage High
0x0518:P0518:Idle air Control Circuit Intermittent Interruption
0x0519:P0519:Idle air Control System Performance
0x0520:P0520:Oil Pressure Sensor/Switch Circuit
0x0521:P0521:Oil Pressure Sensor/Switch Range/Performance
0x0522:P0522:Oil Pressure Sensor/Switch Low Voltage
0x0523:P0523:Oil Pressure Sensor/Switch Voltage High
0x0524:P0524:Oil Pressure Too Low
0x0525:P0525:Cruise Control Servo Control Circuit Range/Performance
0x0526:P0526:Cooling Fan Sensor Circuit
0x0527:P0527:Cooling Fan Sensor Circuit Scope/Performance

0x0528:P0528:Cooling Fan Sensor No Signal
0x0529:P0529:Cooling Fan Sensor Intermittent Interruption
0x0530:P0530:A/C Refrigerant Pressure Sensor A Circuit
0x0531:P0531:A/C Refrigerant Pressure Sensor A Circuit Range/Performance
0x0532:P0532:Refrigerant Pressure Sensor A Circuit Low Voltage
0x0533:P0533:Refrigerant Pressure Sensor A Circuit High Voltage
0x0534:P0534:Refrigerant Volume Loss
0x0535:P0535:Evaporation Temperature Sensor Circuit
0x0536:P0536:Evaporative Temperature Sensor Circuit Range/Performance
0x0537:P0537:Evaporation Temperature Sensor Circuit Voltage Low
0x0538:P0538:Evaporation Temperature Sensor Circuit Voltage High
0x0539:P0539:Evaporation Temperature Sensor Circuit Intermittent Interruption
0x0540:P0540:Preheater A Circuit
0x0541:P0541:Preheater A Circuit Low Voltage
0x0542:P0542:Preheater A Circuit High Voltage
0x0543:P0543:Preheater A Circuit Disconnected
0x0544:P0544:Temperature Sensor Circuit (BANK 1 SENSOR 1)
0x0545:P0545:Temperature Sensor Circuit Low Voltage (BANK 1 SENSOR 1)
0x0546:P0546:Temperature Sensor Circuit Voltage High (BANK 1 SENSOR 1)
0x0547:P0547:Temperature Sensor Circuit (BANK 2 SENSOR 1)
0x0548:P0548:Temperature Sensor Circuit Voltage Low (BANK 2 SENSOR 1)
0x0549:P0549:Temperature Sensor Circuit Voltage High (BANK 2 SENSOR 1)
0x0550:P0550:Steering pressure Sensor/Switch Circuit
0x0551:P0551:Steering Pressure Sensor/Switch Circuit Range/Performance
0x0552:P0552:Steering Pressure Sensor/Switch Circuit Input Voltage Low
0x0553:P0553:Steering Pressure Sensor/Switch Circuit Input Voltage High
0x0554:P0554:Steering Pressure Sensor/Switch Circuit Intermittent Interruption
0x0555:P0555:Boost Sensor Circuit
0x0556:P0556:Boost Sensor Circuit Range/Performance
0x0557:P0557:Boost Sensor Circuit Low Input Voltage

0x0558:P0558:Boost Sensor Circuit Input Voltage High
0x0559:P0559:Boost Sensor Circuit Intermittent Interruption
0x0560:P0560:Voltage
0x0561:P0561:Voltage Instability
0x0562:P0562:Voltage Too Low
0x0563:P0563:Voltage Too High
0x0564:P0564:Control Multifunction Input A Circuitry
0x0565:P0565:Control Signal in Progress
0x0566:P0566:Control signal Broken
0x0567:P0567:Control Regain Signal
0x0568:P0568:Control Set Signal
0x0569:P0569:Control Coast Signal
0x0570:P0570:Control Acceleration Signal
0x0571:P0571:Switch A Circuit
0x0572:P0572:Switch A Circuit Low Voltage
0x0573:P0573:Switch A Circuit high Voltage
0x0574:P0574:Control System - Vehicle Speed Too High
0x0575:P0575:Control Input Circuit
0x0576:P0576:Control Input Circuit Low Voltage
0x0577:P0577:Control Input Circuit Voltage High
0x0578:P0578:Control Multifunction Input A Circuit Stuck
0x0579:P0579:Control Multifunction Input A Circuit Range/Performance
0x0580:P0580:Control Multifunction Input A Circuit Low Voltage
0x0581:P0581:Control Multifunction Input A Circuit High Voltage
0x0582:P0582:Control Vacuum Control Circuit / Disconnected
0x0583:P0583:Cruise Control Vacuum Control Circuit Voltage Low
0x0584:P0584:Cruise control Vacuum Control Circuit voltage high
0x0585:P0585:Cruise control multifunction Input A/B correlation
0x0586:P0586:Cruise Control Ventilation Control Circuit / Disconnect
0x0587:P0587:Cruise Control Ventilation Control Circuit Low Voltage

0x0588:P0588:Cruise Control ventilation control circuit voltage high
0x0589:P0589:Cruise Control multifunction input B circuit
0x0590:P0590:Cruise Control Multi-Function Input B Circuit Stuck
0x0591:P0591:Cruise Control Multi-Function Input B Circuit Range/Performance
0x0592:P0592:Cruise Control Multi-Input B Circuit Low Voltage
0x0593:P0593:Cruise Control Multi-Input B Circuit High Voltage
0x0594:P0594:Cruise Control Servo Control Circuit/Disconnected
0x0595:P0595:Cruise control servo Control circuit voltage low
0x0596:P0596:Cruise control servo Control circuit voltage high
0x0597:P0597:Thermostat Heater Control circuit/disconnect
0x0598:P0598:Thermostat Heater Control circuit voltage low
0x0599:P0599:Thermostat Heater Control circuit voltage high
0x0600:P0600:Serial Communication link
0x0601:P0601:Internal Control Module memory check accumulation error
0x0602:P0602:Control Module Programming Error
0x0603:P0603:Internal Control Module hold memory (KAM) error
0x0604:P0604:Random Memory (RAM) error in control module
0x0605:P0605:Control Module read-only memory (ROM) error
0x0606:P0606:Electronic Control Module/Powertrain Control Module Processor
0x0607:P0607:Control Module Performance
0x0608:P0608:Speed Sensor Control Module Output A
0x0609:P0609:Speed Sensor Control Module Output B
0x060A:P060A:Monitor Processor Performance in Control Module
0x060B:P060B:Analog/Digital Processing Performance in Control Module
0x060C:P060C:Main Processor Performance in Control Module
0x060D:P060D:Accelerator Pedal Position Performance in Control Module
0x060E:P060E: Throttle Position Performance in Control Module
0x060F:P060F:Control Module Coolant Temperature Performance
0x0610:P0610:Control Module Car Selection Error
0x0611:P0611:Fuel Injector Control Module Performance

0x0612:P0612:Fuel Injector Control Module Relay Control

0x0613:P0613:Transmission Control Module Processor

0x0614:P0614:Electronic Control Module/Transmission Control Module
Incompatibility

0x0615:P0615:Starter Relay Circuit

0x0616:P0616:Starter Relay Circuit Low Voltage

0x0617:P0617:Starter Relay Circuit High Voltage

0x0618:P0618:Alternative Fuel Control Module Hold Memory/ Error

0x0619:P0619:Replaceable Fuel Control Module Random Memory/Read Only
Memory Error

0x061A:P061A:Inner Control Module Torque Converter Performance

0x061B:P061B:Inner Control Module Torque Converter Calculation Performance

0x061C:P061C:Internal Control Module Engine RPM Performance

0x061D:P061D:Internal Control Module Engine Air Mass Performance

0x061E:P061E:Internal Control Module Brake Signaling Performance

0x061F:P061F:Internal Control Module Throttle Actuator Controller Performance

0x0620:P0620:Generator Control Circuitry

0x0621:P0621:Generator Indicator Light/L Terminal Circuitry

0x0622:P0622:Generator Indicator/F Terminal Circuit

0x0623:P0623:Generator Indicator Lamp Control Circuit

0x0624:P0624:Fuel Cap Lamp Control Circuit

0x0625:P0625:Generator Magnetic Field/F Terminal Circuit Low Voltage

0x0626:P0626:Generator Magnetic Field/F Terminal Circuit Voltage High

0x0627:P0627:Fuel Pump A Control Circuit/Disconnection

0x0628:P0628:Fuel Pump A Control Circuit Voltage Low

0x0629:P0629:Fuel Pump A Circuit Voltage High

0x062A:P062A:Fuel Pump A Control Circuit Range/Performance

0x062B:P062B:Fuel Injector Control Performance in Control Module

0x062C:P062C:Vehicle Speed Performance in Control Module

0x062D:P062D:Group 1 Cylinder Fuel Injector Driver Circuit Performance

0x062E:P062E:Group II Cylinder Fuel Injector Driver Circuit Performance

0x062F:P062F:EEPROM Error in Control Module

0x0630:P0630:VIN Not programmed or incompatible - Electronic Control
Module/Power train Control Module

0x0631:P0631:VIN Not Programmed or Incompatible-Transmission Control Module

0x0632:P0632:Odometer Not Programmed-Electronic Control Module/Power train
Control Module

0x0633:P0633:Stopper Key Not Programmed-Electronic Control Module/Power train
Control Module

0x0634:P0634:Powertrain Control Module/Electronic Control Module/Transmission
Control Module Internal Temperature Too High

0x0635:P0635:Power Steering Control Circuit

0x0636:P0636:Power Steering Control Circuit Low Voltage

0x0637:P0637:Power Steering Control Circuit Voltage High

0x0638:P0638:Throttle Actuator Control Range/Performance (BANK 1)

0x0639:P0639:Throttle Actuator Control Range/Performance (BANK 2)

0x063A:P063A:Generator Voltage Sensor Circuit

0x063B:P063B:Generator Voltage Sensor Circuit Range/Performance

0x063C:P063C:Generator Voltage Sensor Circuit Low Voltage

0x063D:P063D:Generator Voltage Sensor Circuit Voltage High

0x063E:P063E:Auto Config Throttle Input Not Present

0x063F:P063F:Auto Config Engine Coolant Temperature Input Not present

0x0640:P0640:Intake Preheater Control Circuit

0x0641:P0641:Sensor Reference Voltage A Circuit/Disconnect

0x0642:P0642:Sensor Reference Voltage A Circuit Low Voltage

0x0643:P0643:Sensor Reference Voltage A Circuit Voltage High

0x0644:P0644:Driver Display Serial Communication Circuit

0x0645:P0645:A/C Clutch Relay Control Circuit

0x0646:P0646:A/C Clutch Relay Control Circuit Voltage Low

0x0647:P0647:A/C Clutch Relay Control Circuit Voltage High

0x0648:P0648:Anti-Theft Indicator Control Circuit

0x0649:P0649:Speed Control Indicator Control Circuit

0x0650:P0650:Malfunction Indicator Control Circuit

0x0651:P0651:Sensor Reference Voltage B Circuit/Disconnect

0x0652:P0652:Sensor Reference Voltage B Circuit Low Voltage

0x0653:P0653:Sensor Reference Voltage B Circuit Voltage High

0x0654:P0654:Engine Speed Output Circuit

0x0655:P0655:Engine Temperature Indicator Output Control Circuit

0x0656:P0656:Fuel Level Output Circuit

0x0657:P0657:Actuator Supply Voltage A Circuit/Disconnect

0x0658:P0658:Actuator Supply Voltage A Circuit Low Voltage

0x0659:P0659:Actuator Supply Voltage A Circuit High Voltage

0x0660:P0660:Intake Manifold Tuning Valve Control Circuit/Break (BANK 1 A)

0x0661:P0661:Intake Manifold Tuning Valve Control Circuit Voltage Low (BANK 1 A)

0x0662:P0662:Intake Manifold Tuning Valve Control Circuit Voltage High (BANK 1 A)

0x0663:P0663:Intake Manifold Tuning Valve Control Circuit/Disconnect (BANK 2 A)

0x0664:P0664:Intake Manifold Tuning Valve Control Circuit Low Voltage (BANK 2 A)

0x0665:P0665:Intake Manifold Tuning Valve Control Circuit Voltage High (BANK 2 A)

0x0666:P0666:Power train Control Module/Electronic Control Module/Transmission Control Module Internal Temperature Sensor Circuit

0x0667:P0667:Power train Control Module/Electronic Control Module/Transmission Control Module Internal Temperature Sensor Range/Performance

0x0668:P0668:Power train Control Module/Electronic Control Module/Transmission Control Module Internal Temperature Sensor Circuit Voltage Low

0x0669:P0669:Power train Control Module/Electronic Control Module/Transmission

Control Module Internal Temperature Sensor Circuit Voltage High

0x0670:P0670:Preheat Plug Control Module Control Circuit/Disconnect

0x0671:P0671:1 Cylinder Preheater Plug Circuit

0x0672:P0672:2 Cylinder Preheat Plug Circuit

0x0673:P0673:3 Cylinder Preheat Plug Circuit

0x0674:P0674:4 Cylinder Preheat Plug Circuit

0x0675:P0675:5 Cylinder Preheat Plug Circuit

0x0676:P0676:6 Cylinder Preheat Plug Circuit

0x0677:P0677:7 Cylinder Preheat Plug Circuit

0x0678:P0678:8 Cylinder Preheat Plug Circuit

0x0679:P0679:9 Cylinder Preheater Plug Circuit

0x0680:P0680:10 Cylinder Preheat Plug Circuit

0x0681:P0681:11 Cylinder Preheat Plug Circuit

0x0682:P0682:12 Cylinder Preheat plug Circuit

0x0683:P0683:Preheat Plug Control Module to Powertrain Control Module
Communication Circuit

0x0684:P0684:Preheat Plug Control Module to Power train Control Module

Communication Circuit Range/Performance

0x0685:P0685:Electronic Control Module/Power train Control Module Power Relay
Control Circuit/Break Circuit

0x0686:P0686:Electronic Control Module/Power train Control Module Power Relay
Control Circuit Low Voltage

0x0687:P0687:Electronic Control Module/Power train Control Module Power Relay
Control Circuit High Voltage

0x0688:P0688:Electronic Control Module/Power train Control Module Power Relay
Control Circuit/Disconnected

0x0689:P0689:Electronic Control Module/Power train Control Module Power Relay
Control Circuit Voltage Low

0x0690:P0690:Electronic Control Module/Power train Control Module Power Relay
Control Circuit Voltage High

0x0691:P0691:Cooling Fan 1 Control Circuit Voltage Low
0x0692:P0692:Cooling Fan 1 Control Circuit Voltage High
0x0693:P0693:Cooling Fan 2 Control Circuit Low Voltage
0x0694:P0694:Cooling Fan 2 Control Circuit Voltage High
0x0695:P0695:Cooling Fan 3 Control Circuit Low Voltage
0x0696:P0696:Cooling Fan 3 Control Circuit High Voltage
0x0697:P0697:Sensor Reference Voltage C Circuit/Disconnect
0x0698:P0698:Sensor Reference Voltage C Circuit Voltage Low
0x0699:P0699:Sensor Reference Voltage C Circuit High
0x0700:P0700:Transmission Control System (MIL Request)
0x0701:P0701:Transmission Control System Range/Performance
0x0702:P0702:Transmission Control System Electronics
0x0703:P0703:Brake Switch B Circuit
0x0704:P0704:Clutch Switch Input Circuit Failure
0x0705:P0705:Transmission Range Sensor A Circuit Failure (PRNDL Input)
0x0706:P0706:Transmission Range Sensor A Circuit Range/Performance
0x0707:P0707:Transmission Range Sensor A Circuit Low Voltage
0x0708:P0708:Transmission Range Sensor A Circuit Voltage High
0x0709:P0709:Transmission Range Sensor A Circuit Intermittent Interruption
0x0710:P0710:Transmission Fluid Temperature Sensor A Circuit
0x0711:P0711:Transmission Fluid Temperature Sensor A Circuit Range/Performance
0x0712:P0712:Driveline Temperature Sensor A Circuit Low Voltage
0x0713:P0713:Driveline Temperature Sensor A Circuit High Voltage
0x0714:P0714:Driveline Temperature Sensor A Circuit Intermittent Interruption
0x0715:P0715:Input/Turbo Speed Sensor A Circuit
0x0716:P0716:Input/Turbo Speed Sensor A Circuit Range/Performance
0x0717:P0717:Input/Turbine Speed Sensor A Circuit No Signal
0x0718:P0718:Input/Turbine Speed Sensor A Circuit Intermittent Interruption
0x0719:P0719:Brake Switch B Circuit Low Voltage
0x0720:P0720:Output Speed Sensor Circuit

0x0721:P0721:Output Speed Sensor Circuit Range/Performance
0x0722:P0722:Output Speed Sensor Circuit No Signal
0x0723:P0723:Output Speed Sensor Circuit Intermittent Interruption
0x0724:P0724:Brake Switch B Circuit High Voltage
0x0725:P0725:Engine Speed Input Circuit
0x0726:P0726:Engine Speed Input Circuit Range/Performance
0x0727:P0727:Engine Speed Input Circuit No Signal
0x0728:P0728:Engine Input Circuit Intermittent Interruption
0x0729:P0729:Ratio 6 Incorrect
0x0730:P0730:Ratio Incorrect
0x0731:P0731:Ratio 1 Incorrect
0x0732:P0732:Ratio 2 Is Incorrect
0x0733:P0733:Ratio 3 Incorrect
0x0734:P0734:Ratio 4 Incorrect
0x0735:P0735:Ratio 5 Incorrect
0x0736:P0736:Reverse Ratio Incorrect
0x0737:P0737:Transmission Engine Speed Output Circuit
0x0738:P0738:Transmission Engine Speed Output Circuit Low Voltage
0x0739:P0739:Transmission Engine Speed Output Circuit Voltage High
0x0740:P0740:Torque Converter Clutch Circuit/Disconnect
0x0741:P0741:Torque Converter Clutch Circuit Performance or Stuck Closed
0x0742:P0742:Torque Converter Clutch Circuit Stuck on
0x0743:P0743:Torque Converter Clutch Circuit Electronic Device
0x0744:P0744:Torque Converter Clutch Circuit Intermittent Interruption
0x0745:P0745:Pressure Control Solenoid Valve A
0x0746:P0746:Pressure Control Solenoid Valve A Performance or Stuck Closed
0x0747:P0747:Pressure Control Solenoid A Stuck Open
0x0748:P0748:Pressure Control Solenoid A Electronics
0x0749:P0749:Pressure Control Solenoid A Intermittent Interruptions
0x0750:P0750:Shift Solenoid A

0x0751:P0751:Shift Solenoid A Performance or Stuck Off
0x0752:P0752:Shift Solenoid A Stuck On
0x0753:P0753:Shift Solenoid A Electronics
0x0754:P0754:Shift Solenoid A Intermittent Interruption
0x0755:P0755:Shift Solenoid B
0x0756:P0756:Shift Solenoid B Performance or On-On Off
0x0757:P0757:Shift Solenoid B Stuck in Open
0x0758:P0758:Shift Solenoid B Electronics
0x0759:P0759:Shift Solenoid B Intermittent Interruption
0x075A:P075A:Shift Solenoid G
0x075B:P075B:Shift Solenoid G Performance or Stuck Closed
0x075C:P075C:Shift Solenoid G On at Open
0x075D:P075D:Shift Solenoid G Electronics
0x075E:P075E:Shift Solenoid G Intermittent Interruptions
0x0760:P0760:Shift Solenoid C
0x0761:P0761:Shift Solenoid C Performance or stuck off
0x0762:P0762:Shift solenoid C Stuck On
0x0763:P0763:Shift Solenoid C Electronics
0x0764:P0764:Shift Solenoid C Intermittent Interruptions
0x0765:P0765:Shift Solenoid D
0x0766:P0766:Shift Solenoid D Performance or On in Off
0x0767:P0767:Shift Solenoid Valve D Stuck in On
0x0768:P0768:Shift Solenoid D Electronics
0x0769:P0769:Shift Solenoid D Intermittent Interruption
0x076A:P076A:Shift Solenoid H
0x076B:P076B:Shift Solenoid H Performance or Stuck Closed
0x076C:P076C:Shift Solenoid H Stuck On
0x076D:P076D:Shift Solenoid H Electronics
0x076E:P076E:Shift Solenoid H Intermittent Interruption
0x0770:P0770:Shift Solenoid E

0x0771:P0771:Shift Solenoid E Performance or Stuck Closed
0x0772:P0772:Shift Solenoid E Stuck On
0x0773:P0773:Shift Solenoid E Electronics
0x0774:P0774:Shift Solenoid E Intermittent Interruption
0x0775:P0775:Pressure Control Solenoid B
0x0776:P0776:Pressure Control Solenoid B Performance or Stuck Closed
0x0777:P0777:Pressure Control Solenoid B Stuck Open
0x0778:P0778:Pressure Control Solenoid Valve B Electronics
0x0779:P0779:Pressure Control Solenoid B Intermittent Interruption
0x0780:P0780:Shift Error
0x0781:P0781:1-2 Gears
0x0782:P0782:2-3 Gears
0x0783:P0783:3-4 Gears
0x0784:P0784:4-5 Gears
0x0785:P0785:Shift/Timing Solenoid Valve
0x0786:P0786:Shift/Timing Solenoid Range/Performance
0x0787:P0787:Shift/Timing Solenoid Low Voltage
0x0788:P0788:Shift/Timing Solenoid Voltage High
0x0789:P0789:Shift/timing Solenoid Intermittent Interruption
0x0790:P0790:Normal/Performance Switch Circuit
0x0791:P0791:Intermediate Shaft Speed Sensor A Circuit
0x0792:P0792:Intermediate Shaft Speed Sensor A Circuit Range/Performance
0x0793:P0793:Intermediate Shaft Speed Sensor A Circuit No Signal
0x0794:P0794:Intermediate Shaft Speed Sensor A Circuit Intermittent Interruption
0x0795:P0795:Pressure Control Solenoid C
0x0796:P0796:Pressure Control Solenoid C Performance or Stuck Closed
0x0797:P0797:Pressure Control Solenoid Valve C Open at Open
0x0798:P0798:Pressure Control Solenoid Valve C Electronics
0x0799:P0799:Pressure Control Solenoid C Intermittent Interruption
0x0800:P0800:Distributor Control System (Fault Indicator Request)

0x0801:P0801:Reverse Inhibit Control Circuit
0x0802:P0802:Transmission Control System Malfunction Indicator Request
Circuit/Disconnect
0x0803:P0803:Upshift/Jump Gear Solenoid Control Circuit
0x0804:P0804:Upshift/Jump Gear indicator Control Circuit
0x0805:P0805:Clutch Position Sensor Circuit
0x0806:P0806:Clutch Position Sensor Circuit Range/Performance
0x0807:P0807:Clutch Position Sensor Circuit Low Voltage
0x0808:P0808:Clutch Position Sensor Circuit Voltage High
0x0809:P0809:Clutch Position Sensor Circuit Intermittent Interruption
0x080A:P080A:Clutch Position Not Learned
0x080B:P080B:Upshift/Jump Solenoid Control Circuit range/Performance
0x080C:P080C:Upshift/Jump Gear Solenoid Control Circuit Low Voltage
0x080D:P080D:Upshift/Jump solenoid Control Circuit Voltage high
0x0810:P0810:Clutch position Control Error
0x0811:P0811:Clutch A Slip Too Much
0x0812:P0812:Reverse Gear Input Circuit
0x0813:P0813:Reverse Gear Output Circuit
0x0814:P0814:Transmission Range Display Circuit
0x0815:P0815:Upshift Switch Circuit
0x0816:P0816:Downshift Switch Circuit
0x0817:P0817:Starter Inhibit Circuit/Open Circuit
0x0818:P0818:Disconnect Switch Input System Circuit
0x0819:P0819:Lift Shift Switch Transmission Range Related
0x081A:P081A:Starter Disable Circuit Low Voltage
0x081B:P081B:Starter Disable Circuit Voltage High
0x081C:P081C:Parking Input Circuit
0x081D:P081D:Neutral Input Circuit
0x081E:P081E:Excessive Clutch B Slip
0x0820:P0820:Shifter X-Y Position Sensor Circuit

0x0821:P0821:Shifter X Position Circuit
0x0822:P0822:Shifter Y Position Circuit
0x0823:P0823:Shifter X Position Circuit Intermittent Interrupt
0x0824:P0824:Shifter Y Position Circuit Intermittently Interrupted
0x0825:P0825:Shifter Push-Pull Switch (Advance Shift)
0x0826:P0826:Up/Downshift Switch Circuit
0x0827:P0827:Up/Downshift Switch Circuit Low Voltage
0x0828:P0828:Up/Downshift Switch Circuit Voltage High
0x0829:P0829:5-6 Gears
0x0830:P0830:Clutch Pedal Switch A Circuit
0x0831:P0831:Clutch Pedal Switch A Circuit Low Voltage
0x0832:P0832:Clutch Pedal Switch A Circuit High Voltage
0x0833:P0833:Clutch Pedal Switch B Circuit
0x0834:P0834:Clutch Pedal Switch B Circuit Low Voltage
0x0835:P0835:Clutch Pedal Switch B Circuit High Voltage
0x0836:P0836:4WD Switch Circuit
0x0837:P0837:4WD Switch Circuit Range/Performance
0x0838:P0838:4WD Switch Circuit Low Voltage
0x0839:P0839:4WD Switch Circuit Voltage High
0x083A:P083A:Transmission Fluid Pressure Sensor/Switch G Circuit
0x083B:P083B:Transmission Oil Pressure Sensor/Switch G Circuit
Range/Performance
0x083C:P083C:Transmission Oil Pressure Sensor/Switch G Circuit Low Voltage
0x083D:P083D:Transmission Oil Pressure Sensor/Switch G Circuit High Voltage
0x083E:P083E:Transmission Oil Pressure Sensor/Switch G Circuit Intermittent
Interruption
0x0840:P0840:Transmission Fluid Pressure Sensor/Switch A Circuit
0x0841:P0841:Transmission Oil Pressure Sensor/Switch A Circuit
Range/Performance
0x0842:P0842:Transmission Oil Pressure Sensor/Switch A Circuit Low Voltage

0x0843:P0843:Transmission Oil Pressure Sensor/Switch A Circuit High Voltage

0x0844:P0844:Transmission Oil Pressure Sensor/Switch A Circuit Intermittent
Interruption

0x0845:P0845:Transmission Oil Pressure Sensor/Switch B Circuit

0x0846:P0846:Transmission Oil Pressure Sensor/Switch B Circuit
Range/Performance

0x0847:P0847:Transmission Oil Pressure Sensor/Switch B Circuit Low Voltage

0x0848:P0848:Transmission Oil Pressure Sensor/Switch B Circuit High Voltage

0x0849:P0849:Transmission Oil Pressure Sensor/Switch B Circuit Intermittent
Interruption

0x084A:P084A:Transmission Oil Pressure Sensor/Switch H Circuit

0x084B:P084B:Transmission Oil Pressure Sensor/Switch H Circuit
Range/Performance

0x084C:P084C:Transmission Oil Pressure Sensor/Switch H Circuit Low Voltage

0x084D:P084D:Transmission Oil Pressure Sensor/Switch H Circuit Voltage High

0x084E:P084E:Transmission Oil Pressure Sensor/Switch H Circuit Intermittent
Interrupt

0x0850:P0850:Park/Neutral Switch Input Circuit

0x0851:P0851:Park/Neutral Switch Input Circuit Low Voltage

0x0852:P0852:Park/Neutral Switch Input Circuit Voltage High

0x0853:P0853:Drive Switch Input Circuit

0x0854:P0854:Drive Switch Input Circuit Voltage Low

0x0855:P0855:Drive Switch Input Circuit Voltage High

0x0856:P0856:Traction Control Input Signal

0x0857:P0857:Traction Control Input Signal Range/Performance

0x0858:P0858:Traction Control Input Signal Voltage Low

0x0859:P0859:Traction Control Input Signal Voltage High

0x0860:P0860:Shift Module Communication Circuit

0x0861:P0861:Shift Module Communication Circuit Voltage Low

0x0862:P0862:Shift Module Communication Circuit Voltage High

0x0863:P0863:TCM Communication Circuit

0x0864:P0864:TCM Communication Circuit Range/Performance

0x0865:P0865:TCM Communication Circuit Voltage Low

0x0866:P0866:TCM Communication Circuit Voltage High

0x0867:P0867:Transmission Oil Pressure

0x0868:P0868:Transmission Oil Pressure Voltage Low

0x0869:P0869:Transmission Oil Pressure Voltage High

0x0870:P0870:Transmission Oil Pressure Sensor/Switch C Circuit

0x0871:P0871:Transmission Oil Pressure Sensor/Switch C Circuit
Range/Performance

0x0872:P0872:Transmission Oil Pressure Sensor/Switch C Circuit Low Voltage

0x0873:P0873:Transmission Oil Pressure Sensor/Switch C Circuit High Voltage

0x0874:P0874:Transmission Oil Pressure Sensor/Switch C Circuit Intermittent
Interruption

0x0875:P0875:Transmission Oil Pressure Sensor/Switch D Circuit

0x0876:P0876:Transmission Oil Pressure Sensor/Switch D Circuit
Range/Performance

0x0877:P0877:Transmission Oil Pressure Sensor/Switch D Circuit Low Voltage

0x0878:P0878:Transmission Oil Pressure Sensor/Switch D Circuit High Voltage

0x0879:P0879:Transmission Oil Pressure Sensor/Switch D Circuit Intermittent
Interruption

0x0880:P0880:TCM Power Input Signal

0x0881:P0881:TCM Power Input Signal Range/Performance

0x0882:P0882:TCM Power Input Signal Low Voltage

0x0883:P0883:TCM Power Input Signal Voltage High

0x0884:P0884:TCM Power Input Signal Intermittent Interruption

0x0885:P0885:TCM Power Relay Control Circuit/Disconnection

0x0886:P0886:TCM Power Relay Control Circuit Voltage Low

0x0887:P0887:TCM Power Relay Control Circuit Voltage High

0x0888:P0888:TCM Power Relay Sensor Circuit

0x0889:P0889:TCM Power Relay Sensor Circuit Range/Performance
0x0890:P0890:TCM Power Relay Sensor Circuit Voltage Low
0x0891:P0891:TCM Power Relay Sensor Circuit High Voltage
0x0892:P0892:TCM Power Relay Sensor Circuit Intermittent Interruption
0x0893:P0893:Multiple Gears In Gear
0x0894:P0894:Transmission Component slipping
0x0895:P0895:Shift Time Too Short
0x0896:P0896:Shift Time Too Long
0x0897:P0897:Transmission Fluid Deterioration
0x0898:P0898:Transmission Control System Malfunction Indicator Request Circuit
Voltage Low
0x0899:P0899:Transmission Control System Malfunction indicator request circuit
Voltage High
0x0900:P0900:Clutch Actuator Circuit/Disconnection
0x0901:P0901:Clutch Actuator Circuit Range/Performance
0x0902:P0902:Clutch Actuator Circuit Low Voltage
0x0903:P0903:Clutch Actuator Circuit Voltage High
0x0904:P0904:Locator Plate Selector Position Circuit
0x0905:P0905:Locator Plate Selector Position Circuit Range/Performance
0x0906:P0906:Locator Plate Selector Position Circuit Voltage Low
0x0907:P0907:Positioner Selector Position Circuit High Voltage
0x0908:P0908:Locator Board Select Position Circuit Intermittent Interruption
0x0909:P0909:Positioning Board Select Control Error
0x0910:P0910:Positioning Plate Select Actuator Circuit/Interruption
0x0911:P0911:Positioning Plate Select Actuator Circuit Range/Performance
0x0912:P0912:Positioning Board Selector Actuator Circuit Low Voltage
0x0913:P0913:Positioning Board Selector Actuator Circuit Voltage High
0x0914:P0914:Shift Position Circuit
0x0915:P0915:Shift Position Circuit Range/Performance
0x0916:P0916:Shift Position Circuit Voltage Low

0x0917:P0917:Shift Position Circuit Voltage High
0x0918:P0918:Shift Position Circuit Intermittent Interruption
0x0919:P0919:Shift Position Control Error
0x0920:P0920:Shift Forward Actuator Circuit/Disconnection
0x0921:P0921:Shift Forward Actuator Circuit Range/Performance
0x0922:P0922:Shift Forward Actuator Circuit Voltage Low
0x0923:P0923:Shift forward Actuator Circuit Voltage High
0x0924:P0924:Shift Reverse Actuator Circuit/Disconnect
0x0925:P0925:Shift Reverse Actuator Circuit Range/Performance
0x0926:P0926:Shift Reverse Actuator Circuit Low Voltage
0x0927:P0927:Shift Reverse Actuator Circuit Voltage High
0x0928:P0928:Shift Lock Solenoid Control Circuit/Disconnection
0x0929:P0929:Shift Lockout Solenoid Control Circuit Range/Performance
0x0930:P0930:Shift Lock Solenoid Control Circuit Low Voltage
0x0931:P0931:Shift Lock Solenoid Control Circuit High Voltage
0x0932:P0932:Hydraulic Sensor Circuit
0x0933:P0933:Hydraulic Sensor Range/Performance
0x0934:P0934:Hydraulic Sensor Circuit Low Voltage
0x0935:P0935:Hydraulic Sensor Circuit Voltage High
0x0936:P0936:Hydraulic Sensor Circuit Intermittent Interruption
0x0937:P0937:Hydraulic Oil Temperature Sensor Circuit
0x0938:P0938:Hydraulic Oil Temperature Sensor Range/Performance
0x0939:P0939:Hydraulic Oil Temperature Sensor Circuit Low Voltage
0x0940:P0940:Hydraulic Oil Temperature Sensor Circuit High Voltage
0x0941:P0941:Hydraulic Oil Temperature Sensor Circuit Intermittent Interruption
0x0942:P0942:Hydraulic Unit
0x0943:P0943:Hydraulic Unit Cycle Time Too Short
0x0944:P0944:Hydraulic Unit Pressure Loss
0x0945:P0945:Hydraulic Pump Relay Circuit/Disconnection
0x0946:P0946:Hydraulic Pump Relay Circuit Range/Performance

0x0947:P0947:Hydraulic Pump Relay Circuit Low Voltage
0x0948:P0948:Hydraulic Pump Relay Circuit Voltage High
0x0949:P0949:Auto Shift Manual Adaptation Learning Not Complete
0x0950:P0950:Auto Shift Manual Control Circuit
0x0951:P0951:Auto Shift Manual Control Circuit Range/Performance
0x0952:P0952:Auto Shift Manual Control Circuit Low Voltage
0x0953:P0953:Auto Shift Manual Control Circuit Voltage High
0x0954:P0954:Auto Shift Manual Control Circuit Intermittent Interruption
0x0955:P0955:Auto Shift Manual Mode Circuit
0x0956:P0956:Auto Shift Manual Mode Circuit Range/Performance
0x0957:P0957:Auto Shift Manual Mode Circuit Low Voltage
0x0958:P0958:Auto Shift Manual Mode Circuit Voltage High
0x0959:P0959:Auto Shift Manual Mode Circuit Intermittent Interruption
0x0960:P0960:Pressure Control Solenoid A Control Circuit / Disconnect
0x0961:P0961:Pressure Control Solenoid A Control Circuit Range/Performance
0x0962:P0962:Pressure Control Solenoid Valve A Control Circuit Low Voltage
0x0963:P0963:Pressure Control Solenoid A Control Circuit Voltage High
0x0964:P0964:Pressure Control Solenoid Valve B Control Circuit / Disconnected
0x0965:P0965:Pressure Control Solenoid Valve B Control Circuit Range/Performance
0x0966:P0966:Pressure Control Solenoid B Control circuit low voltage
0x0967:P0967:Pressure Control Solenoid Valve B Control Circuit Voltage High
0x0968:P0968:Pressure Control Solenoid Valve C Control Circuit/Disconnection
0x0969:P0969:Pressure Control Solenoid Valve C Control Circuit Range/Performance
0x0970:P0970:Pressure Control Solenoid Valve C Control Circuit Voltage Low
0x0971:P0971:Pressure Control Solenoid Valve C Control Circuit Voltage High
0x0972:P0972:Shift Solenoid A Control Circuit Range/Performance
0x0973:P0973:Shift Solenoid A Control Circuit Voltage Low
0x0974:P0974:Shift Solenoid A Control Circuit High Voltage
0x0975:P0975:Shift Solenoid B Control Circuit Range/Performance
0x0976:P0976:Shift Solenoid B Control Circuit Voltage Low

0x0977:P0977:Shift Solenoid A Control Circuit High Voltage
0x0978:P0978:Shift Solenoid C Control Circuit Range/Performance
0x0979:P0979:Shift Solenoid C Control Circuit Voltage Low
0x0980:P0980:Shift Solenoid C Control Circuit Voltage High
0x0981:P0981:Shift Solenoid D Control Circuit Range/Performance
0x0982:P0982:Shift Solenoid D Control Circuit Voltage Low
0x0983:P0983:Shift Solenoid D Control Circuit Voltage High
0x0984:P0984:Shift Solenoid E Control Circuit Range/Performance
0x0985:P0985:Shift Solenoid E Control Circuit Voltage Low
0x0986:P0986:Shift Solenoid E Control Circuit Voltage High
0x0987:P0987:Transmission Fluid Pressure Sensor/Switch E Circuit
0x0988:P0988:Transmission Oil Pressure Sensor/Switch E Circuit
Range/Performance
0x0989:P0989:Transmission Oil Pressure Sensor/Switch E Circuit Low Voltage
0x0990:P0990:Transmission Oil Pressure Sensor/Switch E Circuit High Voltage
0x0991:P0991:Transmission Oil Pressure Sensor/Switch E Circuit Gap Interrupted
0x0992:P0992:Transmission oil pressure sensor/switch F circuit
0x0993:P0993:Transmission Oil Pressure Sensor/Switch F Circuit
Range/Performance
0x0994:P0994:Transmission Oil Pressure Sensor/Switch F Circuit Low Voltage
0x0995:P0995:Transmission Oil Pressure Sensor/Switch F Circuit High Voltage
0x0996:P0996:Transmission Oil Pressure Sensor/Switch F Circuit Gap Interrupted
0x0997:P0997:Shift Solenoid F Control Circuit Range/Performance
0x0998:P0998:Shift Solenoid F Control Circuit Low Voltage
0x0999:P0999:Shift Solenoid F Control Circuit Voltage High
0x099A:P099A:Shift Solenoid G Control Circuit Range/Performance
0x099B:P099B:Shift Solenoid G Control Circuit Voltage Low
0x099C:P099C:Shift solenoid G Control circuit voltage high
0x099D:P099D:Shift solenoid H Control circuit range/performance
0x099E:P099E:Shift Solenoid H Control Circuit Voltage Low

0x099F:P099F:Shift Solenoid H Control Circuit Voltage High

0x0A00:P0A00:Motor Electronics Coolant Temperature Sensor Circuit

0x0A01:P0A01:Motor Electronics Coolant Temperature Sensor Circuit
Range/Performance

0x0A02:P0A02:Motor Electronics Coolant Temperature Sensor Circuit Low Voltage

0x0A03:P0A03:Motor Electronics Coolant Temperature Sensor Circuit High Voltage

0x0A04:P0A04:Motor Electronics Coolant Temperature Sensor Circuit Intermittent
Interruption

0x0A05:P0A05:Motor Electronics Coolant Pump Control Circuit/Disconnect

0x0A06:P0A06:Motor Electronics Cooling Pump Control Circuit Low Voltage

0x0A07:P0A07:Motor Electronic Device Cooling Pump Control Circuit Voltage High

0x0A08:P0A08:DC/DC Converter Circuit

0x0A09:P0A09:DC/DC Converter Circuit Low Input

0x0A0A:P0A0A:HV System Interlock Circuit

0x0A0B:P0A0B:HV System Interlock Circuit Performance

0x0A0C:P0A0C:HV System Interlock Circuit Low Voltage

0x0A0D:P0A0D:HV System Interlock Circuit High Voltage

0x0A0E:P0A0E:HV System Interlock Circuit Intermittent Interruption

0x0A0F:P0A0F:Engine Start Failure

0x0A10:P0A10:DC/DC Converter Circuit High Input

0x0A11:P0A11:DC/DC Converter Circuit/Disconnection

0x0A12:P0A12:DC/DC Converter Circuit Voltage Low

0x0A13:P0A13:DC/DC Converter Circuit Voltage High

0x0A14:P0A14:Engine Mount Circuit A/Disconnect

0x0A15:P0A15:Engine Mount Circuit A Low Voltage

0x0A16:P0A16:Engine Mount Circuit A High Voltage

0x0A17:P0A17:Motor Torque Converter Sensor Circuit

0x0A18:P0A18:Motor Torque Converter Sensor Circuit Range/Performance

0x0A19:P0A19:Motor Torque Converter Sensor Circuit Low Voltage

0x0A1A:P0A1A:Generator Control Module

0x0A1B:P0A1B:Drive Motor A Control Circuit
0x0A1C:P0A1C:Drive Motor B Control Circuit
0x0A1D:P0A1D:Hybrid Control Module
0x0A1E:P0A1E:Starter/Generator Control Module
0x0A1F:P0A1F:Battery Energy Control Module
0x0A20:P0A20:Motor Torque Sensor Circuit High Voltage
0x0A21:P0A21:Motor Torque Sensor Circuit Intermittent Interruption
0x0A22:P0A22:Generator Torque Sensor Circuit
0x0A23:P0A23:Generator Torque Sensor Circuit Range/Performance
0x0A24:P0A24:Generator Torque Sensor Circuit Low Voltage
0x0A25:P0A25:Generator Torque Sensor Circuit High Voltage
0x0A26:P0A26:Generator Torque Sensor Circuit Intermittently Interrupted
0x0A27:P0A27:Hybrid Battery Shutdown Circuit
0x0A28:P0A28:Hybrid Battery Shutdown Circuit Low Voltage
0x0A29:P0A29:Hybrid Battery Shutdown Circuit High Voltage
0x0A2A:P0A2A:Drive Motor A Temperature Sensor Circuit
0x0A2B:P0A2B:Drive Motor A Temperature Sensor Circuit Range/Performance
0x0A2C:P0A2C:Drive Motor A Temperature Sensor Circuit Low Voltage
0x0A2D:P0A2D:Drive Motor A Temperature Sensor Circuit High Voltage
0x0A2E:P0A2E:Drive Motor A Temperature Sensor Circuit Intermittent Interruption
0x0A2F:P0A2F:Drive Motor A Temperature Too High
0x0A30:P0A30:Drive Motor B Temperature Sensor Circuitry
0x0A31:P0A31:Drive Motor B Temperature Sensor Circuit Range/Performance
0x0A32:P0A32:Drive Motor B Temperature Sensor Circuit Low Voltage
0x0A33:P0A33:Drive Motor B Temperature Sensor Circuit High Voltage
0x0A34:P0A34:Drive Motor B Temperature Sensor Circuit Intermittent Interruption
0x0A35:P0A35:Drive Motor B temperature too High
0x0A36:P0A36:Generator Temperature Sensor Circuit
0x0A37:P0A37:Generator Temperature Sensor Circuit Range/Performance
0x0A38:P0A38:Generator Temperature Sensor Circuit Low Voltage

0x0A39:P0A39:Generator Temperature Sensor Circuit High Voltage
0x0A3A:P0A3A:Generator Temperature Sensor Circuit Intermittent Interruption
0x0A3B:P0A3B:Generator Temperature Too High
0x0A3C:P0A3C:Drive Motor A Inverter Overheating
0x0A3D:P0A3D:Drive Motor B Inverter Overheat
0x0A3E:P0A3E:Generator Inverter Overheat
0x0A3F:P0A3F:Drive Motor A Position Sensor Circuit
0x0A40:P0A40:Drive Motor A Position Sensor Circuit Range/Performance
0x0A41:P0A41:Drive Motor A Position Sensor Circuit Low Voltage
0x0A42:P0A42:Drive Motor A Position Sensor Circuit High Voltage
0x0A43:P0A43:Drive Motor A Position Sensor Circuit Intermittent Interruption
0x0A44:P0A44:Drive Motor A Position Sensor Circuit Overspeed
0x0A45:P0A45:Drive Motor B Position Sensor Circuit
0x0A46:P0A46:Drive Motor B Position Sensor Circuit Range/Performance
0x0A47:P0A47:Drive Motor B Position Sensor Circuit Low Voltage
0x0A48:P0A48:Drive Motor B Position Sensor Circuit High Voltage
0x0A49:P0A49:Drive Motor B Position Sensor Circuit Intermittent Interruption
0x0A4A:P0A4A:Drive Motor B Position Sensor Circuit Overspeed
0x0A4B:P0A4B:Generator Position Sensor Circuit
0x0A4C:P0A4C:Generator Position Sensor Circuit Range/Performance
0x0A4D:P0A4D:Generator Position Sensor Circuit Low Voltage
0x0A4E:P0A4E:Generator Position Sensor Circuit Voltage High
0x0A4F:P0A4F:Generator Position Sensor Circuit Intermittent Interruption
0x0A50:P0A50:Generator Position Sensor Circuit Overspeed
0x0A51:P0A51:Drive Motor A Current Sensor Circuit
0x0A52:P0A52:Drive Motor A Current Sensor Circuit Range/Performance
0x0A53:P0A53:Drive Motor A Current Sensor Circuit Low Voltage
0x0A54:P0A54:Drive Motor A Current Sensor Circuit High Voltage
0x0A55:P0A55:Drive Motor B Current Sensor Circuit
0x0A56:P0A56:Drive Motor B Current Sensor Circuit Range/Performance

0x0A57:P0A57:Drive Motor B Current Sensor Circuit Low Voltage
0x0A58:P0A58:Drive Motor B Current Sensor Circuit High Voltage
0x0A59:P0A59:Current Sensor Generator Circuit
0x0A5A:P0A5A:Current Sensor Generation Circuit Range/Performance
0x0A5B:P0A5B:Current Sensor Generation Circuit Voltage Low
0x0A5C:P0A5C:Current Sensor Generator Circuit Voltage High
0x0A5D:P0A5D:Drive Motor A phase U Current
0x0A5E:P0A5E:Drive Motor A phase U Current Voltage Low
0x0A5F:P0A5F:Drive Motor A Phase U Current High Voltage
0x0A60:P0A60:Drive Motor A Phase V Current
0x0A61:P0A61:Drive Motor A phase V Current Low Voltage
0x0A62:P0A62:Drive Motor A Phase V Current Voltage High
0x0A63:P0A63:Drive Motor A Phase W Current
0x0A64:P0A64:Drive Motor A Phase W Current Voltage Low
0x0A65:P0A65:Drive Motor A Phase W Current Voltage High
0x0A66:P0A66:Drive Motor B Phase U Current
0x0A67:P0A67:Drive Motor B phase U Current Low Voltage
0x0A68:P0A68:Drive Motor B Phase U Current Voltage High
0x0A69:P0A69:Drive Motor B Phase V Current
0x0A6A:P0A6A:Drive Motor A phase V Current Low Voltage
0x0A6B:P0A6B:Drive Motor A Phase V Current Voltage High
0x0A6C:P0A6C:Drive Motor A Phase W Current
0x0A6D:P0A6D:Drive Motor A Phase W Current Voltage Low
0x0A6E:P0A6E:Drive Motor Phase A W Current Voltage High
0x0A6F:P0A6F:Generator phase U Current
0x0A70:P0A70:Generator Phase U Current Voltage Low
0x0A71:P0A71:Generator Phase U Current Voltage High
0x0A72:P0A72:Generator Phase V Current
0x0A73:P0A73:Generator Phase V Current Voltage Low
0x0A74:P0A74:Generator phase V Current Voltage High

0x0A75:P0A75:Generator Phase W Current
0x0A76:P0A76:Generator Phase W Current Voltage Low
0x0A77:P0A77:Generator Phase W Current Voltage High
0x0A78:P0A78:Drive Motor A Inverter Performance
0x0A79:P0A79:Drive Motor B Inverter Performance
0x0A7A:P0A7A:Generator Inverter Performance
0x0A7B:P0A7B:Fault Indicator Lighting Requested by Battery Energy Control
Module
0x0A7C:P0A7C:Motor Electronics Overheating
0x0A7D:P0A7D:Low State of Charge of mixed battery packs
0x0A7E:P0A7E:Hybrid battery pack overheating
0x0A7F:P0A7F:Hybrid battery pack degradation
0x0A80:P0A80:Replace Hybrid Battery Pack
0x0A81:P0A81:Hybrid Battery Pack Cooling Fan 1 Control Circuit/Disconnect
0x0A82:P0A82:Mixed Battery Pack Cooling Fan 1 Performance or stuck off
0x0A83:P0A83:Mixed Battery Pack Cooling Fan 1 stuck on
0x0A84:P0A84:Hybrid Battery Pack Cooling Fan 1 Control Circuit Low Voltage
0x0A85:P0A85:Mixed Battery Pack Cooling Fan 1 Control Circuit Voltage High
0x0A86:P0A86:14V Power Module Current Sensor Circuit
0x0A87:P0A87:14V Power Module Current Sensor Circuit Range/Performance
0x0A88:P0A88:14V Power Module Current Sensor Circuit Low Voltage
0x0A89:P0A89:14V Power Module Current Sensor Circuit High Voltage
0x0A8A:P0A8A:14V Power Module Current Sensor Circuit Intermittent Interruption
0x0A8B:P0A8B:14V Voltage module system voltage
0x0A8C:P0A8C:14V Voltage module system voltage unstable
0x0A8D:P0A8D:14V Voltage module system voltage low
0x0A8E:P0A8E:14V Voltage module system voltage high
0x0A8F:P0A8F:14V Voltage module system performance
0x0A90:P0A90:Drive motor A performance
0x0A91:P0A91:Drive motor B performance

0x0A92:P0A92:Hybrid Generator Performance

0x0A93:P0A93:Inverter Cooling System Performance

0x0A94:P0A94:DC/DC Converter Performance

0x0A95:P0A95:High Voltage Fuses

0x0A96:P0A96:Hybrid Battery Pack Cooling Fan 2 Control Circuitry

0x0A97:P0A97:Hybrid Battery Pack Cooling Fan 2 performance or stuck off

0x0A98:P0A98:Hybrid Battery Pack Cooling Fan 2 stuck on

0x0A99:P0A99:Mixed Battery Pack Cooling Fan 2 Control Circuit Low Voltage

0x0A9A:P0A9A:Mixed Battery Pack Cooling Fan 2 Control Circuit Voltage High

0x0A9B:P0A9B:Mixed Battery Temperature Sensor A Circuit

0x0A9C:P0A9C:Hybrid Battery Temperature Sensor A Circuit Range/Performance

0x0A9D:P0A9D:Mixed Battery Temperature Sensor A Circuit Low Voltage

0x0A9E:P0A9E:Mixed Battery Temperature Sensor A Circuit High Voltage

0x0A9F:P0A9F:Mixed Battery Temperature Sensor A Circuit Intermittent/Unstable

0x0AA0:P0AA0:Mixed Battery Positive Current Contactor Circuit

0x0AA1:P0AA1:Mixed Battery Positive Current Contactor Circuit Stuck Closed

0x0AA2:P0AA2:Mixed battery positive current contactor Circuit stuck on

0x0AA3:P0AA3:Mixed battery positive current contactor Circuit

0x0AA4:P0AA4:Mixed Battery Positive Current Contactor Circuit stuck closed

0x0AA5:P0AA5:Mixed battery positive current contactor Circuit stuck on

0x0AA6:P0AA6:Mixed battery voltage system isolation error

0x0AA7:P0AA7:Mixed Battery Voltage System Isolation Sensor Circuit

0x0AA8:P0AA8:Hybrid Battery Voltage System Isolation Sensor Circuit
Range/Performance

0x0AA9:P0AA9:Hybrid Battery Voltage System Isolated Sensor Circuit Low Voltage

0x0AAA:P0AAA:Mixed Battery Voltage System Isolated Sensor Circuit Voltage
High

0x0AAB:P0AAB:Mixed Battery Voltage System Isolated Sensor Circuit Intermittent
Interruption/Unstable

0x0AAC:P0AAC:Hybrid Battery Pack Air Temperature Sensor A circuit

0x0AAD:P0AAD:Hybrid Battery Pack Air Temperature Sensor A Circuit

Range/Performance

0x0AAE:P0AAE:Mixed Battery Pack Air Temperature Sensor A Circuit Low Voltage

0x0AAF:P0AAF:Mixed Battery Pack Air Temperature Sensor A Circuit High Voltage

0x0AB0:P0AB0:Mixed Battery Pack Air Temperature Sensor A Circuit Intermittent
Interrupt/Unstable

0x0AB1:P0AB1:Mixed Battery Pack Air Temperature Sensor B Circuit

0x0AB2:P0AB2:Mixed Battery Pack Air Temperature Sensor B Circuit

Range/Performance

0x0AB3:P0AB3:Mixed Battery Pack Air Temperature Sensor B Circuit Low Voltage

0x0AB4:P0AB4:Mixed Battery Pack Air Temperature Sensor B Circuit High Voltage

0x0AB5:P0AB5:Mixed Battery Pack Air Temperature Sensor B circuit intermittent
interrupt/unstable

0x0AB6:P0AB6:Engine Mount Circuit B / Disconnected

0x0AB7:P0AB7:Engine mount circuit B Low voltage

0x0AB8:P0AB8:Engine mount circuit B high voltage

0x0AB9:P0AB9:Hybrid system performance

0x0ABA:P0ABA:Hybrid battery pack voltage sensor circuit

0x0ABB:P0ABB:Hybrid Battery Pack Voltage Sensor Circuit Range/Performance

0x0ABC:P0ABC:Hybrid Battery Pack Voltage Sensor Circuit Low Voltage

0x0ABD:P0ABD:Mixed Battery Pack Voltage Sensor Circuit High Voltage

0x0ABE:P0ABE:Mixed Battery Pack Voltage Sensor Circuit Intermittent
Interruption/Unstable

0x0ABF:P0ABF:Mixed battery pack current sensor circuit

0x0AC0:P0AC0:Mixed Battery Pack Current Sensor Circuit Range/Performance

0x0AC1:P0AC1:Mixed Battery Pack Current Sensor Circuit Low Voltage

0x0AC2:P0AC2:Mixed Battery Pack Current Sensor Circuit High Voltage

0x0AC3:P0AC3:Mixed battery pack current sensor circuit intermittent
interruption/unstable

0x0AC4:P0AC4:Hybrid System Control Module requested fault indicator

illumination

0x0AC5:P0AC5:Hybrid Battery Temperature Sensor B circuit

0x0AC6:P0AC6:Hybrid Battery Temperature Sensor B Circuit Range/Performance

0x0AC7:P0AC7:Mixed Battery Temperature Sensor B Circuit Low Voltage

0x0AC8:P0AC8:Mixed Battery Temperature Sensor B Circuit High Voltage

0x0AC9:P0AC9:Mixed Battery Temperature Sensor B Circuit Intermittent

Interruption/Unstable

0x0ACA:P0ACA:Mixed Battery Temperature Sensor C Circuit

0x0ACB:P0ACB:Hybrid Battery Temperature Sensor C Circuit Range/Performance

0x0ACC:P0ACC:Mixed Battery Temperature Sensor C Circuit Low Voltage

0x0ACD:P0ACD:Mixed Battery Temperature Sensor C Circuit High Voltage

0x0ACE:P0ACE:Mixed Battery Temperature Sensor C Circuit intermittent

interrupt/unstable

0x0ACF:P0ACF:Mixed Battery Pack Cooling Fan 3 Control Circuit

0x0AD0:P0AD0:Hybrid Battery Pack Cooling Fan 3 performance or stuck off

0x0AD1:P0AD1:Mixed Battery Pack Cooling Fan 3 stuck on

0x0AD2:P0AD2:Mixed Battery Pack Cooling Fan 3 Control Circuit Low Voltage

0x0AD3:P0AD3:Mixed Battery Pack Cooling Fan 3 Control Circuit Voltage High

0x0AD4:P0AD4:Mixed battery pack air flow system insufficient air flow

0x0AD5:P0AD5:Mixed Battery Pack Air Flow Valve A Control Circuit/Disconnect

0x0AD6:P0AD6:Mixed Battery Pack Air Flow Valve A Control Circuit

Range/Performance

0x0AD7:P0AD7:Mixed Battery Pack Air Flow Valve A Control Circuit Low Voltage

0x0AD8:P0AD8:Mixed Battery Pack Air Flow Valve A Control Circuit High Voltage

0x0AD9:P0AD9:Mixed Battery Positive Contactor Control Circuit / Disconnect

0x0ADA:P0ADA:Mixed Battery Positive Contactor Control Circuit

Range/Performance

0x0ADB:P0ADB:Mixed cell positive contactor control circuit low voltage

0x0ADC:P0ADC:Mixed Battery Positive Contactor Control Circuit High Voltage

0x0ADD:P0ADD:Mixed battery negative contactor control circuit/disconnection

0x0ADE:P0ADE:Mixed cell negative contactor control circuit range/performance

0x0ADF:P0ADF:Mixed cell negative contactor control circuit voltage low

0x0AE0:P0AE0:Mixed battery negative contactor control circuit high voltage

0x0AE1:P0AE1:Hybrid battery pre-charge contactor circuit

0x0AE2:P0AE2:Hybrid battery pre-charge contactor circuit stuck closed

0x0AE3:P0AE3:Hybrid battery pre-charge contactor circuit stuck open

0x0AE4:P0AE4:Hybrid battery pre-charge contactor control circuit

0x0AE5:P0AE5:Hybrid Battery Precharge Contactor Control Circuit

Range/Performance

0x0AE6:P0AE6:Hybrid Battery Precharge Contactor Control Circuit Low Voltage

0x0AE7:P0AE7:Hybrid Battery Precharge Contactor Control Circuit High Voltage

0x0AE8:P0AE8:Mixed Battery Temperature Sensor D Circuit

0x0AE9:P0AE9:Hybrid Battery Temperature Sensor D Circuit Range/Performance

0x0AEA:P0AEA:Mixed Battery Temperature Sensor D Circuit Low Voltage

0x0AEB:P0AEB:Mixed Battery Temperature Sensor D Circuit High Voltage

0x0AEC:P0AEC:Mixed Battery Temperature Sensor D Circuit intermittent/unstable

0x0AED:P0AED:Drive Motor Inverter Temperature Sensor A Circuit

0x0AEE:P0AEE:Drive Motor Inverter Temperature Sensor A Circuit

Scope/Performance

0x0AEF:P0AEF:Drive Motor Inverter Temperature Sensor A Circuit Low Voltage

0x0AF0:P0AF0:Drive Motor Inverter Temperature Sensor A Circuit Voltage High

0x0AF1:P0AF1:Drive motor inverter temperature sensor A circuit intermittent

interrupt/unstable

0x0AF2:P0AF2:Drive Motor Inverter Temperature Sensor B Circuit

0x0AF3:P0AF3:Drive Motor Inverter Temperature Sensor B Circuit

Range/Performance

0x0AF4:P0AF4:Drive Motor Inverter Temperature Sensor B Circuit Low Voltage

0x0AF5:P0AF5:Drive Motor Inverter Temperature Sensor B Circuit Voltage High

0x0AF6:P0AF6:Drive motor inverter temperature sensor B circuit intermittent

interrupt/unstable

0x0AF7:P0AF7:14V Power Module Internal Temperature Too High

0x0AF8:P0AF8:Mixed battery system voltage

0x0AF9:P0AF9:Mixed battery system voltage unstable

0x0AFA:P0AFA:Mixed battery system voltage low

0x0AFB:P0AFB:Mixed battery system voltage high

0x2000:P2000:NOx capture efficiency below threshold (BANK 1)

0x2001:P2001:NOx capture efficiency below threshold (BANK 2)

0x2002:P2002:Diesel particulate filter efficiency below threshold (BANK 1)

0x2003:P2003: Diesel particulate filter efficiency below threshold (BANK 2)

0x2004:P2004:Intake manifold slide control card is open (BANK 1 A)

0x2005:P2005:Intake Manifold Slide Pipe Control Clip On (BANK 2 A)

0x2006:P2006: Intake manifold slider control card is closing (BANK 1 A)

0x2007:P2007:Intake manifold slider control card is closing (BANK 2 A)

0x2008:P2008:Intake manifold slide control circuit/disconnect (BANK 1 A)

0x2009:P2009:Intake manifold slider control circuit low voltage (BANK 1 A)

0x2010:P2010:Intake manifold slide control circuit voltage high (BANK 1 A)

0x2011:P2011:Intake manifold slider control circuit/disconnection (BANK 2 A)

0x2012:P2012:Intake manifold slider control circuit low voltage (BANK 2 A)

0x2013:P2013:Intake manifold slider control circuit voltage high (BANK 2 A)

0x2014:P2014:Intake manifold rotor position sensor/switch circuit (BANK 1 A)

0x2015:P2015:Intake manifold rotor position sensor/switch circuit range/performance (BANK 1 A)

0x2016:P2016:Intake manifold rotor position sensor/switch circuit voltage low (BANK 1 A)

0x2017:P2017:Intake manifold rotor position sensor/switch circuit voltage high (BANK 1 A)

0x2018:P2018:Intake manifold rotor position sensor/switch circuit intermittently interrupted (BANK 1 A)

0x2019:P2019:Intake manifold rotor position sensor/switch circuit (BANK 2 A)

0x2020:P2020:Intake manifold rotor position sensor/switching circuit

range/performance (BANK 2 A)

0x2021:P2021:Intake Manifold Rotor Position Sensor/Switching Circuit Low Voltage
(BANK 2 A)

0x2022:P2022:Intake manifold rotor position sensor/switch circuit voltage high
(BANK 2 A)

0x2023:P2023:Intake manifold rotor position sensor/switch circuit intermittently
interrupted (BANK 2 A)

0x2024:P2024:Fuel Evaporative Emissions (EVAP) Temperature Sensor Circuit

0x2025:P2025:Fuel evaporative emissions (EVAP) temperature sensor performance

0x2026:P2026:Fuel Evaporative Emissions (EVAP) Temperature Sensor Circuit Low
Voltage

0x2027:P2027:Fuel Evaporative Emissions (EVAP) Temperature Sensor Circuit High
Voltage

0x2028:P2028:Fuel Evaporative Emissions (EVAP) Temperature Sensor Circuit
Intermittent Interruption

0x2029:P2029:Fuel combustion heater inoperable

0x2030:P2030:Fuel combustion heater performance

0x2031:P2031:Exhaust temperature sensor circuit (BANK 1 SENSOR 2)

0x2032:P2032:Exhaust temperature sensor circuit low voltage (BANK 1 SENSOR 2)

0x2033:P2033:Exhaust Temperature Sensor Circuit Voltage High (BANK 1 SENSOR
2)

0x2034:P2034:Exhaust temperature sensor circuit (BANK 2 SENSOR 2)

0x2035:P2035:Exhaust temperature sensor circuit voltage low (BANK 2 SENSOR 2)

0x2036:P2036:Exhaust temperature sensor circuit voltage high (BANK 2 SENSOR 2)

0x2037:P2037:Reductant injection air pressure sensor circuit

0x2038:P2038:Reductant Injection Air Pressure Sensor Circuit Range/Performance

0x2039:P2039:Reductant Injection Air Pressure Sensor Circuit Low Input

0x203A:P203A:Reductant Level Sensor Circuit

0x203B:P203B:Reductant Level Sensor Circuit Range/Performance

0x203C:P203C:Reductant Level Sensor Circuit Low Voltage

0x203D:P203D:Reductant Level Sensor Circuit Voltage High
0x203E:P203E:Reductant Level Sensor Circuit Intermittent Interruption
0x203F:P203F:Reductant level low
0x2040:P2040:Reductant injection air pressure sensor circuit high input
0x2041:P2041:Reductant Injection Air Pressure Sensor Circuit Intermittent Interruption
0x2042:P2042:Reductant temperature sensor circuit
0x2043:P2043:Reductant Temperature Sensor Circuit Range/Performance
0x2044:P2044:Reductant Temperature Sensor Circuit Low Input
0x2045:P2045:Reductant Temperature Sensor Circuit High Input
0x2046:P2046:Reductant Temperature Sensor Circuit Intermittent Interrupt
0x2047:P2047:Reductant Injector Circuit / Disconnect
0x2048:P2048:Reductant injector circuit low voltage
0x2049:P2049:Reductant injector circuit voltage high
0x204A:P204A:Reductant pressure sensor circuit
0x204B:P204B:Reductant Pressure Sensor Circuit Range/Performance
0x204C:P204C:Reductant Pressure Sensor Circuit Low Voltage
0x204D:P204D:Reductant Pressure Sensor Circuit High Voltage
0x204E:P204E:Reductant Pressure Sensor Circuit Intermittent Interruption/Instability
0x2050:P2050:Reductant Injector Circuit/Disconnect (BANK 2 UNIT 1)
0x2051:P2051:Reductant injector circuit low voltage (BANK 2 UNIT 1)
0x2052:P2052:Reductant injector circuit voltage high (BANK 2 UNIT 1)
0x2053:P2053:Reductant injector circuit/disconnect (BANK 1 UNIT 2)
0x2054:P2054:Reductant injector circuit voltage low (BANK 1 UNIT 2)
0x2055:P2055:Reductant injector circuit voltage high (BANK 1 UNIT 2)
0x2056:P2056:Reductant injector circuit/disconnect (BANK 2 UNIT 2)
0x2057:P2057:Reductant injector circuit voltage low (BANK 2 UNIT 2)
0x2058:P2058:Reductant injector circuit voltage high (BANK 2 UNIT 2)
0x2059:P2059:Reductant injection air pump control circuit/disconnection
0x205A:P205A:Reductant fuel tank temperature sensor circuit

0x205B:P205B:Reductant Fuel Tank Temperature Sensor Circuit Range/Performance
0x205C:P205C:Reductant Fuel Tank Temperature Sensor Circuit Low Voltage
0x205D:P205D:Reductant Fuel Tank Temperature Sensor Circuit High Voltage
0x205E:P205E:Reductant Fuel Tank Temperature Sensor Circuit Intermittent
Interruption/Unstable
0x2060:P2060:Reductant injection air pump control circuit low voltage
0x2061:P2061:Reductant Injection Air Pump Control Circuit High Voltage
0x2062:P2062:Reductant supply control circuit/disconnection
0x2063:P2063:Reductant supply control circuit voltage low
0x2064:P2064:Reductant supply control circuit voltage high
0x2065:P2065:Fuel level sensor B circuit
0x2066:P2066:Fuel Level Sensor B Performance
0x2067:P2067:Fuel Level Sensor B circuit low voltage
0x2068:P2068:Fuel Level Sensor B Circuit High Voltage
0x2069:P2069:Fuel Level Sensor B circuit intermittent interruption
0x206E:P206E:Intake Manifold Tuning (IMT) valve stuck open (BANK 2 A)
0x206F:P206F: Intake manifold tuning (IMT) valve stuck closed (BANK 2 A)
0x2070:P2070:Intake manifold tuning (IMT) valve stuck open (BANK 1 A)
0x2071:P2071:Intake manifold tuning (IMT) valve stuck closed (BANK 1 A)
0x2072:P2072:Throttle Actuator Control System - Ice Plugs
0x2073:P2073:Manifold Absolute Pressure/Air Flow Meter-Throttle Position Related
Idle Speed
0x2074:P2074:Manifold Absolute Pressure/Air Flow Meter-Throttle Position Related
High Load
0x2075:P2075:Intake Manifold Tuning (IMT) Valve Position Sensor/Switch Circuit
(BANK 1 A)
0x2076:P2076:Intake manifold tuning (IMT) valve position sensor/switch circuit
(BANK 1 A)
0x2077:P2077:Intake manifold tuning (IMT) valve position sensor/switch circuit low
voltage (BANK 1 A)

0x2078:P2078: Intake Manifold Tuning (IMT) Valve Position Sensor/Switch Circuit
High Voltage (BANK 1 A)

0x2079:P2079: Intake Manifold Tuning (IMT) Valve Position Sensor/Switch Circuit
Intermittent Interrupt (BANK 1 A)

0x207A:P207A: Intake Manifold Tuning (IMT) Valve Position Sensor/Switch Circuit
(BANK 2 A)

0x207B:P207B: Intake Manifold Tuning (IMT) Valve Position Sensor/Switching
Circuit Range/Performance (BANK 2 A)

0x207C:P207C: Intake Manifold Tuning (IMT) Valve Position Sensor/Switching
Circuit Low Voltage (BANK 2 A)

0x207D:P207D: Intake manifold tuning (IMT) valve position sensor/switch circuit
voltage high (BANK 2 A)

0x207E:P207E: Intake manifold tuning (IMT) valve position sensor/switch circuit
intermittent interrupt (BANK 2 A)

0x2080:P2080: Exhaust Temperature Sensor Circuit Range/Performance (BANK 1
SENSOR 1)

0x2081:P2081: Exhaust temperature sensor circuit intermittent interrupt (BANK 1
SENSOR 1)

0x2082:P2082: Exhaust Temperature Sensor Circuit Range/Performance (BANK 2
SENSOR 1)

0x2083:P2083: Exhaust Temperature Sensor Circuit Intermittent Interrupt (BANK 2
SENSOR 1)

0x2084:P2084: Exhaust temperature sensor circuit range/performance (BANK 1
SENSOR 2)

0x2085:P2085: Exhaust temperature sensor circuit intermittent interrupt (BANK 1
SENSOR 2)

0x2086:P2086: Exhaust temperature sensor circuit range/performance (BANK 2
SENSOR 2)

0x2087:P2087: Exhaust Temperature Sensor Circuit Intermittent Interrupt (BANK 2
SENSOR 2)

0x2088:P2088:A Camshaft position actuator control circuit low voltage (BANK 1 B)
 0x2089:P2089:A Camshaft position actuator control circuit voltage high (BANK 1 B)
 0x2090:P2090:B Camshaft position actuator control circuit voltage low (BANK 1 C)
 0x2091:P2091:B Camshaft position actuator control circuit voltage high (BANK 1 C)
 0x2092:P2092:A Camshaft position actuator control circuit voltage low (BANK 2 B)
 0x2093:P2093:A Camshaft position actuator control circuit voltage high (BANK 2 B)
 0x2094:P2094:B Camshaft position actuator control circuit voltage low (BANK 2 C)
 0x2095:P2095:B Camshaft position actuator control circuit voltage high (BANK 2 C)
 0x2096:P2096:Fuel system correction after catalyst too thin (BANK 1)
 0x2097:P2097:Fuel system correction after catalyst too thick (BANK 1)
 0x2098:P2098:Fuel system correction for over-dilution after catalyst (BANK 2)
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 0x2100:P2100:Throttle Actuator Control Motor Circuit/Disconnect
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 0x2104:P2104:Throttle Actuator Control System - Forced Idling
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 0x2106:P2106:Throttle Actuator Control System-Forced Power Limit
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 0x2108:P2108:Throttle Actuator Control Module Performance
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 0x2111:P2111:Throttle Actuator Control System-On at Open
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 0x2113:P2113:Throttle/Pedal Position Sensor B Minimum Braking Performance
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 0x2115:P2115:Throttle/pedal position sensor D Minimum braking performance
 0x2116:P2116:Throttle/pedal position sensor E Minimum braking performance
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0x2118:P2118:Throttle actuator control motor current range/performance
 0x2119:P2119:Throttle actuator control throttle body range/performance
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 0x2121:P2121:Throttle/Pedal Position Sensor/Switch D Circuit Range/Performance
 0x2122:P2122:Throttle/Pedal Position Sensor/Switch D Circuit Low Input
 0x2123:P2123:Throttle/Pedal Position Sensor/Switch D Circuit High Input
 0x2124:P2124:Throttle/Pedal Position Sensor/Switch D circuit intermittent interruption
 0x2125:P2125:Throttle/Pedal Position Sensor/Switch E Circuit
 0x2126:P2126:Throttle/Pedal Position Sensor/Switch E Circuit Range/Performance
 0x2127:P2127:Throttle/Pedal Position Sensor/Switch E Circuit Low Input
 0x2128:P2128:Throttle/Pedal Position Sensor/Switch E Circuit High Input
 0x2129:P2129:Throttle/Pedal Position Sensor/Switch E Circuit Intermittent Interrupt
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 0x2131:P2131:Throttle/Pedal Position Sensor/Switch F Circuit Range/Performance
 0x2132:P2132:Throttle/Pedal Position Sensor/Switch F Circuit Low Input
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 0x2140:P2140:Throttle/pedal position sensor/switch E/F Voltage dependence
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 0x2142:P2142:Exhaust gas recirculation throttle control circuit high voltage
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0x2146:P2146:Group A injector power circuit/disconnect
0x2147:P2147:Group A Injector Power Circuit Low Voltage
0x2148:P2148:A Group Injector Power Circuit High Voltage
0x2149:P2149:B Group injector power circuit/disconnect
0x2150:P2150:B Group Injector Power Circuit Low Voltage
0x2151:P2151:B Group Injector Power Circuit High Voltage
0x2152:P2152:C Group injector power circuit/disconnect
0x2153:P2153:C Group injector power circuit low voltage
0x2154:P2154:C Group Injector Power Circuit High Voltage
0x2155:P2155:D Group injector power circuit/disconnect
0x2156:P2156:D Group injector power circuit low voltage
0x2157:P2157:D Group injector power circuit voltage high
0x2158:P2158:Vehicle speed sensor B
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0x2160:P2160:Vehicle Speed Sensor B Circuit Low Voltage
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0x2165:P2165:Throttle/Pedal Position Sensor C Maximum Braking Performance
0x2166:P2166:Throttle/pedal position sensor D Maximum braking performance
0x2167:P2167:Throttle/pedal position sensor E Maximum braking performance
0x2168:P2168:Throttle/pedal position sensor F Maximum braking performance
0x2169:P2169:Exhaust regulator vent solenoid control circuit/disconnection
0x2170:P2170:Exhaust regulator vent solenoid control circuit voltage low
0x2171:P2171:Exhaust regulator ventilation solenoid control circuit voltage high
0x2172:P2172:Throttle actuator control system - sudden high airflow detected
0x2173:P2173:Throttle actuator control system - high airflow detected
0x2174:P2174:Throttle actuator control system - sudden low airflow detected
0x2175:P2175:Throttle actuator control system - low airflow detected

0x2176:P2176:Throttle Actuator Control System-Idle Position Not Learned
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0x2178:P2178:System too rich shutdown idle (BANK 1)
0x2179:P2179:Idle off with system too thin (BANK 2)
0x2180:P2180:System rich idle off (BANK 2)
0x2181:P2181:Cooling System Performance
0x2182:P2182:Engine Coolant Temperature Sensor 2 Circuit
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0x2184:P2184:Engine Coolant Temperature Sensor 2 Circuit Low Voltage
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0x2187:P2187:System over thin idle (BANK 1)
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0x2189:P2189:System over-dilute idle (BANK 2)
0x2190:P2190:System over idle (BANK 2)
0x2191:P2191:System over-dilute high load (BANK 1)
0x2192:P2192:System overdensity high load (BANK 1)
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0x2195:P2195:Oxygen sensor signal deviation/stuck overdilution (BANK 1 SENSOR 1)
0x2196:P2196:Oxygen sensor signal deviation/stuck too thick (BANK 1 SENSOR 1)
0x2197:P2197:Oxygen sensor signal deviation/stuck too thin (BANK 2 SENSOR 1)
0x2198:P2198:Oxygen sensor signal deviation/stuck too thick (BANK 2 SENSOR 1)
0x2199:P2199:Intake air temperature sensor 1/2 correlation
0x2200:P2200:NOx Sensor Circuit (BANK 1)
0x2201:P2201:NOx Sensor Circuit Range/Performance (BANK 1)
0x2202:P2202:NOx Sensor Circuit Low Input (BANK 1)
0x2203:P2203:NOx Sensor Circuit High Input (BANK 1)
0x2204:P2204:NOx sensor circuit input intermittent interrupt (BANK 1)

0x2205:P2205:NOx sensor heater control circuit/interrupt (BANK 1)
 0x2206:P2206:NOx sensor heater control circuit low voltage (BANK 1)
 0x2207:P2207:NOx sensor heater control circuit voltage high (BANK 1)
 0x2208:P2208:NOx sensor heater sense circuit (BANK 1)
 0x2209:P2209:NOx sensor heater sense circuit range/performance (BANK 1)
 0x2210:P2210:NOx Sensor Heater Sense Circuit Low Input (BANK 1)
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 0x2212:P2212:NOx sensor heater sense circuit intermittent interrupt (BANK 1)
 0x2213:P2213:NOx Sensor Circuit (BANK 2)
 0x2214:P2214:NOx Sensor Circuit Range/Performance (BANK 2)
 0x2215:P2215:NOx Sensor Circuit Low Input (BANK 2)
 0x2216:P2216:NOx Sensor Circuit High Input (BANK 2)
 0x2217:P2217:NOx sensor circuit input intermittent interrupt (BANK 2)
 0x2218:P2218:NOx Sensor Heater Control Circuit/Interrupt (BANK 2)
 0x2219:P2219:NOx sensor heater control circuit low voltage (BANK 2)
 0x2220:P2220:NOx sensor heater control circuit voltage high (BANK 2)
 0x2221:P2221:NOx sensor heater sense circuit (BANK 2)
 0x2222:P2222:NOx sensor heater sense circuit range/performance (BANK 2)
 0x2223:P2223:NOx Sensor Heater Sense Circuit Low Voltage (BANK 2)
 0x2224:P2224:NOx sensor heater sense circuit voltage high (BANK 2)
 0x2225:P2225:NOx sensor heater sense circuit intermittent interrupt (BANK 2)
 0x2226:P2226:Atmospheric pressure circuit
 0x2227:P2227:Atmospheric Pressure Circuit Scope/Performance
 0x2228:P2228:Atmospheric pressure circuit low voltage
 0x2229:P2229:Atmospheric Pressure Circuit Voltage High
 0x2230:P2230:Atmospheric pressure circuit intermittent interruption
 0x2231:P2231:Oxygen sensor signal circuit shorted to heater circuit (BANK 1
 SENSOR 1)
 0x2232:P2232:Oxygen sensor signal circuit shorted to heater circuit (BANK 1
 SENSOR 2)

0x2233:P2233:Oxygen sensor signal circuit to heater circuit short circuit (BANK 1 SENSOR 3)

0x2234:P2234:Oxygen sensor signal circuit to heater circuit short circuit (BANK 2 SENSOR 1)

0x2235:P2235:Oxygen sensor signal circuit to heater circuit short circuit (BANK 2 SENSOR 2)

0x2236:P2236:Oxygen sensor signal circuit to heater circuit short circuit (BANK 2 SENSOR 3)

0x2237:P2237:Oxygen sensor positive current control circuit/disconnect (BANK 1 SENSOR 1)

0x2238:P2238:Oxygen sensor positive current control circuit low voltage (BANK 1 SENSOR 1)

0x2239:P2239:Oxygen sensor positive current control circuit voltage high (BANK 1 SENSOR 1)

0x2240:P2240:Oxygen sensor positive current control circuit/disconnection (BANK 2 SENSOR 1)

0x2241:P2241:Oxygen sensor positive current control circuit voltage low (BANK 2 SENSOR 1)

0x2242:P2242:Oxygen sensor positive current control circuit voltage high (BANK 2 SENSOR 1)

0x2243:P2243:Oxygen sensor reference voltage circuit/disconnection (BANK 1 SENSOR 1)

0x2244:P2244:Oxygen sensor reference voltage performance (BANK 1 SENSOR 1)

0x2245:P2245:Oxygen sensor reference circuit voltage low (BANK 1 SENSOR 1)

0x2246:P2246:Oxygen sensor reference circuit voltage high (BANK 1 SENSOR 1)

0x2247:P2247:Oxygen sensor reference voltage circuit/disconnect (BANK 2 SENSOR 1)

0x2248:P2248:Oxygen sensor reference voltage performance (BANK 2 SENSOR 1)

0x2249:P2249:Oxygen sensor reference circuit voltage low (BANK 2 SENSOR 1)

0x2250:P2250:Oxygen sensor reference circuit voltage high (BANK 2 SENSOR 1)

0x2251:P2251:Oxygen sensor negative current control circuit/disconnection (BANK 1 SENSOR 1)

0x2252:P2252:Oxygen sensor negative current control circuit voltage low (BANK 1 SENSOR 1)

0x2253:P2253:Oxygen sensor negative current control circuit voltage high (BANK 1 SENSOR 1)

0x2254:P2254:Oxygen sensor negative current control circuit/disconnection (BANK 2 SENSOR 1)

0x2255:P2255:Oxygen sensor negative current control circuit voltage low (BANK 2 SENSOR 1)

0x2256:P2256:Oxygen sensor negative current control circuit voltage high (BANK 2 SENSOR 1)

0x2257:P2257:Secondary Air Injection Control A Circuit Low Voltage

0x2258:P2258:Secondary Air Injection Control A Circuit High Voltage

0x2259:P2259:Secondary Air Injection Control B Circuit Voltage Low

0x2260:P2260:Secondary Air Injection Control B Circuit High Voltage

0x2261:P2261:Exhaust Turbocharger/Mechanical Supercharger Bypass Valve - Mechanical

0x2262:P2262:Exhaust gas turbocharger/mechanical supercharger boost not detected-mechanical

0x2263:P2263:Exhaust gas turbocharger/mechanical supercharger boost system performance

0x2264:P2264:Water-laden fuel sensor circuitry

0x2265:P2265:Water-laden fuel sensor circuit range/performance

0x2266:P2266:Watery Fuel Sensor Circuit Low Voltage

0x2267:P2267:Watery Fuel Sensor Circuit High Voltage

0x2268:P2268:Watery Fuel Sensor Circuit Intermittent Interruption

0x2269:P2269:Fuel with water status

0x2270:P2270:Oxygen sensor signal stuck thin (BANK 1 SENSOR 2)

0x2271:P2271:Oxygen sensor signal stuck thick (BANK 1 SENSOR 2)

0x2272:P2272:Oxygen sensor signal stalled (BANK 2 SENSOR 2)
0x2273:P2273:Oxygen sensor signal stuck thick (BANK 2 SENSOR 2)
0x2274:P2274:Oxygen sensor signal stalled thin (BANK 1 SENSOR 3)
0x2275:P2275:Oxygen sensor signal stuck thick (BANK 1 SENSOR 3)
0x2276:P2276:Oxygen Sensor Signal Stuck Thin (BANK 2 SENSOR 3)
0x2277:P2277:Oxygen sensor signal stuck thick (BANK 2 SENSOR 3)
0x2278:P2278:BANK 1 SENSOR 3 / BANK 2 SENSOR 3 Signal Exchange
0x2279:P2279:Intake system leakage
0x2280:P2280:Airflow resistance/air leak between air filter and air flow meter
0x2281:P2281:Air leak between air flow meter and throttle body, intake valve
0x2282:P2282:Air leakage between air filter and throttle valve
0x2283:P2283:Injector control pressure sensor circuit
0x2284:P2284:Injector Control Pressure Sensor Circuit Range/Performance
0x2285:P2285:Injector Control Pressure Sensor Circuit Low Voltage
0x2286:P2286:Injector Control Pressure Sensor Circuit Voltage High
0x2287:P2287:Injector Control Pressure Sensor Circuit Intermittent Interruption
0x2288:P2288:Injector control pressure too high
0x2289:P2289:Injector control pressure too high - engine shutdown
0x2290:P2290:Injector control pressure too low
0x2291:P2291:Injector control pressure too low-engine startup
0x2292:P2292:Injector control pressure unstable
0x2293:P2293:Fuel pressure regulator 2 performance
0x2294:P2294:Fuel pressure regulator 2 control circuit
0x2295:P2295:Fuel pressure regulator 2 control circuit low voltage
0x2296:P2296:Fuel Pressure Regulator 2 Control Circuit High Voltage
0x2297:P2297:Oxygen sensor out of range in deceleration (BANK 1 SENSOR 1)
0x2298:P2298:Oxygen sensor out of range in deceleration (BANK 2 SENSOR 1)
0x2299:P2299:Brake Pedal Position / Gas Pedal Position Incompatible
0x2300:P2300:Ignition Coil A Primary Control Circuit Low Voltage
0x2301:P2301:Ignition Coil A Primary Control Circuit High Voltage

0x2302:P2302:Ignition Coil A Secondary Circuit
0x2303:P2303:Ignition coil B Primary control circuit voltage low
0x2304:P2304:Ignition Coil B Primary Control Circuit High Voltage
0x2305:P2305:Ignition Coil B Secondary Circuit
0x2306:P2306:Ignition coil C Primary control circuit voltage low
0x2307:P2307:Ignition coil C Primary control circuit voltage high
0x2308:P2308:Ignition Coil C Secondary Circuit
0x2309:P2309:Ignition coil D Primary control circuit voltage low
0x2310:P2310:Ignition coil D Primary control circuit voltage high
0x2311:P2311:Ignition coil D Secondary circuit
0x2312:P2312:Ignition coil E Primary control circuit voltage low
0x2313:P2313:Ignition coil E Primary control circuit voltage high
0x2314:P2314:Ignition coil E Secondary circuit
0x2315:P2315:Ignition coil F Primary control circuit voltage low
0x2316:P2316:Ignition coil F Primary control circuit voltage high
0x2317:P2317:Ignition Coil F Secondary Circuit
0x2318:P2318:Ignition coil G Primary control circuit voltage low
0x2319:P2319:Ignition coil G Primary control circuit voltage high
0x2320:P2320:Ignition Coil G Secondary Circuit
0x2321:P2321:Ignition coil H Primary control circuit voltage low
0x2322:P2322:Ignition coil H Primary control circuit voltage high
0x2323:P2323:Ignition coil H Secondary circuit
0x2324:P2324:Ignition coil I Primary control circuit low voltage
0x2325:P2325:Ignition coil I Primary control circuit voltage high
0x2326:P2326: Ignition Coil I Secondary Circuit
0x2327:P2327:Ignition coil J Primary control circuit voltage low
0x2328:P2328:Ignition coil J Primary control circuit voltage high
0x2329:P2329:Ignition coil J Secondary circuit
0x2330:P2330:Ignition coil K Primary control circuit voltage low
0x2331:P2331:Ignition coil K Primary control circuit voltage high

0x2332:P2332: Ignition coil K Secondary circuit

0x2333:P2333:Ignition coil L Primary control circuit voltage low

0x2334:P2334:Ignition coil L Primary control circuit voltage high

0x2335:P2335: Ignition coil L Secondary circuit

0x2336:P2336:Cylinder #1 above deflagration threshold

0x2337:P2337:Cylinder #2 above deflagration threshold

0x2338:P2338:Cylinder #3 above deflagration threshold

0x2339:P2339:Cylinder #4 above deflagration threshold

0x2340:P2340:Cylinder #5 above deflagration threshold

0x2341:P2341:Cylinder #6 above deflagration threshold

0x2342:P2342:Cylinder #7 above deflagration threshold

0x2343:P2343:Cylinder #8 above deflagration threshold

0x2344:P2344:9 Cylinder above deflagration threshold

0x2345:P2345:Cylinder #10 above deflagration threshold

0x2346:P2346:11 Cylinder above deflagration threshold

0x2347:P2347:Cylinder #12 above deflagration threshold

0x2400:P2400:Evaporative Drain System Leak Pump Detection Circuit / Disconnect

0x2401:P2401:Evaporative emissions system leak pump detection circuit low voltage

0x2402:P2402:Evaporative emissions system leak pump detection circuit voltage high

0x2403:P2403:Evaporative emissions system leak pump detection sense
circuit/disconnect

0x2404:P2404:Evaporative emissions system leak pump detection sense circuit
range/performance

0x2405:P2405:Evaporative emission system leak pump detection sense circuit low
voltage

0x2406:P2406:Evaporative emissions system leak pump detection sense circuit
voltage high

0x2407:P2407:Evaporative emissions system leak pump detection sense circuit
intermittently interrupted/unstable

0x2408:P2408:Fuel Cap Sensor/Switch Circuit

0x2409:P2409:Fuel cap sensor/switch circuit range/performance
0x2410:P2410:Fuel cap sensor/switch circuit low voltage
0x2411:P2411:Fuel cap sensor/switch circuit high voltage
0x2412:P2412:Fuel cap sensor/switch circuit intermittent interruption/unstable
0x2413:P2413:Exhaust gas recirculation system performance
0x2414:P2414:Oxygen sensor exhaust sample error (BANK 1 SENSOR 1)
0x2415:P2415:Oxygen sensor exhaust sample error (BANK 2 SENSOR 1)
0x2416:P2416:BANK 1 Cylinder 2 Oxygen Sensor / BANK 1 Cylinder 3 Oxygen
Sensor signal exchange.
0x2417:P2417:BANK 2 Oxygen Sensor / BANK 3 Oxygen Sensor Signal Exchange
0x2418:P2418:Evaporative emission system switching valve control
circuit/disconnection
0x2419:P2419:Evaporative emission system switching valve control circuit low
voltage
0x2420:P2420:Evaporative emission system switching valve control circuit voltage
high
0x2421:P2421:Evaporative emissions system vent valve stuck open
0x2422:P2422:Evaporative emission system vent valve stuck closed
0x2423:P2423:Hydrocarbon adsorption catalytic unit performance below limits
(BANK 1)
0x2424:P2424:Hydrocarbon adsorption catalytic unit performance below limits
(BANK 2)
0x2425:P2425:Exhaust gas recirculation cooling valve control circuit/disconnect
0x2426:P2426:Exhaust gas recirculation cooling valve control circuit low voltage
0x2427:P2427:Exhaust gas recirculation cooling valve control circuit voltage high
0x2428:P2428:Exhaust gas recirculation temperature too high (BANK 1)
0x2429:P2429:Exhaust gas recirculation temperature too high (BANK 2)
0x242A:P242A:Exhaust gas recirculation temperature sensor circuit (BANK 1
SENSOR 3)
0x242B:P242B:Exhaust gas recirculation temperature sensor circuit

range/performance (BANK 1 SENSOR 3)

0x242C:P242C:Exhaust gas recirculation temperature sensor circuit low voltage
(BANK 1 SENSOR 3)

0x242D:P242D:Exhaust gas recirculation temperature sensor circuit voltage high
(BANK 1 SENSOR 3)

0x242E:P242E:Exhaust gas recirculation temperature sensor circuit intermittent
interrupt/unstable (BANK 1 SENSOR 3)

0x242F:P242F:Diesel Particulate Filter Limit-Ash Accumulation

0x2430:P2430:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
(BANK 1)

0x2431:P2431:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
Range/Performance (BANK 1)

0x2432:P2432:Secondary Air Injection System Air Flow/Pressure Sensor Circuit Low
Voltage (BANK 1)

0x2433:P2433:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
Voltage High (BANK 1)

0x2434:P2434:Secondary air injection system air flow/pressure sensor circuit
intermittently interrupted/unstable (BANK 1)

0x2435:P2435:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
(BANK 2)

0x2436:P2436:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
Range/Performance (BANK 2)

0x2437:P2437:Secondary Air Injection System Air Flow/Pressure Sensor Circuit Low
Voltage (BANK 2)

0x2438:P2438:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
Voltage High (BANK 2)

0x2439:P2439:Secondary Air Injection System Air Flow/Pressure Sensor Circuit
Intermittent Interruption/Unstable (BANK 2)

0x2440:P2440:Secondary air injection system switching valve stuck open (BANK 1)

0x2441:P2441:Secondary air injection system switching valve stuck closed (BANK

1)

0x2442:P2442:Secondary air injection system switching valve stuck open (BANK 2)

0x2443:P2443:Secondary air injection system switching valve stuck closed (BANK

2)

0x2444:P2444:Secondary air injection system pump blocked open (BANK 1)

0x2445:P2445:Secondary air injection system pump stuck closed (BANK 1)

0x2446:P2446:Secondary air injection system pump blockage on (BANK 2)

0x2447:P2447:Secondary air injection system pump blockage OFF (BANK 2)

0x2448:P2448:Secondary Air Injection System High Air Flow (BANK 1)

0x2449:P2449:Secondary Air Injection System High Air Flow (BANK 2)

0x2450:P2450:Evaporative vent system switching valve performance or stuck open

0x2451:P2451:Evaporative emissions system switching valve stuck closed

0x2452:P2452:Diesel Particulate Filter Differential Pressure Sensor Circuit

0x2453:P2453:Diesel particulate filter differential pressure sensor circuit

range/performance

0x2454:P2454:Diesel Particulate Filter Differential Pressure Sensor Circuit Low

Voltage

0x2455:P2455:Diesel Particulate Filter Differential Pressure Sensor Circuit High

Voltage

0x2456:P2456:Diesel Particulate Filter Differential Pressure Sensor Circuit

Intermittent Interruption/Unstable

0x2457:P2457:Exhaust gas recirculation cooling system performance

0x2458:P2458:Diesel Particulate Filter Regeneration Duration

0x2459:P2459:Diesel particulate filter regeneration frequency

0x245A:P245A:Exhaust gas recirculation cooler bypass control circuit/disconnect

0x245B:P245B:Exhaust gas recirculation cooler bypass control circuit

range/performance

0x245C:P245C:Exhaust gas recirculation cooler bypass control circuit low voltage

0x245D:P245D:Exhaust gas recirculation cooler bypass control circuit voltage high

0x2500:P2500:Generator Indicator Light/L Terminal Circuit Voltage Low

0x2501:P2501:Generator Indicator/L end circuit high voltage

0x2502:P2502:Charging system voltage

0x2503:P2503:Charging system voltage low

0x2504:P2504:Charging system voltage high

0x2505:P2505:Electronic Control Module/Powertrain Control Module power input signal

0x2506:P2506:Electronic Control Module/Powertrain Control Module power input signal range/performance

0x2507:P2507:Electronic Control Module/Powertrain Control Module Power Input Signal Low

0x2508:P2508:Electronic Control Module/Powertrain Control Module power input signal high

0x2509:P2509:Electronic Control Module/Powertrain Control Module power input signal intermittent interruption

0x250A:P250A:Oil level sensor circuit

0x250B:P250B:Oil level sensor circuit range/performance

0x250C:P250C:Oil level sensor circuit low voltage

0x250D:P250D:Oil level sensor circuit voltage high

0x250E:P250E:Oil level sensor circuit intermittent interrupt/unstable

0x250F:P250F:Low oil level

0x2510:P2510:Electronic Control Module/Powertrain Control Module Power Relay Sense Circuit Range/Performance

0x2511:P2511:Electronic Control Module/Powertrain Control Module Power Relay Sense Circuit Intermittent Interruption

0x2512:P2512:Event Data Recorder Request Circuit/Interruptions

0x2513:P2513:Event Data Recorder Request Circuit Low Voltage

0x2514:P2514:Event Data Logger Request Circuit Voltage High

0x2515:P2515:A/C Refrigerant Pressure Sensor B Circuit

0x2516:P2516:A/C Refrigerant Pressure Sensor B Circuit Range/Performance

0x2517:P2517:A/C Refrigerant Pressure Sensor B Circuit Low Voltage

0x2518:P2518:A/C Refrigerant Pressure Sensor B Circuit High Voltage
0x2519:P2519:A/C Request A Circuit
0x251A:P251A:PTO Disable Switch Circuit/Disconnect
0x251B:P251B:PTO Disable Switch Circuit Low Voltage
0x251C:P251C:PTO Disable Switch Circuit High Voltage
0x251D:P251D:PTO Engine shutdown circuit/disconnect
0x251E:P251E:PTO Engine shutdown circuit voltage low
0x251F:P251F:PTO Engine shutdown circuit voltage high
0x2520:P2520:A/C Request A circuit low voltage
0x2521:P2521:A/C Request A Circuit Voltage High
0x2522:P2522:A/C Request B Circuit High
0x2523:P2523:A/C Request B Circuit Low Voltage
0x2524:P2524:A/C Request B Circuit High
0x2525:P2525:Vacuum Tank Pressure Sensor Circuit
0x2526:P2526:Vacuum Tank Pressure Sensor Circuit Range/Performance
0x2527:P2527:Vacuum Tank Pressure Sensor Circuit Low Voltage
0x2528:P2528:Vacuum Tank Pressure Sensor Circuit High Voltage
0x2529:P2529:Vacuum Tank Pressure Sensor Circuit Intermittent Interruption
0x252A:P252A:Oil quality sensor circuit
0x252B:P252B:Oil Quality Sensor Circuit Range/Performance
0x252C:P252C:Oil Quality Sensor Circuit Low Voltage
0x252D:P252D:Oil Quality Sensor Circuit High Voltage
0x252E:P252E:Oil quality sensor circuit intermittent interrupt/unstable
0x252F:P252F:Oil level too high
0x2530:P2530:Ignition switch run position circuit
0x2531:P2531:Ignition switch operation position circuit low voltage
0x2532:P2532:Ignition switch operation position circuit voltage high
0x2533:P2533:Ignition switch run/initial position circuit
0x2534:P2534:Ignition switch run/initial position circuit voltage low
0x2535:P2535:Ignition switch operation/initial position circuit voltage high

0x2536:P2536:Ignition switch accessory position circuit
0x2537:P2537:Ignition switch accessory position circuit voltage low
0x2538:P2538:Ignition switch accessory position circuit voltage high
0x2539:P2539:Low pressure fuel system sensor circuit
0x253A:P253A:PTO sense circuit/disconnect
0x253B:P253B:PTO Sense Circuit Range/Performance
0x253C:P253C:PTO Sense Circuit Low Voltage
0x253D:P253D:PTO Sense Circuit High Voltage
0x253E:P253E:PTO sense circuit intermittent interrupt/unstable
0x253F:P253F:Oil has deteriorated
0x2540:P2540:Low pressure fuel system sense circuit range/performance
0x2541:P2541:Low pressure fuel system sense circuit low voltage
0x2542:P2542:Low Pressure Fuel System Sense Circuit High Voltage
0x2543:P2543:Low pressure fuel system sense circuit intermittent interruption
0x2544:P2544:Torque manipulation request input signal A
0x2545:P2545:Torque Manipulation Request Input Signal A Range/Performance
0x2546:P2546:Torque manipulation request input signal A Low voltage
0x2547:P2547:Torque manipulation request input signal A High voltage
0x2548:P2548:Torque control request input signal B
0x2549:P2549:Torque manipulation request input signal B Range/performance
0x254A:P254A:PTO Speed Selector Sensor/Switch 1 Circuit/Broken
0x254B:P254B:PTO Speed Selector Sensor/Switch 1 Range/Performance
0x254C:P254C:PTO Speed Selector Sensor/Switch 1 Circuit Voltage Low
0x254D:P254D:PTO Speed Selector Sensor/Switch 1 Circuit Voltage High
0x254E:P254E:PTO Speed Selector Sensor/Switch 1 Circuit intermittent/unstable
0x254F:P254F:Hood Switch Circuit
0x2550:P2550:Torque Manipulation Request Input Signal B Low Voltage
0x2551:P2551:Torque steering request input signal B High voltage
0x2552:P2552:Throttle/fuel inhibit circuit
0x2553:P2553:Throttle/fuel cutoff circuit range/performance

0x2554:P2554:Throttle/Fuel Inhibit Circuit Voltage Low
0x2555:P2555:Throttle/Fuel Inhibit Circuit Voltage High
0x2556:P2556:Engine coolant level sensor/switch circuit
0x2557:P2557:Engine coolant level sensor/switch circuit range/performance
0x2558:P2558:Engine Coolant Level Sensor/Switch Circuit Low Voltage
0x2559:P2559:Engine Coolant Level Sensor/Switch Circuit High Voltage
0x255A:P255A:PTO Speed Select Sensor/Switch 2 Circuit/Disconnected
0x255B:P255B:PTO Speed Select Sensor/Switch 2 Range/Performance
0x255C:P255C:PTO Speed Select Sensor/Switch 2 Circuit Low Voltage
0x255D:P255D:PTO Speed Select Sensor/Switch 2 Circuit Voltage High
0x255E:P255E:PTO Speed Select Sensor/Switch 2 Circuit Intermittent/Unstable
0x2560:P2560:Engine coolant level low
0x2561:P2561:A/C Control Module Request Fault Indicator Lighting
0x2562:P2562:Exhaust Turbocharger Boost Control Position Sensor A Circuit
0x2563:P2563:Exhaust Turbocharger Boost Control Position Sensor A Circuit
Range/Performance
0x2564:P2564:Exhaust Turbocharger Boost Control Position Sensor A Circuit Low
Voltage
0x2565:P2565:Exhaust Turbocharger Boost Control Position Sensor A Circuit High
Voltage
0x2566:P2566:Exhaust Turbocharger Boost Control Position Sensor A Circuit
Intermittent Interruption
0x2567:P2567:Direct ozone depletion catalyst temperature sensor circuit
0x2568:P2568:Direct Ozone Depletion Catalyst Temperature Sensor Circuit
Range/Performance
0x2569:P2569:Direct ozone depletion catalyst temperature sensor circuit low voltage
0x256A:P256A:Engine Idle Selector Sensor/Switch Circuit/Disconnect
0x256B:P256B:Engine idle selector sensor/switch range/performance
0x256C:P256C:Engine idle selector sensor/switch circuit low voltage
0x256D:P256D:Engine idle selector sensor/switch circuit voltage high

0x256E:P256E:Engine idle selector sensor/switch circuit intermittently interrupted/unstable

0x2570:P2570:Direct ozone depletion catalyst temperature sensor circuit high voltage

0x2571:P2571:Direct ozone depletion catalyst temperature sensor circuit intermittently interrupted/unstable

0x2572:P2572:Direct ozone depletion catalyst degradation sensor circuit

0x2573:P2573:Direct ozone depletion catalyst degradation sensor circuit range/performance

0x2574:P2574:Direct Ozone Depletion Catalyst Deterioration Sensor Circuit Low Voltage

0x2575:P2575:Direct Ozone Depletion Catalyst Deterioration Sensor Circuit High Voltage

0x2576:P2576:Direct ozone depletion catalyst degradation sensor circuit intermittently interrupted/unstable

0x2577:P2577:Direct ozone depletion catalyst efficiency below limit

0x2578:P2578:Turbocharger speed sensor circuit

0x2579:P2579:Turbocharger Speed Sensor Circuit Range/Performance

0x2580:P2580:Turbocharger speed sensor circuit low voltage

0x2581:P2581:Turbocharger speed sensor circuit high voltage

0x2582:P2582:Turbocharger speed sensor circuit intermittent interruption

0x2583:P2583:Cruise control front distance range sensor

0x2584:P2584:Fuel Additive Control Module fault indicator request

0x2585:P2585:Fuel Additive Control Module warning indicator request

0x2586:P2586:Exhaust Turbocharger Boost Control Position Sensor B circuit

0x2587:P2587:Exhaust Turbocharger Boost Control Position Sensor B Circuit Range/Performance

0x2588:P2588:Exhaust Turbocharger Boost Control Position Sensor B Circuit Low Voltage

0x2589:P2589:Exhaust Turbocharger Boost Control Position Sensor B Circuit High Voltage

0x2590:P2590:Exhaust Turbocharger Boost Control Position Sensor B Circuit

Intermittent Interruption/Unstable

0x2600:P2600:Cooling Pump Control Circuit / Disconnected

0x2601:P2601:Cooling Pump Control Circuit Range/Performance

0x2602:P2602:Cooling pump control circuit low voltage

0x2603:P2603:Cooling pump control circuit voltage high

0x2604:P2604:Intake Preheater A Circuit Range/Performance

0x2605:P2605:Intake Preheater A Circuit/Disconnect

0x2606:P2606:Intake Preheater B Circuit Range/Performance

0x2607:P2607:Intake Preheater B Circuit Low Voltage

0x2608:P2608:Intake Preheater B Circuit High Voltage

0x2609:P2609:Intake Preheater System Performance

0x260A:P260A:PTO control circuit/disconnect

0x260B:P260B:PTO control circuit voltage low

0x260C:P260C:PTO control circuit voltage high

0x260D:P260D:PTO Engagement indicator control circuit

0x260F:P260F:Evaporative emissions system monitoring processor performance

0x2610:P2610:ECM/PCM Internal Timer Engine Performance

0x2611:P2611:A/C refrigerant distribution valve control circuit/disconnection

0x2612:P2612:A/C refrigerant distribution valve control circuit voltage low

0x2613:P2613:A/C refrigerant distribution valve control circuit voltage high

0x2614:P2614:Camshaft position signal output circuit/disconnection

0x2615:P2615:Camshaft position signal output circuit voltage low

0x2616:P2616:Camshaft position signal output circuit voltage high

0x2617:P2617:Crankshaft position signal output circuit/disconnection

0x2618:P2618:Crankshaft position signal output circuit voltage low

0x2619:P2619:Crankshaft position signal output circuit voltage high

0x2620:P2620:Throttle position output circuit/disconnection

0x2621:P2621:Throttle position output circuit voltage low

0x2622:P2622:Throttle position output circuit voltage high

0x2623:P2623:Injector pressure regulator control circuit/disconnection
0x2624:P2624:Injector pressure regulator control circuit voltage low
0x2625:P2625:Injector pressure regulator control circuit voltage high
0x2626:P2626:Oxygen sensor pump current correction circuit/disconnection (BANK 1 SENSOR 1)
0x2627:P2627:Oxygen sensor pump current correction circuit voltage low (BANK 1 SENSOR 1)
0x2628:P2628:Oxygen sensor pump current correction circuit voltage high (BANK 1 SENSOR 1)
0x2629:P2629:Oxygen sensor pump current correction circuit/disconnection (BANK 2 SENSOR 1)
0x2630:P2630:Oxygen sensor pump current correction circuit voltage low (BANK 2 SENSOR 1)
0x2631:P2631:Oxygen sensor pump current correction circuit voltage high (BANK 2 SENSOR 1)
0x2632:P2632:Fuel pump B control circuit/disconnection
0x2633:P2633:Fuel pump B control circuit voltage low
0x2634:P2634:Fuel pump B control circuit voltage high
0x2635:P2635:Fuel Pump A low flow/performance
0x2636:P2636:Fuel Pump B low flow/performance
0x2637:P2637:Torque Management Feedback Signal A
0x2638:P2638:Torque Management Feedback Signal A Range/Performance
0x2639:P2639:Torque Management Feedback Signal A Low Voltage
0x2640:P2640:Torque management feedback signal A High voltage
0x2641:P2641:Torque management feedback signal B
0x2642:P2642:Torque Management Feedback Signal B Range/Performance
0x2643:P2643:Torque management feedback signal B Low voltage
0x2644:P2644:Torque management feedback signal B High voltage
0x2645:P2645:A Rocker arm actuator control circuit/disconnection (BANK 1)
0x2646:P2646:A Rocker arm actuator system performance or stuck closed (BANK 1)

0x2647:P2647:A Rocker arm actuator system stuck on (BANK 1)

0x2648:P2648:A Rocker arm actuator control circuit low voltage (BANK 1)

0x2649:P2649:A Rocker arm actuator control circuit voltage high (BANK 1)

0x264A:P264A:A Rocker arm actuator position sensor circuit (BANK 1)

0x264B:P264B:A Rocker arm actuator position sensor circuit range/performance (BANK 1)

0x264C:P264C:A Rocker arm actuator position sensor circuit low voltage (BANK 1)

0x264D:P264D:A Rocker arm actuator position sensor circuit voltage high (BANK 1)

0x264E:P264E:A Rocker arm actuator position sensor circuit intermittently interrupted/unstable (BANK 1)

0x2650:P2650:B Rocker arm actuator control circuit/disconnect (BANK 1)

0x2651:P2651:B Rocker arm actuator system performance or stuck off (BANK 1)

0x2652:P2652:B Rocker arm actuator system stuck on (BANK 1)

0x2653:P2653:B Rocker arm actuator control circuit low voltage (BANK 1)

0x2654:P2654:B Rocker arm actuator control circuit voltage high (BANK 1)

0x2655:P2655:A Rocker arm actuator control circuit/disconnection (BANK 2)

0x2656:P2656:A Rocker arm actuator system performance or stuck closed (BANK 2)

0x2657:P2657:A Rocker arm actuator system stuck on (BANK 2)

0x2658:P2658:A Rocker arm actuator control circuit low voltage (BANK 2)

0x2659:P2659:A Rocker arm actuator control circuit voltage high (BANK 2)

0x265A:P265A:B Rocker arm actuator position sensor circuit (BANK 1)

0x265B:P265B:B Rocker arm actuator position sensor circuit range/performance (BANK 1)

0x265C:P265C:B Rocker arm actuator position sensor circuit low voltage (BANK 1)

0x265D:P265D:B Rocker arm actuator position sensor circuit voltage high (BANK 1)

0x265E:P265E:B Rocker arm actuator position sensor circuit intermittent interrupt/unstable (BANK 1)

0x2660:P2660:B Rocker arm actuator control circuit/disconnection (BANK 2)

0x2661:P2661:B Rocker arm actuator system performance or stuck closed (BANK 2)

0x2662:P2662:B Rocker arm actuator system stuck on (BANK 2)

0x2663:P2663:B Rocker arm actuator control circuit low voltage (BANK 2)
0x2664:P2664:B Rocker arm actuator control circuit voltage high (BANK 2)
0x2665:P2665:Fuel shutoff valve B control circuit/disconnection
0x2666:P2666:Fuel shutoff valve B control circuit voltage low
0x2667:P2667:Fuel shutoff valve B control circuit voltage high
0x2668:P2668:Fuel mode indicator light control circuit
0x2669:P2669:Actuator power supply B circuit/disconnect
0x266A:P266A:A Rocker arm actuator position sensor circuit (BANK 2)
0x266B:P266B:A Rocker Arm Actuator Position Sensor Circuit Range/Performance (BANK 2)
0x266C:P266C:A Rocker arm actuator position sensor circuit low voltage (BANK 2)
0x266D:P266D:A Rocker arm actuator position sensor circuit voltage high (BANK 2)
0x266E:P266E:A Rocker arm actuator position sensor circuit intermittently interrupted/unstable (BANK 2)
0x2670:P2670:Actuator power supply B circuit circuit voltage low
0x2671:P2671:Actuator power supply B circuit circuit voltage high
0x2672:P2672: Injection pump timing shift
0x2673:P2673:Injector pump timing calibration not learned
0x2674:P2674:Injector pump fuel calibration unlearned
0x2675:P2675:Air filter intake circuit/disconnect
0x2676:P2676:Air filter intake circuit low voltage
0x2677:P2677:Air filter intake circuit voltage high
0x2678:P2678:Coolant exhaust valve control circuit/disconnect
0x2679:P2679:Coolant exhaust valve control circuit voltage low
0x267A:P267A:B Rocker arm actuator position sensor circuit (BANK 2)
0x267B:P267B:B Rocker arm actuator position sensor circuit range/performance (BANK 2)
0x267C:P267C:B Rocker arm actuator position sensor circuit low voltage (BANK 2)
0x267D:P267D:B Rocker actuator position sensor circuit high voltage (BANK 2)
0x267E:P267E:B Rocker arm actuator position sensor circuit intermittently

interrupted/unstable (BANK 2)

0x2680:P2680:Coolant exhaust valve control circuit high voltage

0x2681:P2681:Engine coolant bypass valve control circuit/disconnection

0x2682:P2682:Engine coolant bypass valve control circuit voltage low

0x2683:P2683:Engine coolant bypass valve control circuit voltage high

0x2684:P2684:Actuator Power Supply C Circuit/Disconnect

0x2685:P2685:Actuator Power Supply C Circuit Low Voltage

0x2686:P2686:Actuator Power Supply C Circuit Voltage High

0x2700:P2700:Transmission Friction Element A Time Application

Range/Performance

0x2701:P2701:Transmission Friction Element B Time Application

Range/Performance

0x2702:P2702:Transmission Friction Element C Time Application

Range/Performance

0x2703:P2703:Friction Elements for Transmissions C Time Range of
Application/Performance

0x2704:P2704:Friction Elements for Transmissions E Time Range of
Application/Performance

0x2705:P2705:Friction Elements for Transmissions F Time Range of
Application/Performance

0x2706:P2706:Shift Solenoid F

0x2707:P2707:Shift Solenoid F Performance or Stuck Closed

0x2708:P2708:Shift Solenoid F Stuck Open

0x2709:P2709:Shift Solenoid F Electrical

0x2710:P2710:Shift Solenoid F Intermittent Interruption

0x2711:P2711:Unexpected Mechanical Gear Disengagement

0x2712:P2712:Hydraulic Power Unit Drain

0x2713:P2713:Pressure Control Solenoid D

0x2714:P2714:Pressure Control Solenoid Valve D Performance or Stuck Closed

0x2715:P2715:Pressure Control Solenoid Valve D Stuck Open

0x2716:P2716:Pressure Control Solenoid Valve D Electrical

0x2717:P2717:Pressure Control Solenoid Valve D Intermittent Interruption

0x2718:P2718:Pressure Control Solenoid Valve D Control Circuit/Disconnect

0x2719:P2719:Pressure Control Solenoid Valve D Control Circuit

Range/Performance

0x2720:P2720:Pressure Control Solenoid Valve D Control Circuit Voltage Low

0x2721:P2721:Pressure Control Solenoid Valve D Control Circuit Voltage High

0x2722:P2722:Pressure Control Solenoid Valve E

0x2723:P2723:Pressure Control Solenoid Valve E Performance or Stuck Closed

0x2724:P2724:Pressure Control Solenoid Valve E Stuck On

0x2725:P2725:Pressure Control Solenoid E Electrical

0x2726:P2726:Pressure Control Solenoid E Intermittent Interrupt

0x2727:P2727:Pressure Control Solenoid Valve E Control Circuit/Disconnect

0x2728:P2728:Pressure Control Solenoid Valve E Control Circuit Range/Performance

0x2729:P2729:Pressure Control Solenoid Valve E Control Circuit Low Voltage

0x2730:P2730:Pressure Control Solenoid Valve E Control Circuit Voltage High

0x2731:P2731:Pressure Control Solenoid Valve F

0x2732:P2732:Pressure Control Solenoid Valve F Performance or Stuck Closed

0x2733:P2733:Pressure Control Solenoid Valve F Stuck Open

0x2734:P2734:Pressure Control Solenoid Valve F Electrical

0x2735:P2735:Pressure Control Solenoid Valve F Intermittent Interruption

0x2736:P2736:Pressure Control Solenoid Valve F Control Circuit/Break Circuit

0x2737:P2737:Pressure Control Solenoid Valve F Control Circuit Range/Performance

0x2738:P2738:Pressure Control Solenoid Valve F Control Circuit Low Voltage

0x2739:P2739:Pressure Control Solenoid Valve F Control Circuit Voltage High

0x273A:P273A:Transmission Friction Element G Time Application

Range/Performance

0x273B:P273B:Transmission Friction Element H Time Application

Range/Performance

0x2740:P2740:Transmission Fluid Temperature Sensor B Circuit

0x2741:P2741:Transmission Fluid Temperature Sensor B Circuit Range/Performance

0x2742:P2742:Driveline Temperature Sensor B Circuit Low Voltage

0x2743:P2743:Driveline Temperature Sensor B Circuit High Voltage

0x2744:P2744:Driveline Temperature Sensor B Circuit Intermittent Interruption

0x2745:P2745:Intermediate Shaft Speed Sensor B Circuit

0x2746:P2746:Intermediate Shaft Speed Sensor B Circuit Range/Performance

0x2747:P2747:Intermediate Shaft Speed Sensor B Circuit No Signal

0x2748:P2748:Intermediate Axis Speed Sensor B Circuit Intermittent Interruption

0x2749:P2749:Intermediate Axis Speed Sensor C Circuit

0x2750:P2750:Intermediate Shaft Speed Sensor C Circuit Range/Performance

0x2751:P2751:Intermediate Shaft Speed Sensor C Circuit No Signal

0x2752:P2752:Intermediate Axis Speed Sensor C Circuit Intermittent Interruption

0x2753:P2753:Transmission Fluid Cooler Control Circuit/Disconnect

0x2754:P2754:Transmission fluid Cooler Control Circuit Low Voltage

0x2755:P2755:Transmission fluid Cooler Control Circuit Voltage High

0x2756:P2756:Torque Converter Clutch Pressure Control solenoid Valve

0x2757:P2757:Torque Converter Clutch Pressure Control solenoid Control Circuit
Performance or Stuck Closed

0x2758:P2758:Torque Converter Clutch Pressure Control solenoid Control Circuit
Stuck On

0x2759:P2759:Torque Converter Clutch Pressure Control Solenoid Control Circuit
Electrical

0x2760:P2760:Torque Converter Clutch Pressure Control Solenoid Control Circuit
Intermittently Interrupted

0x2761:P2761:Torque Converter Clutch Pressure Control Solenoid Control Circuit w/
Interruption

0x2762:P2762:Torque Converter Clutch Pressure Control Solenoid Control Circuit
Range/Performance

0x2763:P2763:Torque Converter Clutch Pressure Control Solenoid Control Circuit
High Voltage

0x2764:P2764:Torque Converter Clutch Pressure Control Solenoid Control Circuit
Voltage Low

0x2765:P2765:Input/Turbo Speed Sensor B circuit

0x2766:P2766:Input/Turbo Speed Sensor B Circuit Range/Performance

0x2767:P2767:Input/Turbo Speed Sensor B Circuit No Signal

0x2768:P2768:Input/Turbine Speed Sensor B Circuit Intermittent Interruption

0x2769:P2769:Torque Converter Clutch Circuit Low Voltage

0x2770:P2770:Torque Converter Clutch Circuit Voltage High

0x2771:P2771:4WD Low Gear Switch Circuit

0x2772:P2772:Four-Wheel Drive (4WD) Low Gear Switch Circuit
Range/Performance

0x2773:P2773:Four-Wheel Drive (4WD) Low Gear Switch Circuit Low Voltage

0x2774:P2774:Four-Wheel Drive (4WD) Low Gear Switch Circuit Voltage High

0x2775:P2775:Upshift Switch Circuit Range/performance

0x2776:P2776:Upshift Switch Circuit Voltage Low

0x2777:P2777:Upshift Switch Circuit Voltage High

0x2778:P2778:Upshift Switch Circuit Intermittent Interruptions/Instability

0x2779:P2779:Downshift Switch Circuit Range/Performance

0x2780:P2780:Downshift Switch Circuit Low Voltage

0x2781:P2781:Downshift Switch Circuit Voltage High

0x2782:P2782:Downshift Switch Circuit Intermittent Interrupt/Unstable

0x2783:P2783:Torque Converter Temperature Too High

0x2784:P2784:Input/Turbo Speed Sensor A/B Correlation

0x2785:P2785:Clutch Actuator Temperature Too High

0x2786:P2786:Shift Actuator Temperature Too High

0x2787:P2787:Clutch Temperature Too High

0x2788:P2788:Auto Shift Manual Adaptive Learning at Limit

0x2789:P2789:Clutch Adaptive Learning at Limit

0x278A:P278A:Low Gear Switch Circuit

0x278B:P278B:Low-Speed Gear Switch Circuit Range/Performance

0x278C:P278C:Low Voltage Low Gear Switch Circuit
0x278D:P278D:Low Gear Switch Circuit Voltage High
0x278E:P278E:Low Gear Switch Circuit Intermittent Interrupt/Unstable
0x278F:P278F:SAE/ISO Reserved
0x2790:P2790:Orientation Plate Selector Direction Circuit
0x2791:P2791:Positioning Plate Selector Direction Circuit Low Voltage
0x2792:P2792:Positioning Plate Selector Direction Circuit Voltage High
0x2793:P2793:Shift Direction Circuit
0x2794:P2794:Shift Direction Circuit Voltage Low
0x2795:P2795:Shift Direction Circuit Voltage High
0x2796:P2796:Subtransmission Fluid Pump Control Circuit/Disconnection
0x2797:P2797:Subtransmission Fluid Pump Performance
0x2798:P2798:Subtransmission Fluid Pump Control Circuit Voltage Low
0x2799:P2799:Sub-Drive Fluid Pump Control Circuit Voltage High
0x279A:P279A:SAE/ISO Reserved
0x279B:P279B:SAE/ISO Reserved
0x279C:P279C:SAE/ISO Reserved
0x279D:P279D:SAE/ISO Reserved
0x279E:P279E:SAE/ISO Reserved
0x279F:P279F:SAE/ISO Reserved
0x2800:P2800:Transmission Range Sensor B Circuit Failure (PRNDL Input)
0x2801:P2801:Transmission Range Sensor B Circuit Range/Performance
0x2802:P2802:Transmission Range Sensor B Circuit Low Voltage
0x2803:P2803:Transmission Range Sensor B Circuit Voltage High
0x2804:P2804:Transmission Range Sensor B Circuit Intermittent Interruption
0x2805:P2805:Transmission Range Sensor A/B Correlation
0x2806:P2806:Transmission Range Sensor Calibration
0x2807:P2807:Pressure Control Solenoid G
0x2808:P2808:Pressure Control Solenoid G Performance or Stuck Closed
0x2809:P2809:Pressure Control Solenoid G Stuck On

0x2810:P2810:Pressure Control Solenoid Valve G Electrical

0x2811:P2811:Pressure Control Solenoid Valve G Intermittent Interruption

0x2812:P2812:Pressure Control Solenoid Valve G Control Circuit/Break Circuit

0x2813:P2813:Pressure Control Solenoid Valve G Control Circuit

Range/Performance

0x2814:P2814:Pressure Control Solenoid Valve G Control Circuit Voltage Low

0x2815:P2815:Pressure Control Solenoid Valve G Control Circuit Voltage High

0x2816:P2816:Pressure Control Solenoid Valve H

0x2817:P2817:Pressure Control Solenoid Valve H Performance or Stuck Closed

0x2818:P2818:Pressure Control Solenoid Valve H Stuck Open

0x2819:P2819:Pressure Control Solenoid H Electrical

0x281A:P281A:Pressure Control Solenoid Valve H Intermittent Interruptions

0x281B:P281B:Pressure Control Solenoid Valve H Control Circuit/Disconnect

0x281C:P281C:Pressure Control Solenoid Valve H Control Circuit

Range/Performance

0x281D:P281D:Pressure Control Solenoid H Control Circuit Voltage Low

0x281E:P281E:Pressure Control Solenoid Valve H Control Circuit Voltage High

0x281F:P281F:Pressure Control Solenoid Valve J

0x2820:P2820:Pressure Control Solenoid Valve J Performance or Stuck Closed

0x2821:P2821:Pressure Control Solenoid Valve J Stuck Open

0x2822:P2822:Pressure Control Solenoid Valve J Electrical

0x2823:P2823:Pressure Control Solenoid J Intermittent Interrupt

0x2824:P2824:Pressure Control Solenoid Valve J Control Circuit/Disconnect

0x2825:P2825:Pressure Control Solenoid Valve J Control Circuit Range/Performance

0x2826:P2826:Pressure Control Solenoid J Control Circuit Voltage Low

0x2827:P2827:Pressure Control Solenoid Valve J Control Circuit Voltage High

0x2828:P2828:Pressure Control Solenoid Valve K

0x2829:P2829:Pressure Control Solenoid Valve K Performance or Stuck Closed

0x282A:P282A:Pressure Control Solenoid Valve K Stuck Open

0x282B:P282B:Pressure Control Solenoid K Electrical

0x282C:P282C:Pressure Control Solenoid K Intermittent Interruptions
 0x282D:P282D:Pressure Control Solenoid K Control Circuit/Disconnect
 0x282E:P282E:Pressure Control Solenoid K Control Circuit Range/Performance
 0x282F:P282F:Pressure Control Solenoid K Control Circuit Voltage Low
 0x2830:P2830:Pressure Control Solenoid K Control Circuit Voltage High
 0x2A00:P2A00:Oxygen Sensor Circuit Range/Performance (BANK 1 SENSOR 1)
 0x2A01:P2A01:Oxygen Sensor Circuit Range/Performance (BANK 1 SENSOR 2)
 0x2A02:P2A02:Oxygen Sensor Circuit Range/Performance (BANK 1 SENSOR 3)
 0x2A03:P2A03:Oxygen Sensor Circuit Range/Performance (BANK 2 SENSOR 1)
 0x2A04:P2A04:Oxygen Sensor Circuit Range/Performance (BANK 2 SENSOR 2)
 0x2A05:P2A05:Oxygen Sensor Circuit Range/Performance (BANK 2 SENSOR 3)
 0x2A06:P2A06:Oxygen Sensor Negative Voltage (BANK 1 SENSOR 1)
 0x2A07:P2A07:Oxygen Sensor Negative Voltage (BANK 1 SENSOR 2)
 0x2A08:P2A08:Oxygen Sensor Negative Voltage (BANK 1 SENSOR 3)
 0x2A09:P2A09:Oxygen Sensor Negative Voltage (BANK 2 SENSOR 1)
 0x2A10:P2A10:Oxygen Sensor Negative Voltage (BANK 2 SENSOR 2)
 0x2A11:P2A11:Oxygen Sensor Negative Voltage (BANK 2 SENSOR 3)
 0x3400:P3400:Cylinder Break (BANK 1)
 0x3401:P3401:Cylinder 1 Disconnect/Intake Valve Control Circuit/Disconnect
 0x3402:P3402:Cylinder 1 Disconnect/Inlet Valve Control Performance
 0x3403:P3403:Cylinder 1 Disconnect/Inlet Valve Control Circuit Low Voltage
 0x3404:P3404:Cylinder 1 Break/Intake Valve Control Circuit Voltage High
 0x3405:P3405:Cylinder 1 Exhaust Valve Control Circuit/Disconnect
 0x3406:P3406:Cylinder 1 Exhaust Valve Control Performance
 0x3407:P3407:Cylinder 1 Exhaust Valve Control Circuit Voltage Low
 0x3408:P3408:Cylinder 1 Exhaust Valve Control Circuit Voltage High
 0x3409:P3409:Cylinder 2 Disconnect/Inlet Valve Control Circuit/Disconnect
 0x3410:P3410:Cylinder 2 Disconnect/Inlet Valve Control Performance
 0x3411:P3411:Cylinder 2 Disconnect/Intake Valve Control Circuit Low Voltage
 0x3412:P3412:Cylinder 2 Disconnect/Intake Valve Control Circuit High Voltage

0x3413:P3413:Cylinder 2 Exhaust Valve Control Circuit/Disconnect
0x3414:P3414:Cylinder 2 Exhaust Valve Control Performance
0x3415:P3415:Cylinder 2 Exhaust Valve Control Circuit Low Voltage
0x3416:P3416:Cylinder 2 Exhaust Valve Control Circuit Voltage High
0x3417:P3417:Cylinder 3 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3418:P3418:Cylinder 3 Disconnect/Inlet Valve Control Performance
0x3419:P3419:Cylinder 3 Disconnect/Inlet Valve Control Circuit Low Voltage
0x3420:P3420:Cylinder 3 Break/Intake Valve Control Circuit High Voltage
0x3421:P3421:Cylinder 3 Exhaust Valve Control Circuit/Disconnect
0x3422:P3422:Cylinder 3 Exhaust Valve Control Performance
0x3423:P3423:Cylinder 3 Exhaust Valve Control Circuit Low Voltage
0x3424:P3424:Cylinder 3 Exhaust Valve Control Circuit Voltage High
0x3425:P3425:Cylinder 4 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3426:P3426:Cylinder 4 Disconnect/Inlet Valve Control Performance
0x3427:P3427:Cylinder 4 Disconnect/Intake Valve Control Circuit Low Voltage
0x3428:P3428:Cylinder 4 Disconnect/Intake Valve Control Circuit Voltage High
0x3429:P3429:Cylinder 4 Exhaust Valve Control Circuit/Disconnect
0x3430:P3430:Cylinder 4 Exhaust Valve Control Performance
0x3431:P3431:Cylinder 4 Exhaust Valve Control Circuit Low Voltage
0x3432:P3432:Cylinder 4 Exhaust Valve Control Circuit Voltage High
0x3433:P3433:Cylinder 5 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3434:P3434:Cylinder 5 Disconnect/Inlet Valve Control Performance
0x3435:P3435:Cylinder 5 Disconnect/Inlet Valve Control Circuit Low Voltage
0x3436:P3436:Cylinder 5 Disconnect/Intake Valve Control Circuit Voltage High
0x3437:P3437:Cylinder 5 Exhaust Valve Control Circuit/Disconnect
0x3438:P3438:Cylinder 5 Exhaust Valve Control Performance
0x3439:P3439:Cylinder 5 Air Exhaust Control Circuit Low Voltage
0x3440:P3440:Cylinder 5 Air Exhaust Valve Control Circuit High Voltage
0x3441:P3441:Cylinder 6 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3442:P3442:Cylinder 6 Disconnect/Inlet Valve Control Performance

0x3443:P3443:Cylinder 6 Cutoff/Inlet Valve Control Circuit Low Voltage
0x3444:P3444:Cylinder 6 Disconnect/Inlet Valve Control Circuit High Voltage
0x3445:P3445:Cylinder 6 Exhaust Valve Control Circuit/Disconnect
0x3446:P3446:Cylinder 6 Exhaust Valve Control Performance
0x3447:P3447:Cylinder 6 Air Exhaust Control Circuit Low Voltage
0x3448:P3448:Cylinder 6 Exhaust Valve Control Circuit Voltage High
0x3449:P3449:Cylinder 7 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3450:P3450:Cylinder 7 Disconnect/Inlet Valve Control Performance
0x3451:P3451:Cylinder 7 Disconnect/Intake Valve Control Circuit Low Voltage
0x3452:P3452:Cylinder 7 Disconnect/Intake Valve Control Circuit Voltage High
0x3453:P3453:Cylinder 7 Exhaust Valve Control Circuit/Disconnect
0x3454:P3454:Cylinder 7 Exhaust Valve Control Performance
0x3455:P3455:Cylinder 7 Exhaust Valve Control Circuit Low Voltage
0x3456:P3456:Cylinder 7 Exhaust Valve Control Circuit Voltage High
0x3457:P3457:Cylinder 8 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3458:P3458:Cylinder 8 Disconnect/Inlet Valve Control Performance
0x3459:P3459:Cylinder 8 Cutoff/Intake Valve Control Circuit Low Voltage
0x3460:P3460:Cylinder 8 Cutoff/Intake Valve Control Circuit High Voltage
0x3461:P3461:Cylinder 8 Exhaust Valve Control Circuit/Disconnect
0x3462:P3462:Cylinder 8 Exhaust Valve Control Performance
0x3463:P3463:Cylinder 8 Air Exhaust Control Circuit Low Voltage
0x3464:P3464:Cylinder 8 Air Exhaust Control Circuit High Voltage
0x3465:P3465:Cylinder 9 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3466:P3466:Cylinder 9 Disconnect/Inlet Valve Control Performance
0x3467:P3467:Cylinder 9 Disconnect/Intake Valve Control Circuit Low Voltage
0x3468:P3468:Cylinder 9 Break/Intake Valve Control Circuit High Voltage
0x3469:P3469:Cylinder 9 Exhaust Valve Control Circuit/Disconnect
0x3470:P3470:Cylinder 9 Exhaust Valve Control Performance
0x3471:P3471:Cylinder 9 Exhaust Valve Control Circuit Low Voltage
0x3472:P3472:Cylinder 9 Exhaust Valve Control Circuit Voltage High

0x3473:P3473:Cylinder 10 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3474:P3474:Cylinder 10 Disconnect/Inlet Valve Control Performance
0x3475:P3475:Cylinder 10 Disconnect/Inlet Valve Control Circuit Low Voltage
0x3476:P3476:Cylinder 10 Disconnect/Intake Valve Control Circuit High Voltage
0x3477:P3477:Cylinder 10 Exhaust Valve Control Circuit/Disconnect
0x3478:P3478:Cylinder 10 Exhaust Valve Control Performance
0x3479:P3479:Cylinder 10 Air Exhaust Control Circuit Low Voltage
0x3480:P3480:Cylinder 10 Exhaust Valve Control Circuit Voltage High
0x3481:P3481:Cylinder 11 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3482:P3482:Cylinder 11 Disconnect/Inlet Valve Control Performance
0x3483:P3483:Cylinder 11 Disconnect/Inlet Valve Control Circuit Low Voltage
0x3484:P3484:Cylinder 11 Disconnect/Intake Valve Control Circuit High Voltage
0x3485:P3485:Cylinder 11 Exhaust Valve Control Circuit/Disconnect
0x3486:P3486:Cylinder 11 Exhaust Valve Control Performance
0x3487:P3487:Cylinder 11 Exhaust Valve Control Circuit Low Voltage
0x3488:P3488:Cylinder 11 Exhaust Valve Control Circuit High Voltage
0x3489:P3489:Cylinder 12 Disconnect/Inlet Valve Control Circuit/Disconnect
0x3490:P3490:Cylinder 12 Disconnect/Inlet Valve Control Performance
0x3491:P3491:Cylinder 12 Disconnect/Intake Valve Control Circuit Low Voltage
0x3492:P3492:Cylinder 12 Disconnect/intake Valve Control Circuit voltage high
0x3493:P3493:Cylinder 12 Exhaust Valve Control Circuit/Disconnect
0x3494:P3494:Cylinder 12 Exhaust Valve Control Performance
0x3495:P3495:Cylinder 12 Exhaust Control Circuit Low Voltage
0x3496:P3496:Cylinder 12 Exhaust Control Circuit high Voltage
0x3497:P3497:Disconnect (BANK 2)